modeling system

Developed by Dynamic Solutions International LLC

CVLGrid

2D GRID GENERATOR

User's Manual

Version 1.1

Version 1.1
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1 Introduction

CVLGrid is 2-D grid generation software developed by Dynamic Solutions International, LLC for use with curvilinear orthogonal models, and in particular for use with EFDC_DSI, EFDC_SGZ, EFDC_EPA, and EFDC_Hydro models. As all EFDC models require 2D curvilinear orthogonal grids this tool has been developed to meet this need for DSI and our clients and customers. This tool has been further optimized for us with the EFDC_DSI/EFDC_Explorer Modeling System.

Grids are a key component when generating models using Navier-Stokes equations with the numerical method. This software uses Laplace equations to generate curvilinear grids that allow for more accurate modeling calculations than a standard rectangular grid. In order to achieve optimal accuracy of a model, the grid should be shaped to the model domain and maintain orthogonality. The average orthogonal deviation for grid computation should be less than 3° to maintain the accuracy of the numerical model.

This program uses the method of Successive Over-Relaxations (SOR) to solve the Laplace equations. To maximize calculation efficiency and reduce the number of iterations required, an optimum relaxation factor (ω) must be determined and used in the SOR equations. CVLGrid automatically calculates and implements this optimum relaxation factor for the current grid.

1.1 Summary of Capabilities

The following list provides a summary of the major features of CVLGrid. It is recognized that there are other capabilities and improvements that can be made to CVLGrid to increase the ease of use and effectiveness of the software. It is anticipated that new features and improvements will be made as resources are available.

- Optimized for use with EFDC_DSI, EFDC_SGZ, EFDC_EPA, and EFDC_Hydro
- Import existing EFDC Model Grids and export CVL Grids to EFDC Model format
- Load and edit an EFDC grid without changing the I,J mapping, thereby allowing boundary conditions to be maintained
- Display polygon data files in the workspace
- Define and edit polylines in the workspace
- Create, view and edit layers
- Create and edit splines which are the basis for creating curvilinear (CVL) grids
- Generate CVL grids based on the defined splines
- Once a grid is generated, the user is able to:
    - Improve orthogonality, globally or locally
    - Refine or coarsen, globally or locally
    - Regularize the cells
    - Smooth the lines
    - Reverse indices
    - Adjust grid lines
    - Add single cells to the grid
    - Move cell nodes
    - Delete cells
- Connect multiple grids
- The user can view
    - Orthogonal deviation
    - IJ coordinates
    - Cell size (Dx, Dy)
- Vary cell colors

- Label corners
- Show N, S, E, W of cells
- Measure distances
- Load geo-referenced background images
- Change legends, labels and titles
- Export metafile data

1.2 New Features in CVLGrid1.1

A number of new features have been added to CVLGrid1.1 as improvements over CVLGrid1.0. These include:

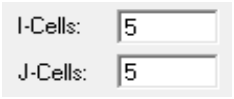
- Improved orthogonalization methodology and improved user controls. New orthogonalization options include use of Sorenson method and user defined coefficients.
- Improved and simplified layer controls
- Load digital terrain models to use topographic data in developing grids.
- Telescoping: ability to create a distributed space between grid cells based on a user defined distance weighting factor.
- Split grid: ability to split a grid domain into two grid domains, with new grid layer automatically generated on the Layer Control.
- Import and write various new grid types: load and write grid types such as ECOMSED, SEAGRID, and CH3D.
- Edit selected polyline by RMC on the polyline: add RM/RK labels,
- Many hotkeys for editing polylines
- Ability to copy polylines to other layers.
- User-interface is now simpler and friendlier to use, with streamlined Side Menu among many changes.

1.3 Conventions

1.3.1 Windows Interface

While the general use of CVLGrid is fairly standard with respect to a user interface for the Windows® operating system, some basic conventions will be explained to help the user.

Table 1-1 CVLGrid User Interface Conventions

	A white box with black text is a field that is definable by the user.
	Any Raised square or rectangular icon can be selected and each button enables a certain tool or function.
	Any icon that has a black triangle pointing down on the right side indicates that there are several options available from this icon. Click the arrow to view the functions.

1.3.2 Message Boxes

When using CVLGrid, message boxes are used to display information about the generated grid cells. To view these message boxes, click on any cell in the current grid. Immediately a yellow box appears in the workspace with information about that particular cell. This message box displays four pieces of information:

- L – The cell number starting with # 2
- I,J – Cell indices
- Cell Area – in m^2
- Corner Order – Clockwise or Counterclockwise

Only one cell's information can be viewed at a time. The message box can me moved by clicking within its border and dragging it to another spot.

1.3.3 Tooltips

CVLGrid allows the user to see the function of many icons by resting the mouse over the icon. A small box appears explaining the function of that icon.

1.3.4 Units

The base unit of CVLGrid is meters. One of the functions of CVLGrid allows the user to create legends which can be accessed from the *Display Options* menu. These legends give a visual representation of distance and can be displayed in several different units such as kilometers and miles. However, this function is just a conversion tool to visually represent different units of distance. If you click on a grid cell, the area shown is always in meters squared.

1.4 Terms and Abbreviations

The following is a description of common terms and abbreviations used in this report.

Orthogonality	A measure of how perpendicular grid cells are, 90° is optimal.
Orthogonal Deviation	The deviation angle away from 90°.
Orthogonalize	The act of “improving” a grid so that its grid lines are more perpendicular.
Refine	Increasing the cell resolution of the grid.
Coarsen	Decreasing the cell resolution of the grid.
Regularize	Applies uniform spacing between grid cells.
Telescoping	Regularize the grid with uneven spacing between grid cells.
Global	Modifications are made to the entire grid.
Local	Modifications are made to a user-specified area only.
Spline	A line that is used to define the boundaries of a grid, four splines are required to generate a grid.
Polyline	The P2D shape file that the user uploads into CVLGrid, usually to serve as an outline to draw the splines.
CVL	Curvilinear, referring to the grid
LMC	Left mouse click
RMC	Right mouse click

1.5 Coordinate System

CVLGrid uses the Cartesian coordinate system to specify cell locations. The Cartesian coordinate system specifies a point on a plane relative to two perpendicular directed lines that split the cell into four quadrants. In CVLGrid “I” represents the index of cell along the X-axis and “J” represents the one along the Y-axis, in which X, Y is the curves in general. The base unit for this coordinate system is meters.

CVLGrid also allows the user to upload a shape file of the model domain. The base coordinate system used for these files is the Universal Transverse Mercator (UTM) with base units in meters as well.

1.6 File Types

The following files are generated by, and compatible with CVLGrid.

POLYFILE.P2D	Data file of polygon coordinates based on the UTM coordinate system. This file is usually used to display the border of a geographic region (see appendix for file format).
SPLINE.SPL	This file contains the user-defined spline coordinates within the CVLGrid workspace (see appendix for file format).
GRID.CVL	The grid file that is generated by CVLGrid. These files can be loaded and edited in CVLGrid or imported into EFDC Explorer.
GRID.GRD	The grid file that is generated by Delft3D-RGFGRID. These files can be loaded and edited in CVLGrid or imported into EFDC Explorer.
PROJECT.CVP	The project file created by CVLGrid.
METAFILE.EMF	The metadata file that can be exported by CVLGrid
BITMAP.PNG	Bitmap files (*JPG, BMP, GIF, TIF)* can be exported by CVLGrid
GEOREFERENCE	CVLGrid can upload geo-referenced files such as GIS or Google earth images that display the geographic region of the model domain.
POSTING FILES	These files contain coordinate data that are linked to specific locations or features. They include P2D, PLT, DXF, DAT, and TXT files.
LABELS FILES	These files contain the names of stations or features that are linked to specific coordinates. This allows the user to label the features added by the Posting file. They include P2D, DAT, TXT and GPX files.
KML FILE (.kml)	KML is a file format used to display geographic data in an Earth browser such as Google Earth.
SHAPEFILE (.shp)	The shapefile is spatial data format and are created directly by lists of the tools available in Environmental Systems Research Institute, Inc. (ESRI), software.
LAND BOUNDARY (.ldb)	The land boundary file contains two blocks with X and Y co-ordinates of two separate pieces of a land boundary. It is user input for Delft3D-RGFGRID.
DTM FILE (.dat)	The file contains three blocks with X, Y co-ordinates and Z value for terrain elevation.
DTM FILE (.tb2)	The file is created by the Digital Terrain Model Tool (DTM_Tool), contains terrain elevation data.

2 Installation and License Management

2.1 Installation

1) If you received CVLGrid in a zip file (CVLGrid.ZIP), unzip the zip file into a *temporary directory* on your hard drive. To simplify the cleanup of files later it is recommended that the *temporary directory* be empty before unzipping.

2) You should use the CVLGrid Installation program to install CVLGrid (CVLGrid1.1_Setup_Ver160622.exe). After opening the setup program the CVLGrid Install Wizard will run as shown in Figure 2-1 below.

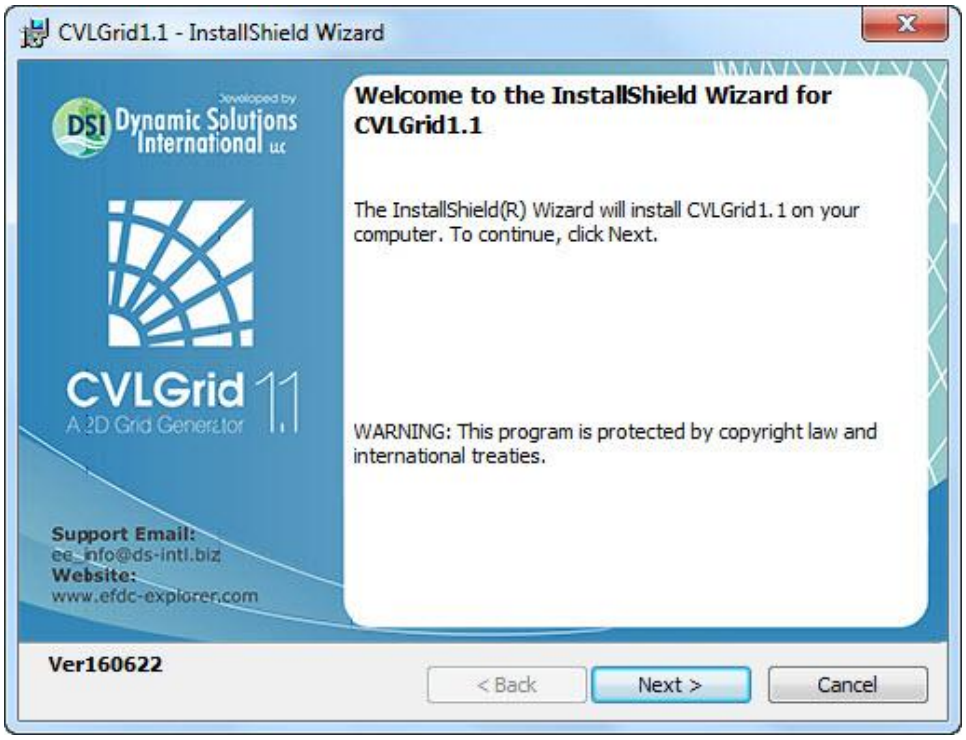


Figure 2-1 Install Wizard for CVLGrid

3) If you used a *temporary directory* you should delete the files that were unzipped. Keep a copy of the zip file in case you need to install the program again.

4) For Vista™ and Windows 7™ users, you may need to change the CVLGrid1.1.exe to "Run as Administrator" before you can use CVLGrid to run models.

5) As part of the conditions of use of CVLGrid, the user must accept the License Agreement before installation.

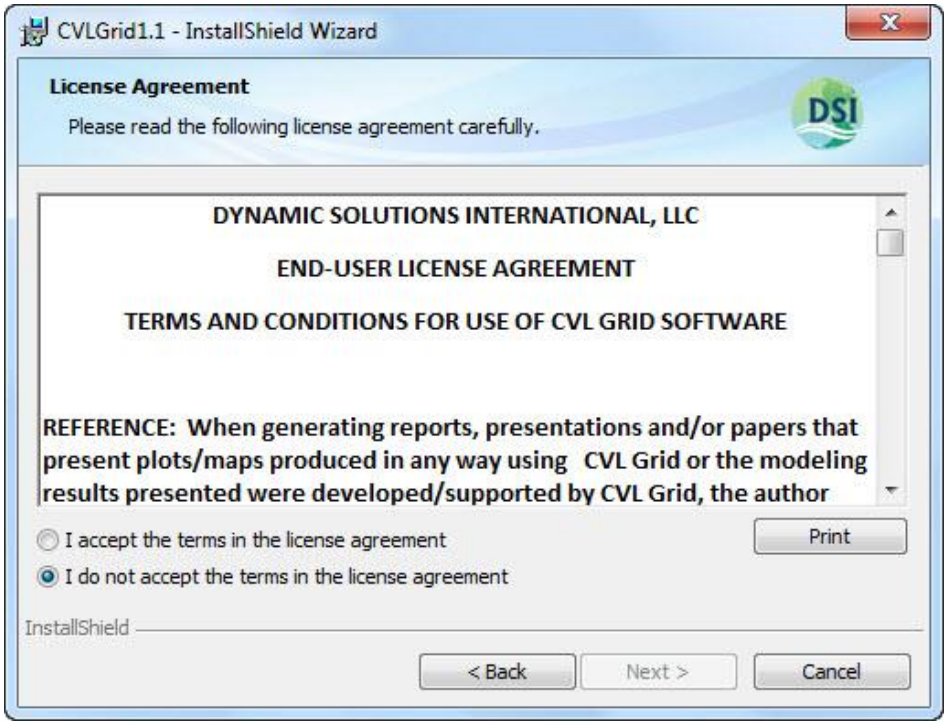


Figure 2-2 License Agreement Prompt

6) The user is then prompted for customer information. Please enter your User Name and Organization in the fields provided.

7) The user is then prompted for the Setup Option desired. A choice of “Complete” or “Custom” setup options are available. “Complete” means that all setup features will be installed, requiring more disc space. The “Custom” setup option allows the user to install just one or a combination of files, including CVLGrid program file, Datafiles, Documents and Extra files.

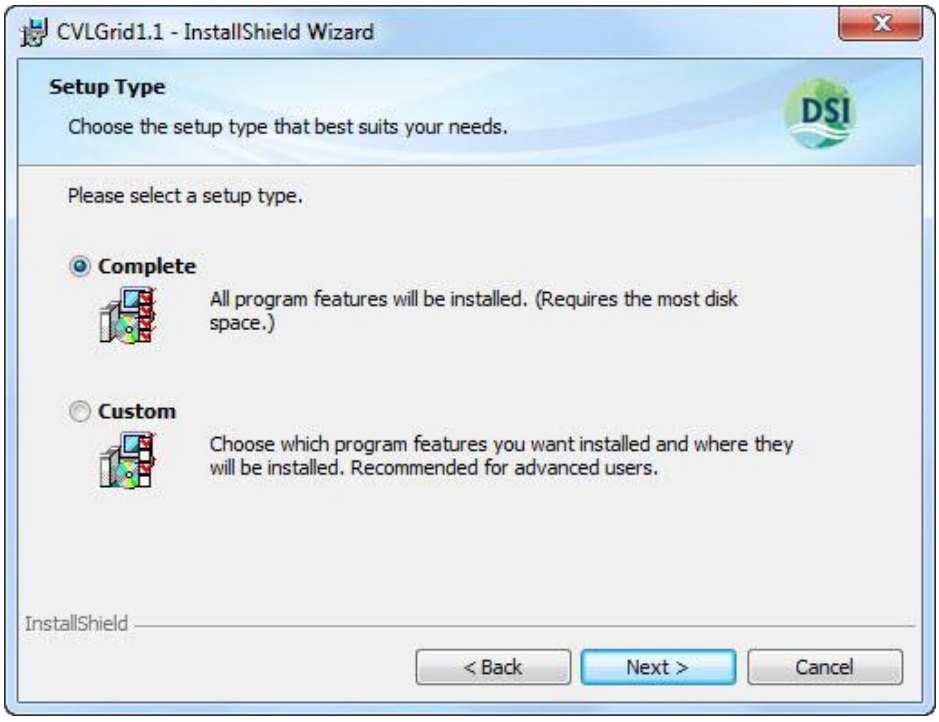


Figure 2-3 CVLGrid Install Wizard – Custom Setup

8) If the “Custom” option is used a number of features may be selected to be installed at shown in are available . Here the user may also decide on the destination folder for the setup by selecting the “Change” button. The default folder is: C:\Program Files(x86)\DSI\CVLGrid1.1\. A CVLGrid example will be saved to C:\Program Files\DSI\CVLGRID1.1\Example.

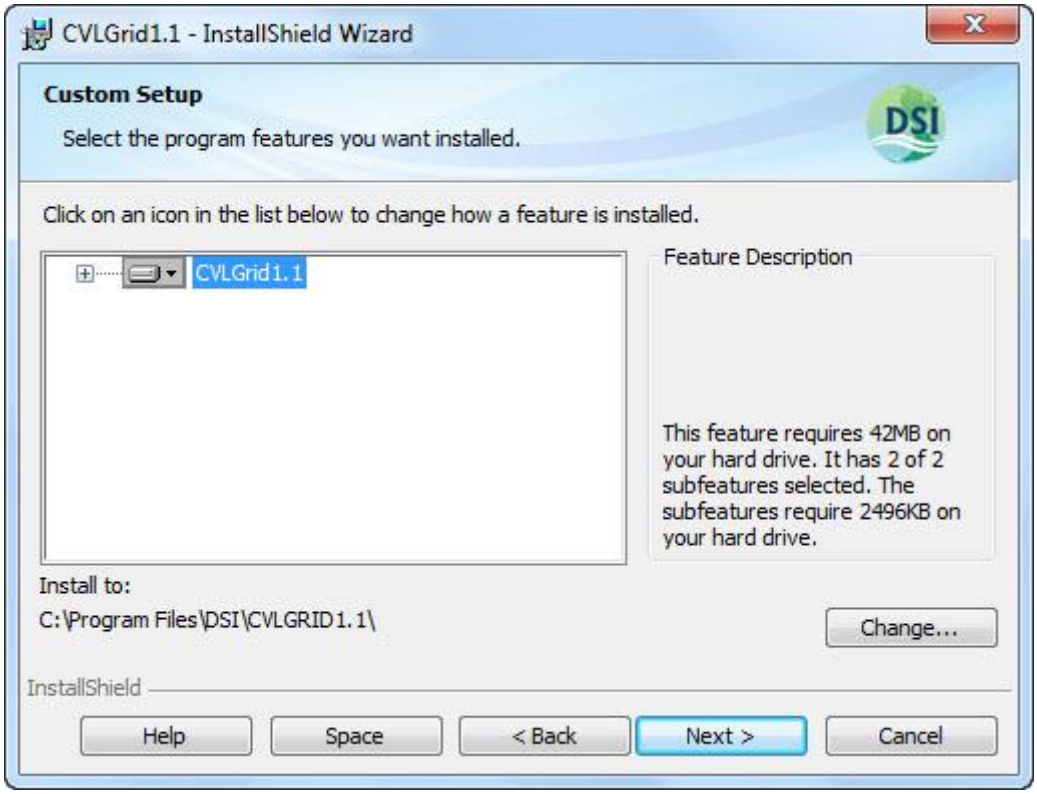


Figure 2-4 Destination folder for the setup

9) Upon completion of the Installation Wizard, the user is prompted with the message “Install Wizard Complete”. After selecting “Finish”, the user is returned to the desktop.



Figure 2-5 CVLGrid Installation Wizard Completion Screen.

2.2 License Models

CVLGrid1.1 utilizes a new licensing system. This system provides increased stability and ease of use for the users. The following license models are now supported by CVLGrid:

2.2.1 Trial License

Trial licenses are provided to people who want to assess the features of the CVLGrid. The *Trial* license does not provide the full capability of CVLGrid.

The *Trial* license is limited to 30 days duration after which the user must contact DSI to purchase a *Perpetual* license. The CVLGrid License Manager informs the user with the *Trial* license about the expiration as shown in Figure 2-6. It should be noted that the *Trial* license cannot be transferred to another machine once the license has been activated.

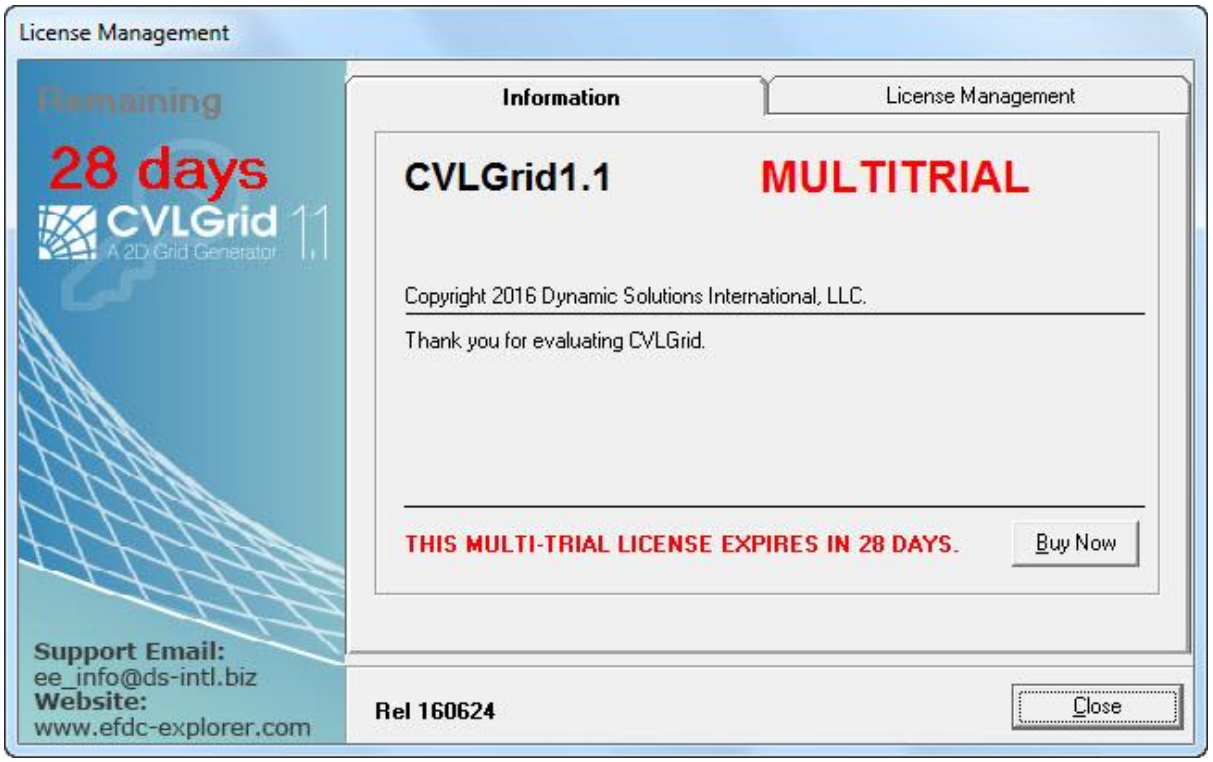


Figure 2-6 CVLGrid License Manager: Trial License.

2.2.2 Perpetual License

*Perpetual l*icenses have no time limit in their use and provide the full capability of the CVLGrid without any kind of limitation. However, users will only be able to update their software to the latest version of CVLGrid within a certain time-frame agreed to in the sales arrangement. After that period of time, CVLGrid will still function normally but any bug fixes or new features which are added to CVLGrid after this date will not be accessible to the user. In order to access these updates the user will be required to have a valid maintenance contract with DSI. The *CVLGrid License Manager* will inform the user when this maintenance period is finished as shown in Figure 2-7. The user may then contact DSI to have their maintenance period extended if required.

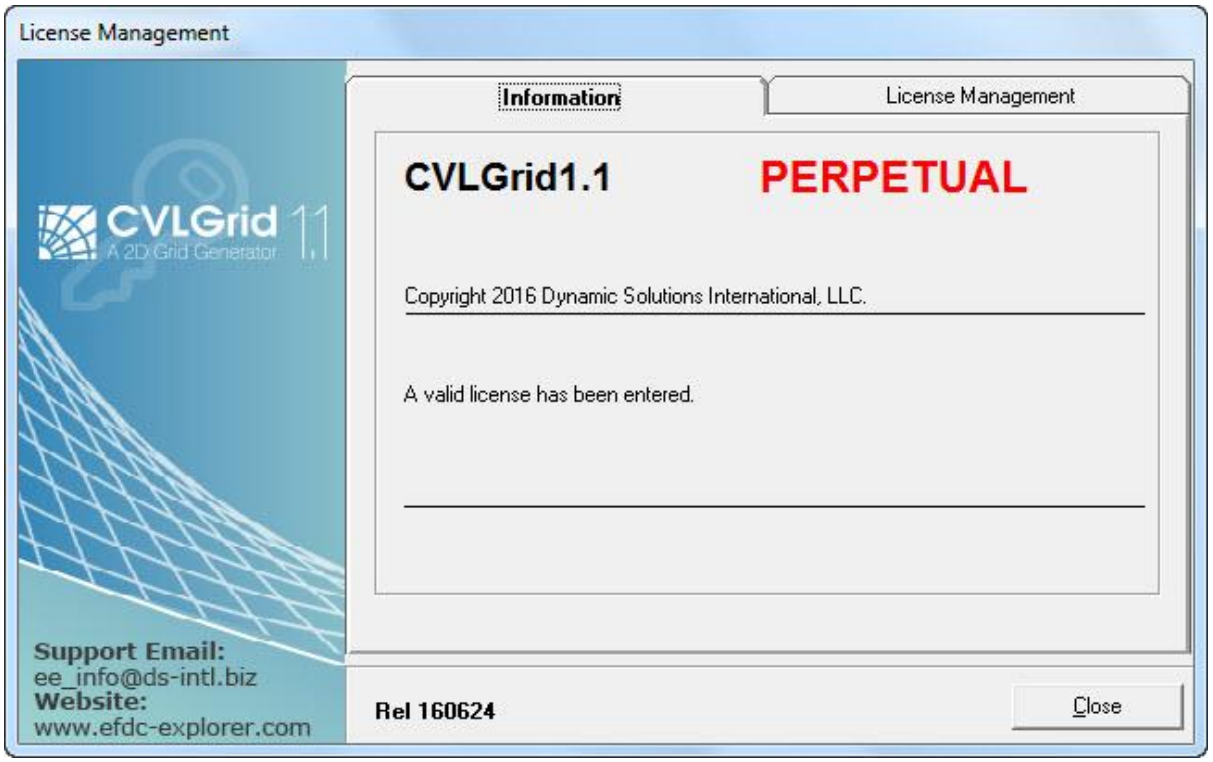


Figure 2-7 CVLGrid License Manager: Perpetual License.

2.3 Online Activation Process

After installation, the user must activate the software before first use. Sometimes problems occur for users at the activation stage. Please note that to activate Perpetual and Trial versions of CVLGrid, the user **must be connected to the internet unless using offline activation.** If the user is not connected to the internet, or if there is a firewall on the network or installation computer, an error message may be given. In this case, please adjust your internet virus checker or firewall settings and try activating again. If this fails please contact support.

1) To activate CVLGrid the user must start CVLGrid from the CVLGrid desktop icon or using the Start menu:

   "Start\All Programs\DSI\CVLGrid1.1\CVLGrid1.1".

2) If this is the first time you have run CVLGrid on your computer you will be see the CVLGrid be prompted for your license name and license code, as shown in Figure

2-8. CVLGrid will automatically activate the Perpetual or Trial version based on your license code.

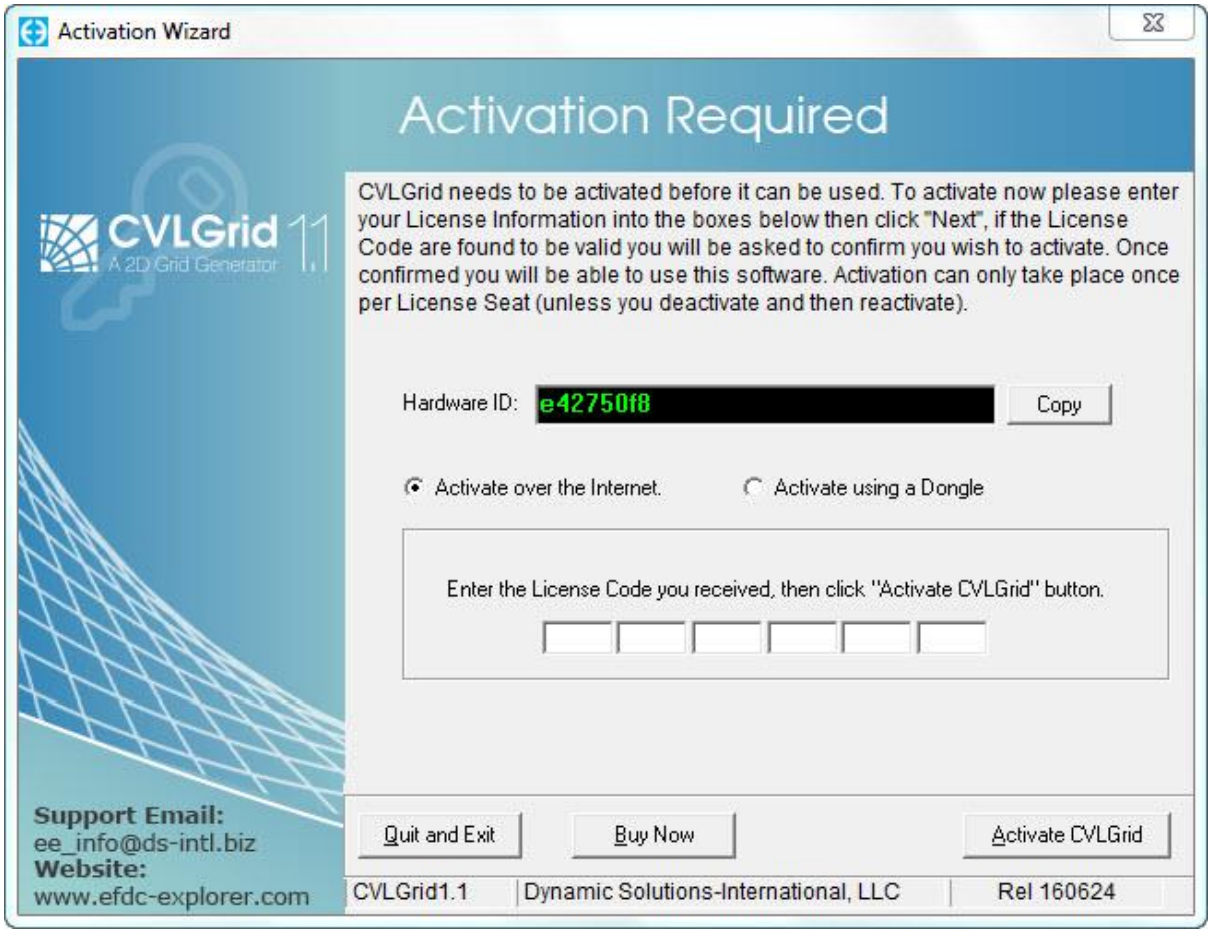


Figure 2-8 CVLGrid Version Options

3) After activation of your software, you will see the main CVLGrid form, as shown in Figure 2-9.

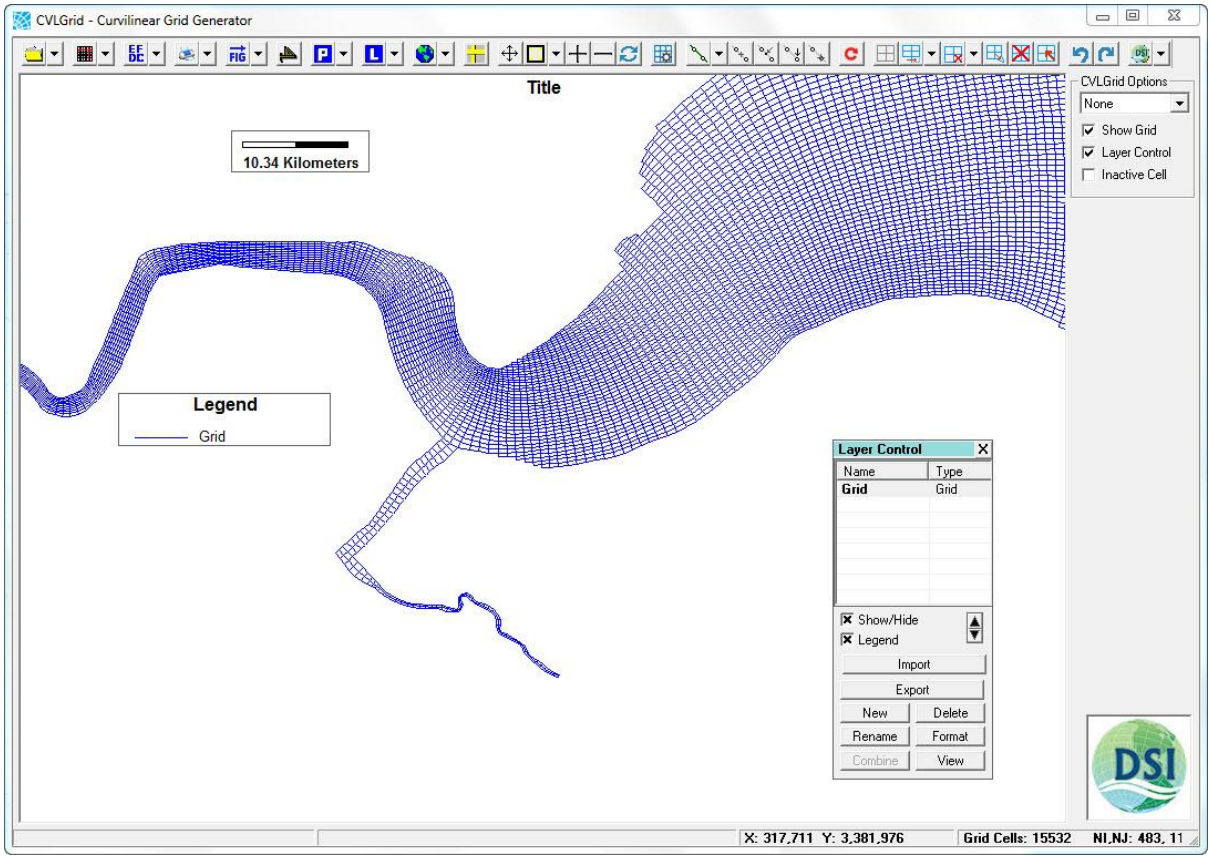


Figure 2-9 Main CVLGrid form (with an grid loaded).

2.4 Online Deactivation and License Management

Deactivation of CVLGrid is required when the user wishes to move their seat to a different computer, or plan to make some major changes to the hardware or operating system of their computer. Deactivation allows the seat to be retained and then reactivated on the same computer or a different computer later.

As for the activation process, deactivation of CVLGrid requires **connection to the internet unless using the offline license.** If the user is not connected to the internet, or if there is a firewall on the network or installation computer, an error message may be given. In this case, please adjust your internet virus checker or firewall settings and try activating again. If this fails please contact support.

1) To deactivate CVLGrid the user must click on the DSI button on the top right of the user form to the drop-down menu and select *License Manager* as shown in Figure 2-10. The License Manager provides the user with basic license information, including the user of the total number of seats available, the number of allowed activations and deactivations remaining, license name and code.
2) In some cases the user settings for screen resolution can cause the DSI button to be hidden. In this case use the F3 button to display *License Manager* shown in Figure 2-10.

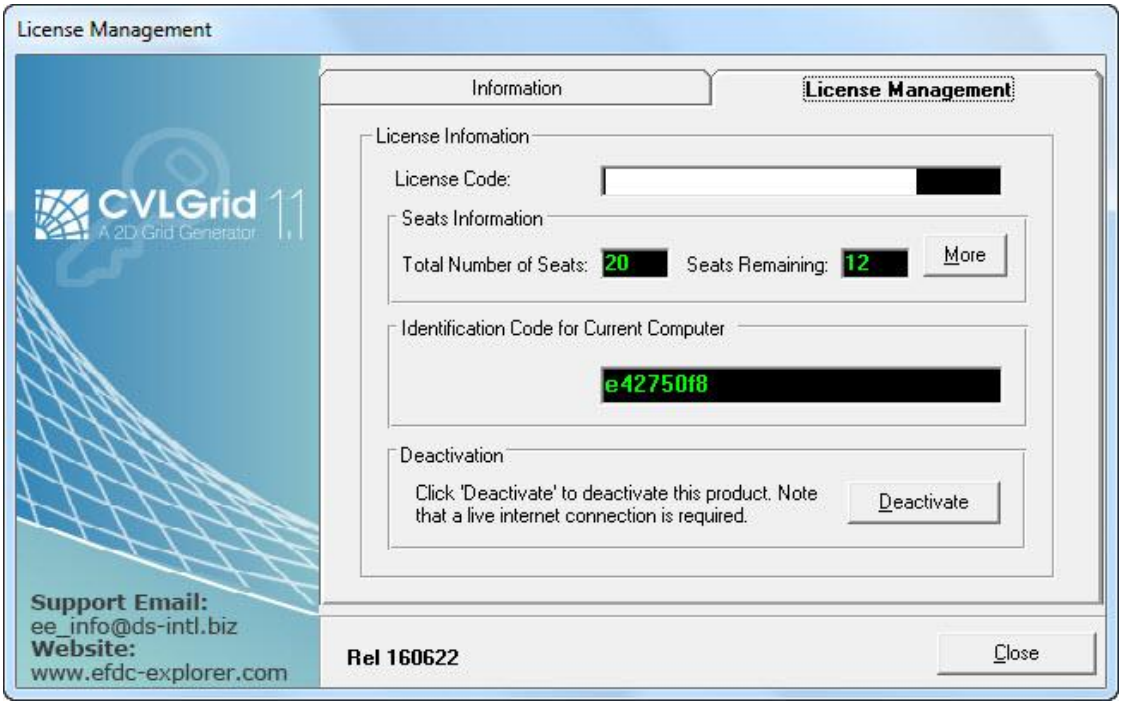


Figure 2-10 CVLGrid License Manager.

3) At the bottom of the *License Manager Form* is the *Deactivation* button. Selecting this button will bring up the *Deactivation* form. Selecting *Next*, and selecting *OK* at the prompt will deactivate your current license seat, and the number of license seats available for activation will increase by one.

   DSI usually provides the license code via email. The best way to enter the License Name and License Code is to copy and paste these from the DSI email (sometimes you may need to copy and paste into a text editor first). When pasting the license code, click on the 1st code field from the left and then paste the entire code. CVLGrid will parse the code into the appropriate blocks.

2.5 Offline Activation

Users who wish to have a license of CVLGrid without requiring a connection to the internet can do so with an Offline License. DSI will provide the user with a USB dongle which can be plugged into their computer to provide the license file required for operation. Offline licenses can be provided for both long term rentals as well as Perpetual Licenses. The steps for activation using a dongle are described below.

1) To use dongle, the Flexnet USB dongle driver may need to be installed for older versions of MS Windows. This will have been provided by email or can be found on the USB dongle. Plug the dongle into the USB port.

2) Open CVLGrid1.1 from shortcut on desktop or from the Start button then the Activation Wizard Form will appear as shown in Figure 2-11. The user should select the *Activation using a Dongle* option.

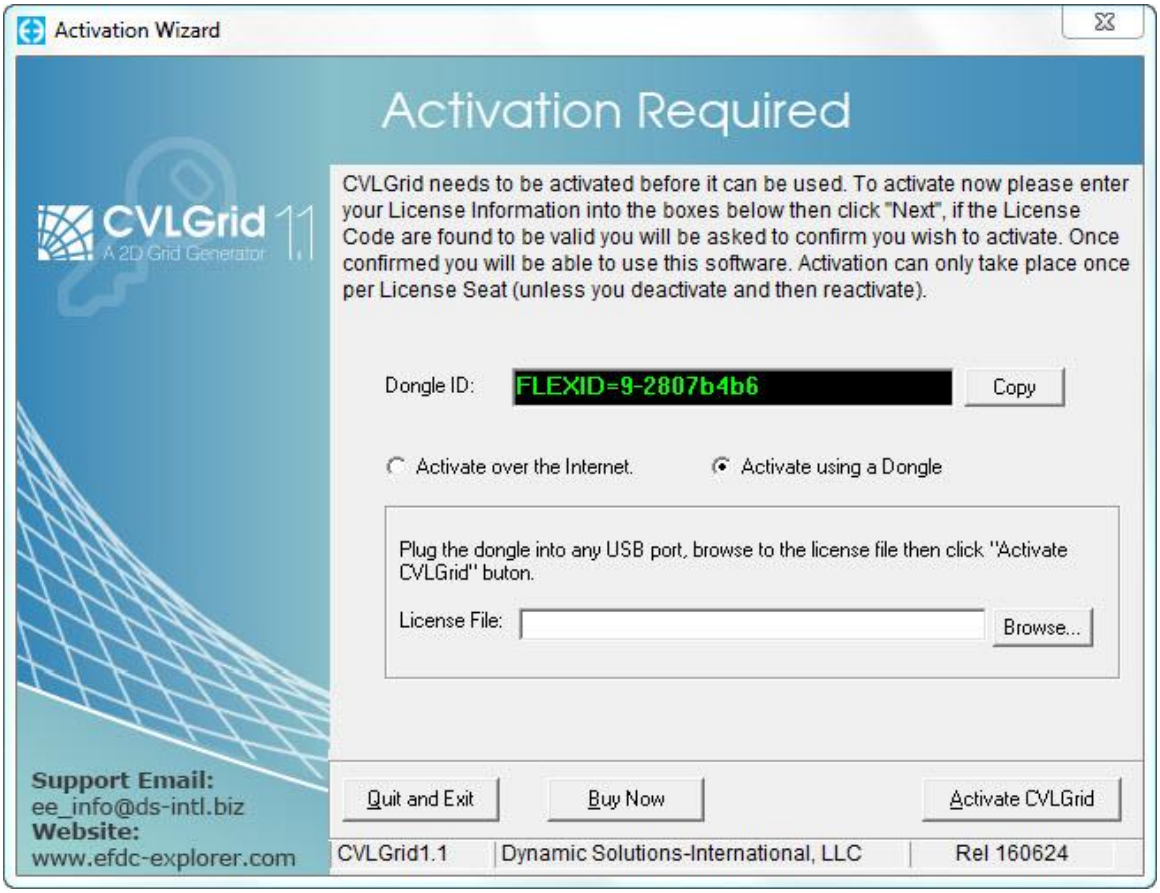


Figure 2-11 CVLGrid Activation Wizard Form for dongle

3) Select the Browse button to browse to license file (*.lic) in on the USB dongle drive as shown in Figure 2-12 and select *Open*.

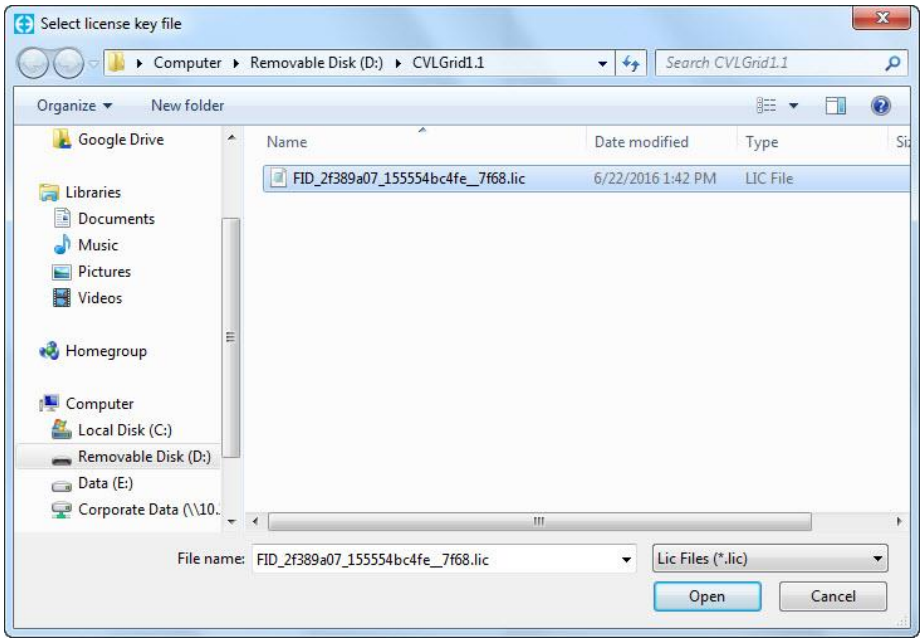


Figure 2-12 License File Selection Browse

4) Select the *Activate CVLGrid* button. The EFDC_Explorer Main form will then be opened.

5) If the user unplugs the USB dongle during use of CVLGrid, a warning form will appear as shown in Figure 2-13. The user has two options: to save the current work and close CVLGrid or keep using CVLGrid by reinserting the dongle into the USB port.

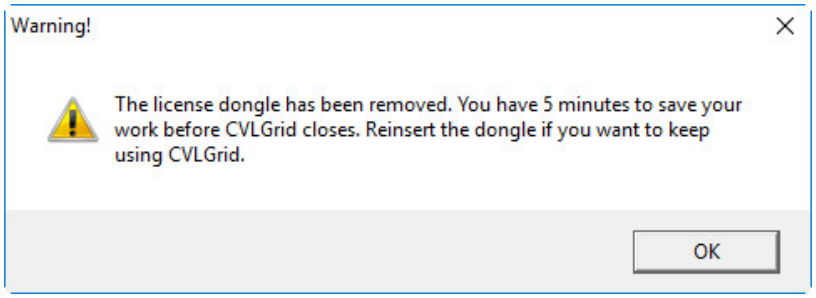


Figure 2-13 USB dongle license removed warning.

2.6 Offline Deactivation

To remove the Dongle key, the user must click on the DSI button on the top right of the CVLGrid main form. In the drop-down menu select *License Manager* to bring up the form shown in Figure 2-14. Select the *Remove* button. A confirmation form will be shown; select *Yes* to proceed. If the license has been removed successfully, the CVLGrid Main Form will close immediately and user will need to reactivate.

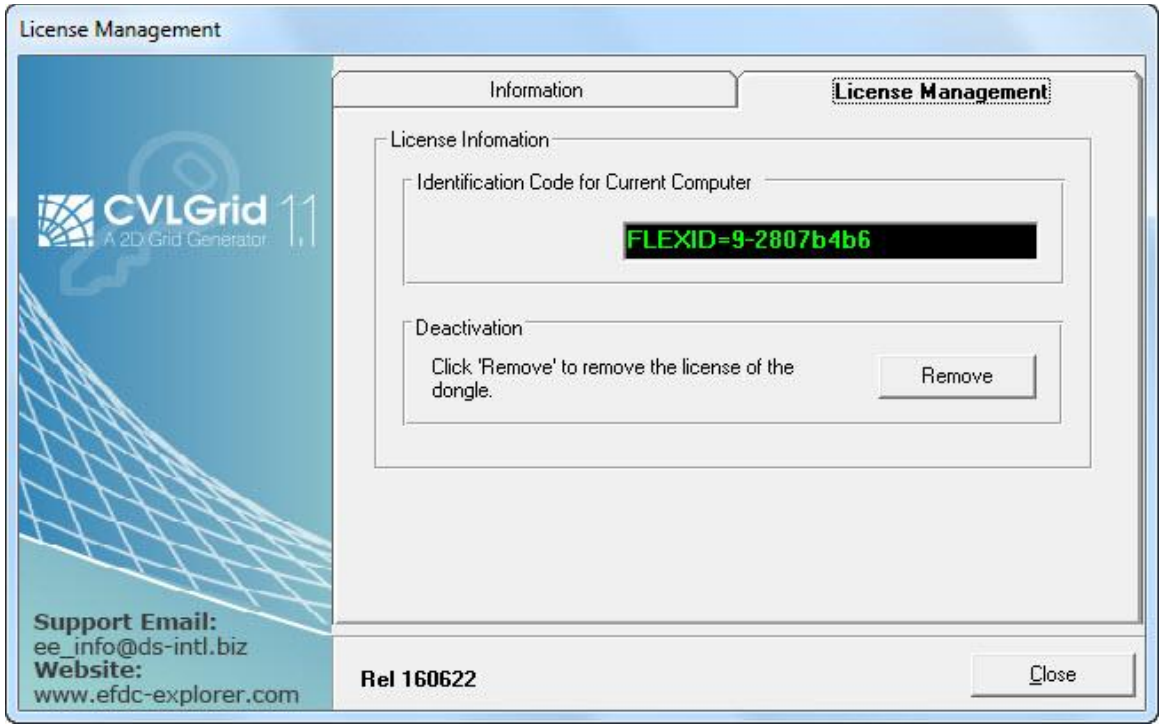


Figure 2-14 License Management Form

3 General Program Operation

3.1 Starting up CVLGrid

Upon starting CVLGrid the user will be presented with the form shown in Figure 3-1. The white window is the workspace for grid generation. On the top of the workspace window is the “main toolbar” where different tools, settings and functions can be accessed. The main toolbar includes the primary tools for creating and editing splines and grids.

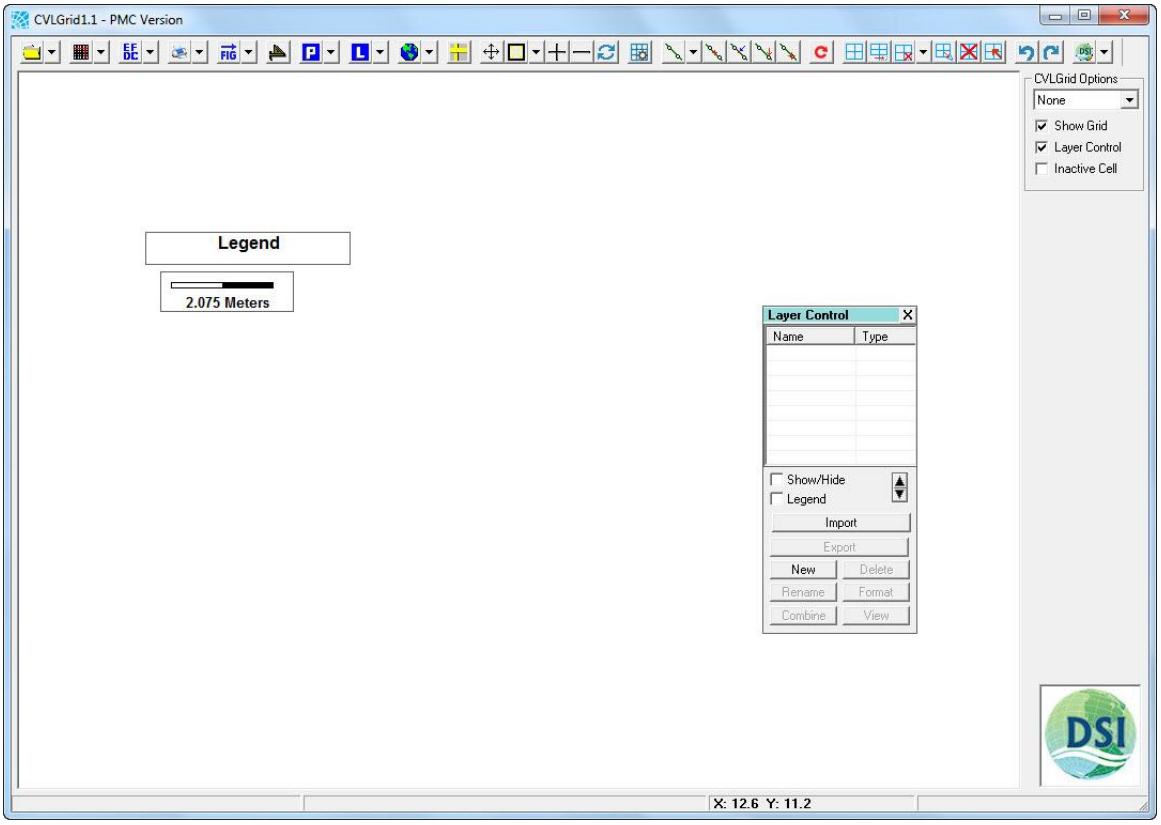


Figure 3-1 Interface of CVLGrid

On the right hand side of the workspace is an easy-access toolbar. These tools are explained in Section 5

3.2 Grid Generation Basics

When generating grids in CVLGrid there are generally three steps in the process: first load a file with which to shape the model domain, next create splines around the model domain and finally create the grid. Often when a grid is being created it is because the user needs to make a model of some geographic domain. Thus the grid should reflect the shape of that domain. CVLGrid allows the user to upload a polygon shape file that outlines the model domain, shown in Figure 3-2. Once this outline is visible in the workspace of CVLGrid, the user must create splines to outline the desired domain. Splines are the most essential part of generating grids correctly and accurately. Four splines enclosing a region are required to generate one grid. If the model domain is a simple shape, the user may be able to accurately model the area with only one grid. However, if the domain is complex with branches and heavy curves, the user will have to create several sub domains and connect multiple grids.

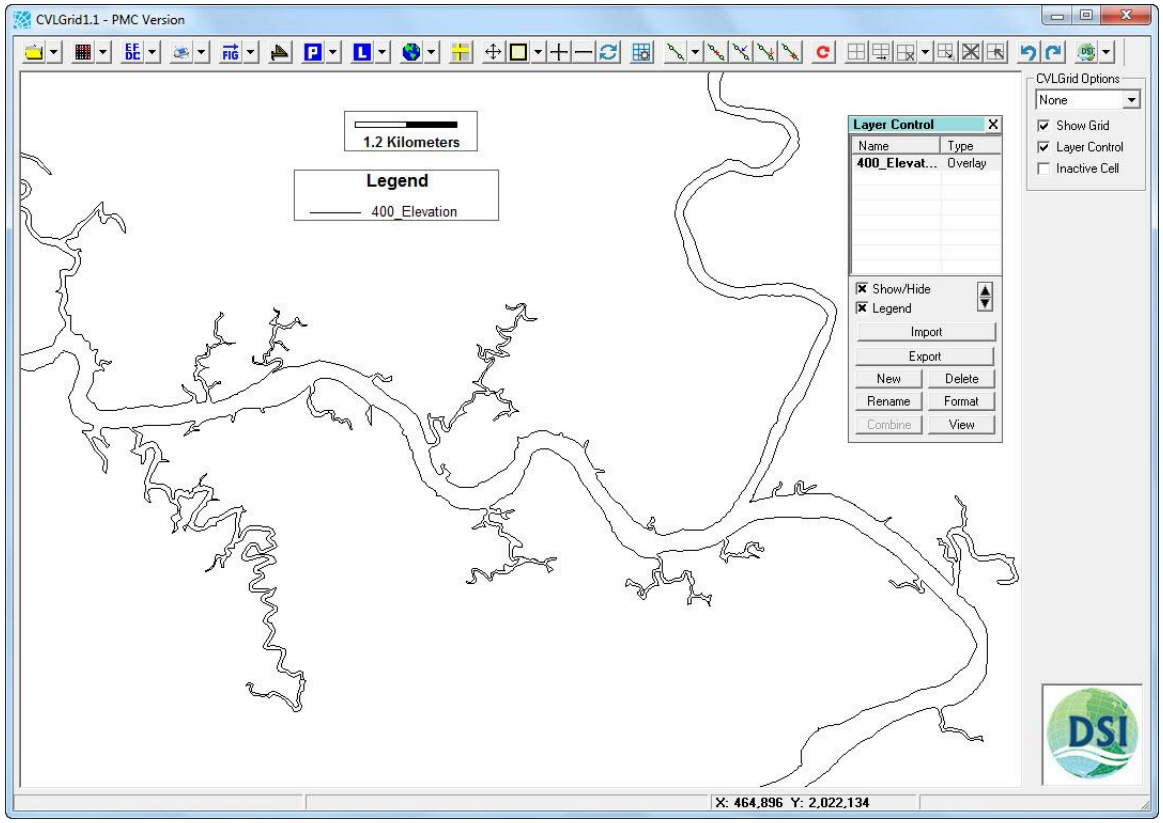


Figure 3-2 P2D Polygon shape file of the model domain.

To create a grid, four splines must be created and must completely enclose the whole or part of the model domain. The first spline will define "I index" of cells in grid. The spline intersects the first spline will define "J index" of cells in grid. The method for drawing the splines will depend on the particular features of the domain. For a very simple domain the first spline is drawn along the bottom of the domain from left to right. The second spline is drawn along the top of the domain. The last two splines are drawn along the sides of the domain and **must intersect** the previous two splines. Figure 3-3 shows how these splines are drawn.

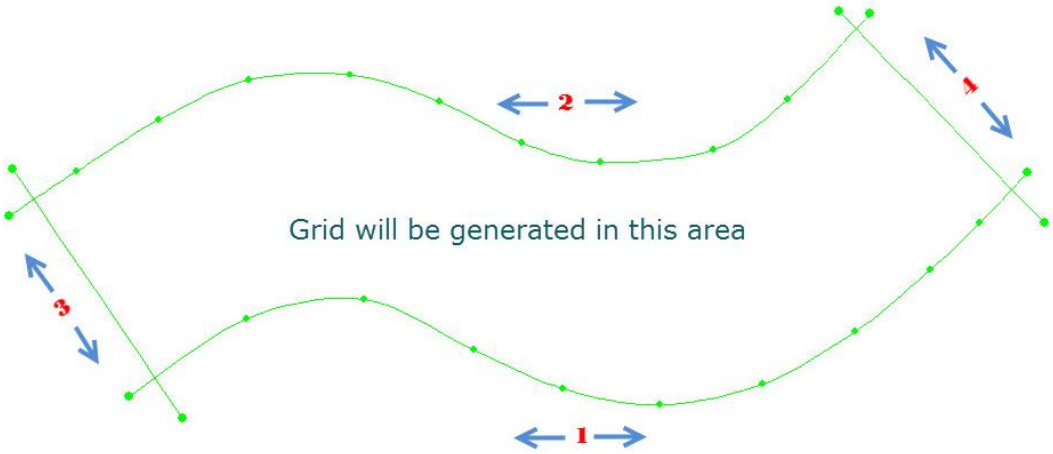


Figure 3-3 Spline drawing guide.

In order to improve orthogonality it is good practice to make the intersecting lines as close to perpendicular as possible. This will simplify the computation and create a more orthogonal grid.

Once four splines are created, fully enclosing the desired area, the user may then generate the grid using the *Generate Grid* button . After clicking the *Generate Grid* icon, the user must specify the number of cells in the I and J (x and y) axis from the pop-up box, as shown in Figure 3-4. The first spline defines the I axis and the side splines define the J axis.

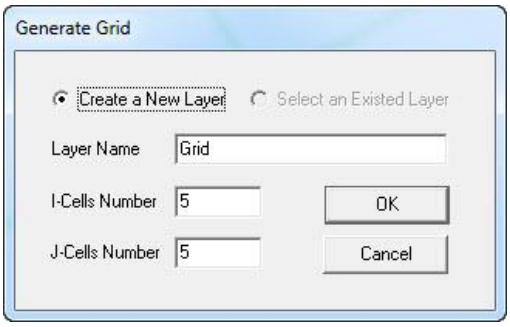


Figure 3-4 Grid cell specification on I and J axis.

Once the grid is generated the user can view properties of the grid, such as orthogonal deviation or the coordinates, by using the *CVLGrid Options* menu in the sidebar. The user can RMC (right mouse click) in the workspace to see options for modifying and improving the grid.

Only one grid with a specific of cells number is generated with a layer of splines. This means if the user wants to generate a new grid they need to create a new layer of splines. To create a new spline layer, click on the *New* button on *Layer Control* menu shown in Figure 3-5.



Figure 3-5 New layer button.

A new pop-up menu will appear, choosing *Layer Type as Splines* and a new name for the layer before clicking the *OK* button to create a new spline layer as shown in Figure 3-6.



Figure 3-6 Defining a new layer.

3.3 Shortcut and Navigation Keys

In the CVLGrid workspace there are several ways to navigate the space and get a better view of the grid. By pressing F2 on the keyboard to displays the full list of shortcut and combination keys as shown in the “Shortcut Keys” in Figure 3-7.

Scroll Wheel	When the workspace is active the mouse scroll wheel allows the user to zoom in and out. Scrolling up zooms in while scrolling down zooms out.
RMC	By right mouse clicking (RMC) and dragging the rectangle over the desired area, the workspace will zoom in on the area encompassed by the rectangle.
Ctrl + E	By pressing Ctrl + E on the keyboard the workspace will automatically "zoom to extents" which puts everything in the workspace in view.
Ctrl + R	By pressing Ctrl + R on the keyboard the workspace will repeat the last CVLGrid operation.
Ctrl + V	Go to a user specified cell by entering the L or I, J coordinates.
Alt + W	By pressing Alt + W on the keyboard the pop-up window will display to allow the Users to enter horizontal scale number. (See Figure 3-8).
Arrow Keys	Using the arrow keys on the keyboard, the user is able to pan through the workspace and move to view different parts of the grid.
- Key	Zoom out
+ Key	Zoom in
Spacebar	End current selected operation.

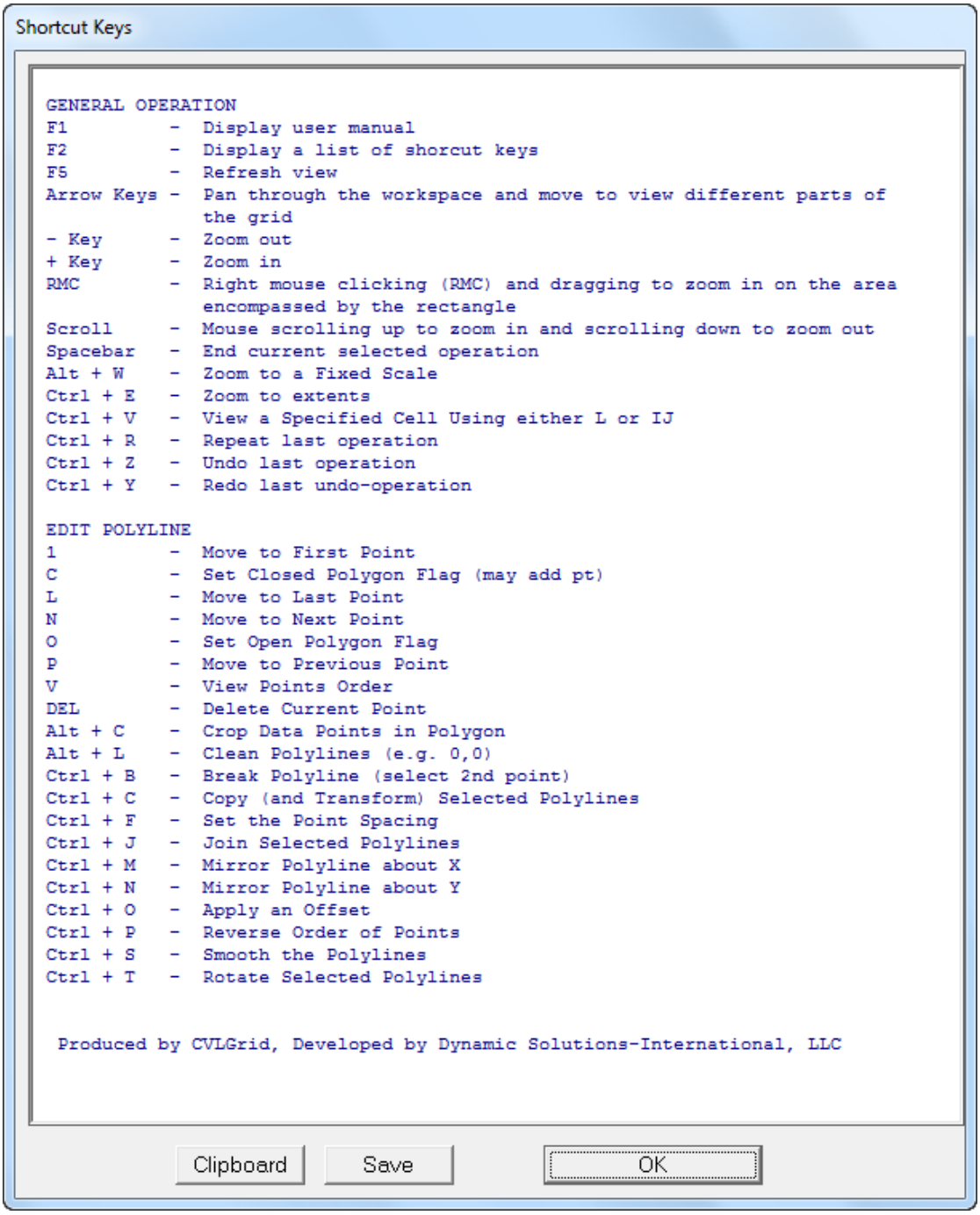


Figure 3-7 Shortcut keys.

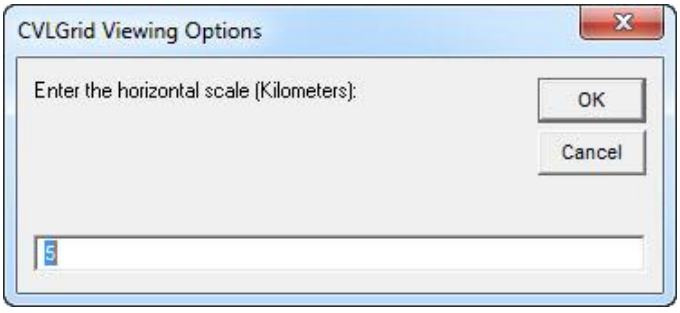


Figure 3-8 CVLGrid viewing options.

4 Using CVLGrid (Menu Options)

There are two toolbars used in CVLGrid, one that runs along the top of the workspace and another that is on the side of the workspace. This section covers the tools and function available on the top bar which will be referred to as the *Main toolbar*. Section 6 covers the tools and functions available in the side toolbar which will be referred to as the *Sidebar*.

4.1 File Menu

The *Files Menu* in the main toolbar is denoted by  icon. When the down arrow on the right of this button is selected the menu shown in Figure 4-1 is displayed.

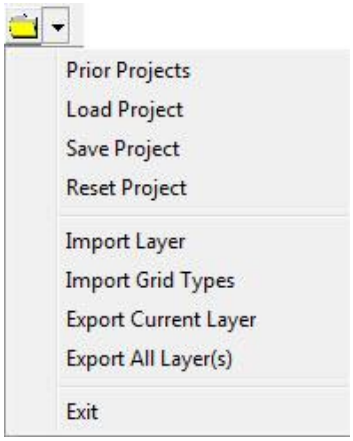


Figure 4-1 File menu options.

4.1.1 Prior Projects

This option in the *File Menu* allows the user to quickly load a project that has been created or edited recently. When this option is selected a window, shown in Figure 4-2, appears in the workspace. The window lists all the recent projects and the user can select the desired project by double clicking on it.

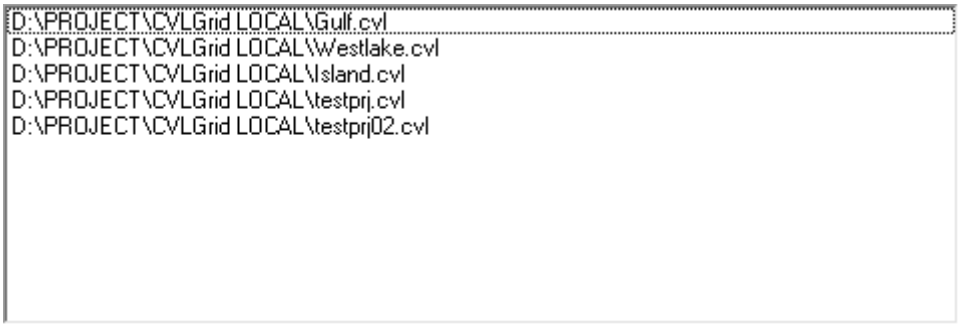


Figure 4-2 Recent projects window.

4.1.2 Load Project

The *Load Project* option in the File Menu loads a project saved on the computer's hard drive or in some external memory. When this option is selected a separate window appears, shown in Figure 4-3 which allows the user to browse to the project file. The *.cvp file may also be dragged and dropped onto the workspace to open the project. Note that the *.cvp extension is a CVLGrid project format whereas *.cvl extension is the grid format.

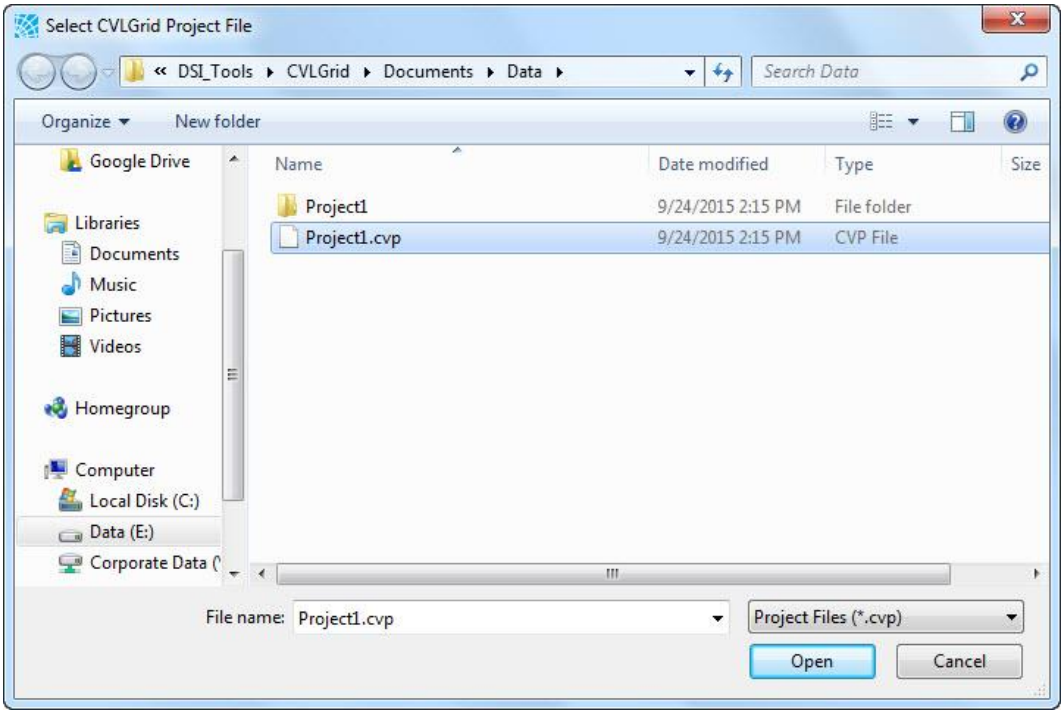


Figure 4-3 Load project browsing window.

4.1.3 Save Project

Save Project allows the user to save the entire project to one .CVP file. This includes the polygon shape file, the splines, and the grid. When the user saves a project a browsing window similar to Figure 4-3 allows the user to specify the desired save location. The project file is saved as a .CVP file, mentioned in Section 1.6

4.1.4 Reset Project

Reset Project simply clears all the data in the workspace. When this option is selected a confirmation box appears, shown in Figure 4-4.

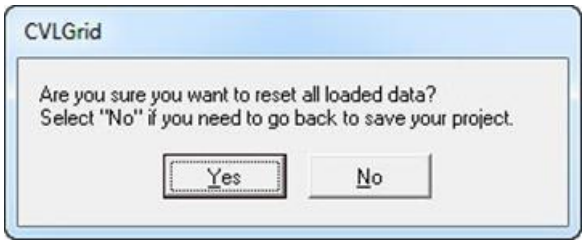


Figure 4-4 Reset project confirmation.

Once “Yes” is selected CVLGrid automatically clears the workspace of everything and starts anew. If the user does not want to lose their previous work then the work must be saved before selecting *Reset Project.*

4.1.5 Import Layer

The *Import Layer* option in the *File Menu* loads a polyline file as spline, overlay, kml and grid files saved on the computer’s hard drive or in some external memory. When this option is selected a separate window appears, shown in Figure 4-5 and Figure 4-6 which allows the user to browse to the project file.

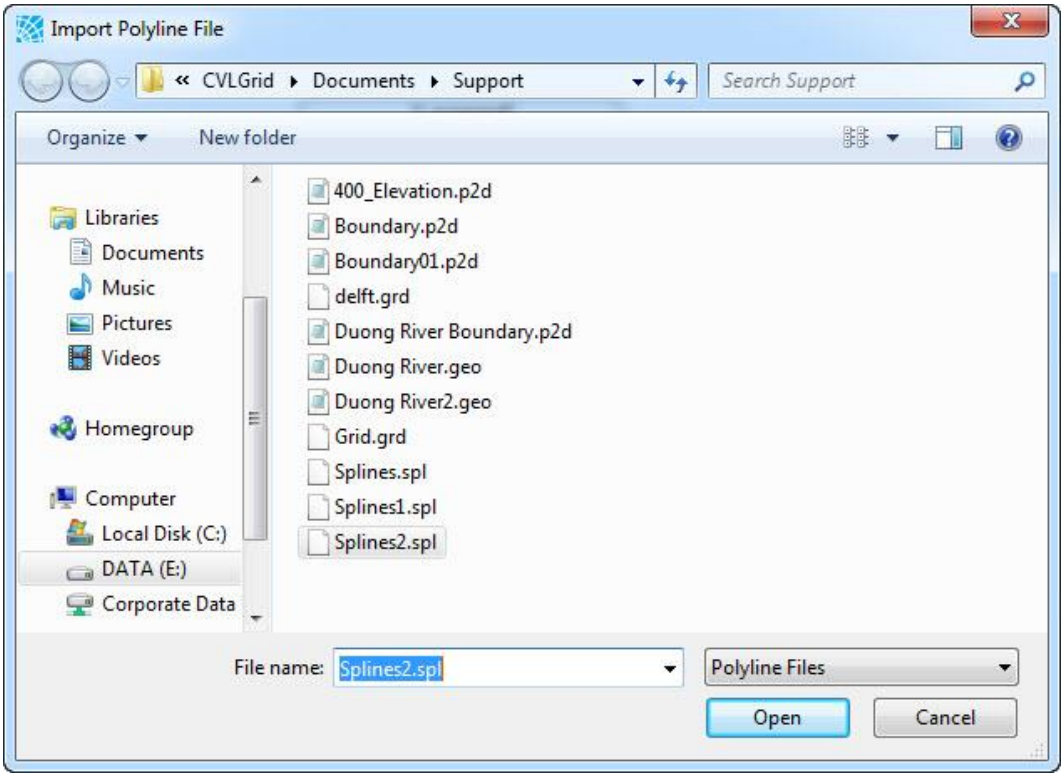


Figure 4-5 Import spline layer browsing window.

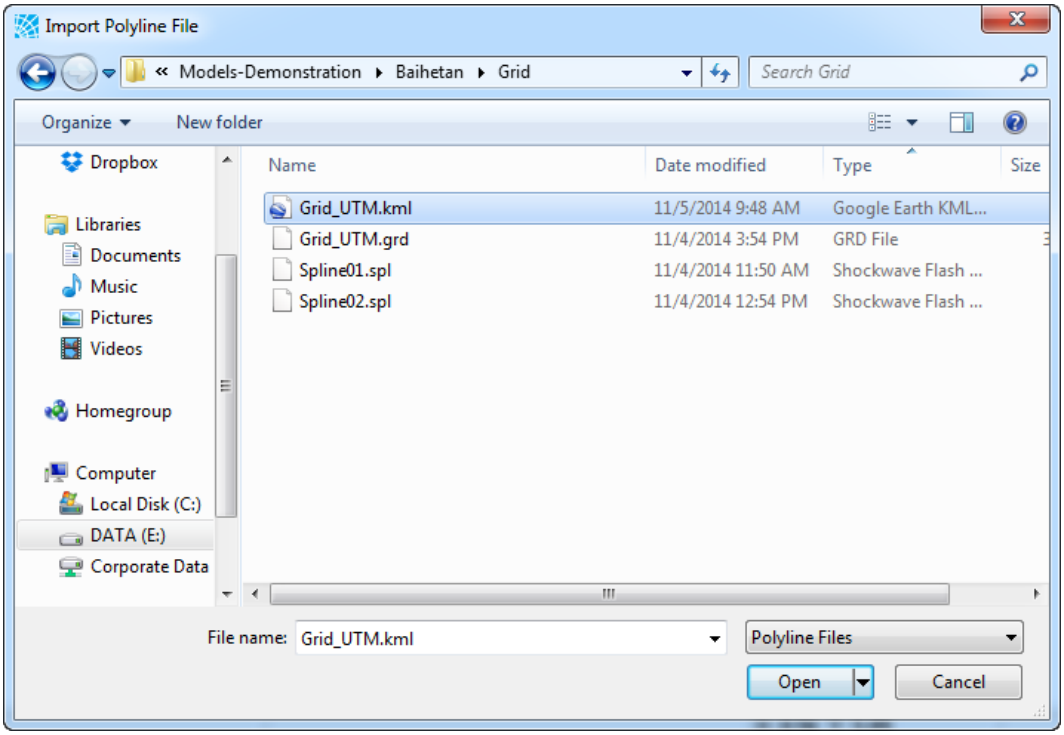


Figure 4-6 Import KML layer browsing window.

4.1.6 Import Grid Types

This option allows the user to import different grid types as DXDY/LXLY, ECOMSED, Grid95, Delft RGFGrid, SEAGRID, CH3D, Corners(4). When this option is selected a separate window appears, shown from Figure 4-7 to Figure 4-13 which allows the user to browse to the project file.

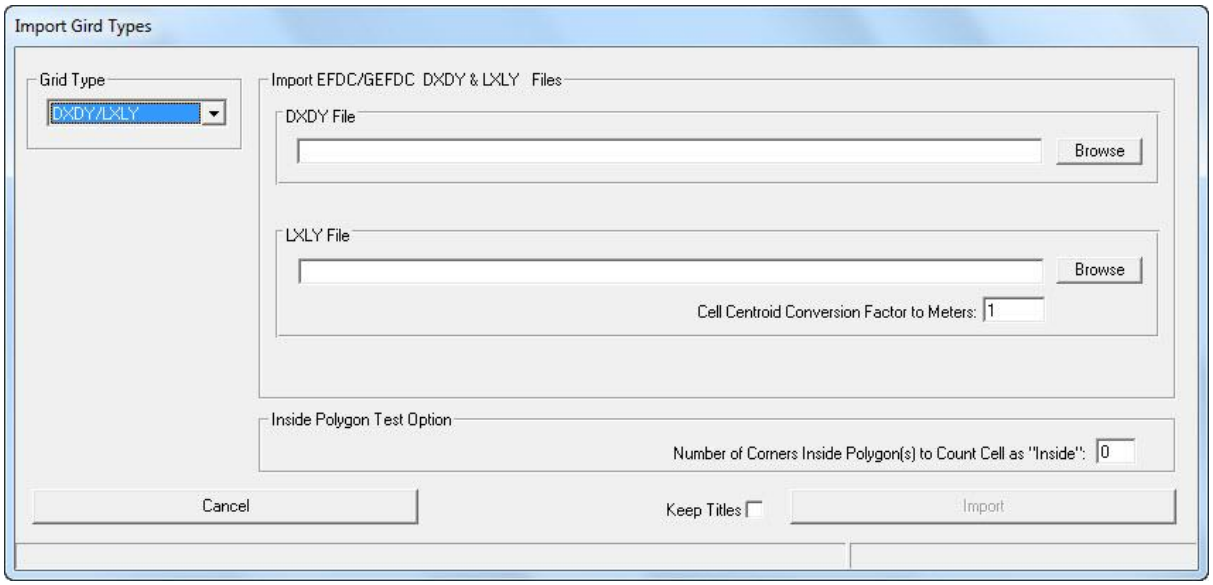


Figure 4-7 Import DXDY/LXLY browsing window.

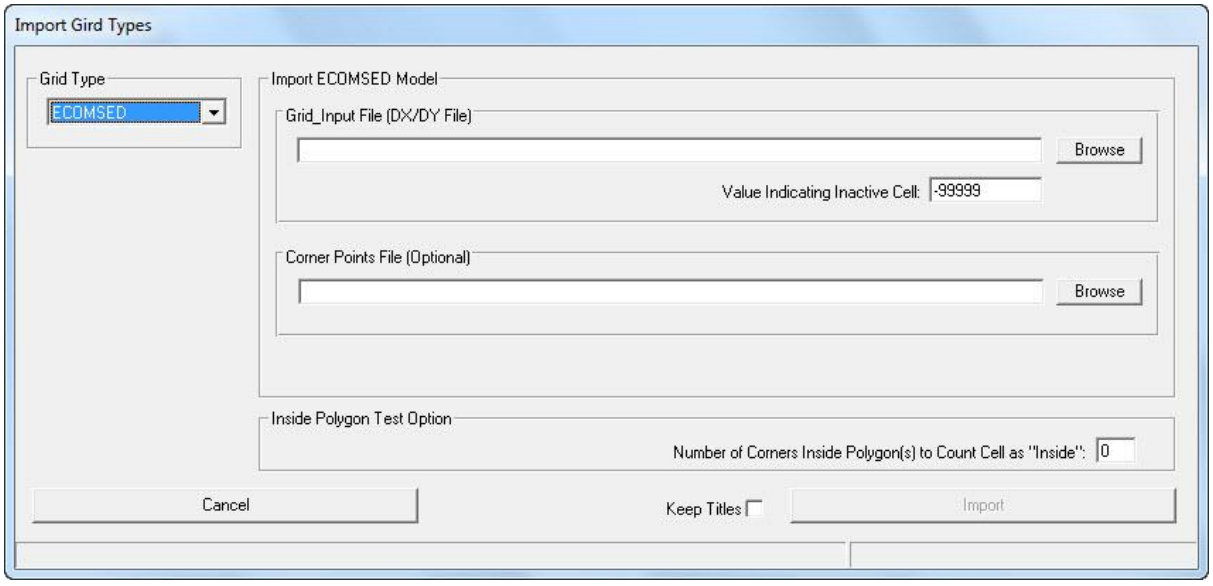


Figure 4-8 Import ECOMSED browsing window.

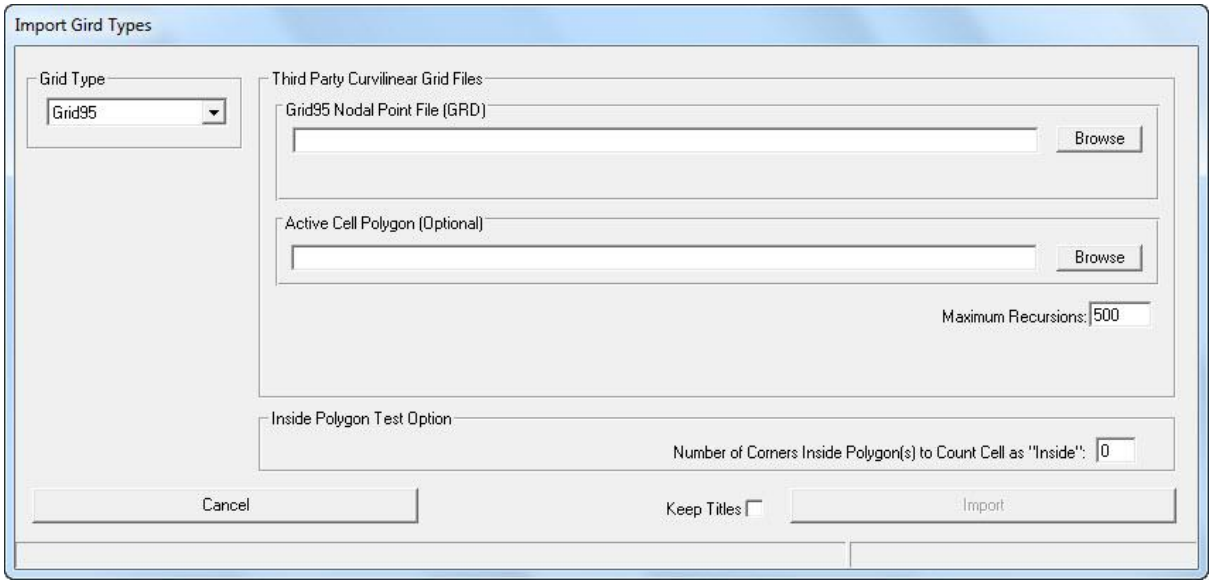


Figure 4-9 Import Grid95 browsing window.

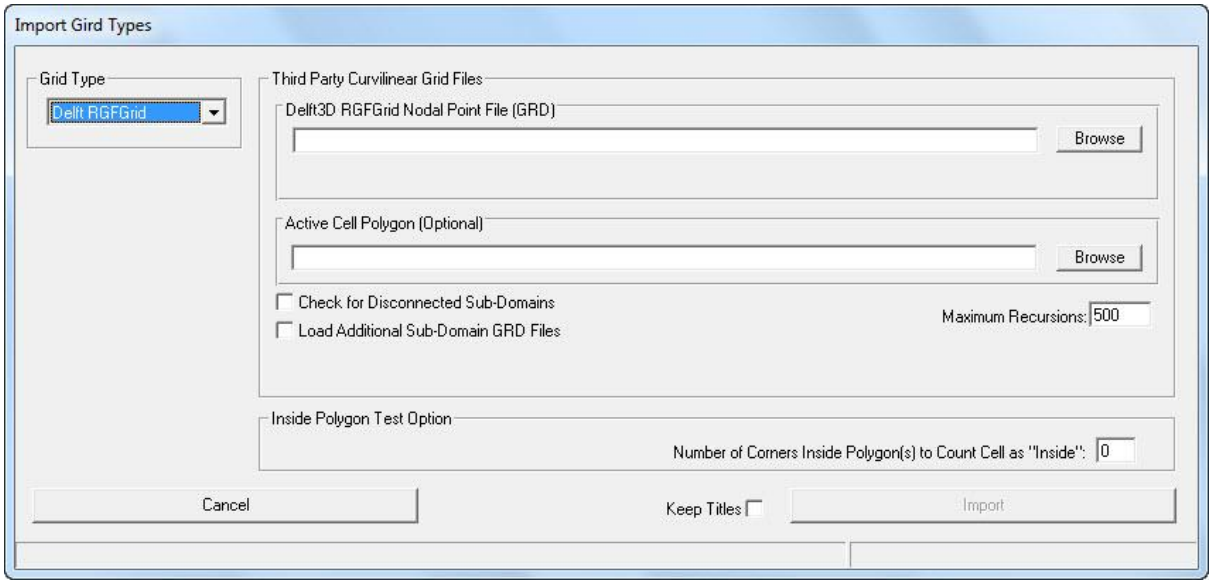


Figure 4-10 Import Delft RGFGrid browsing window.

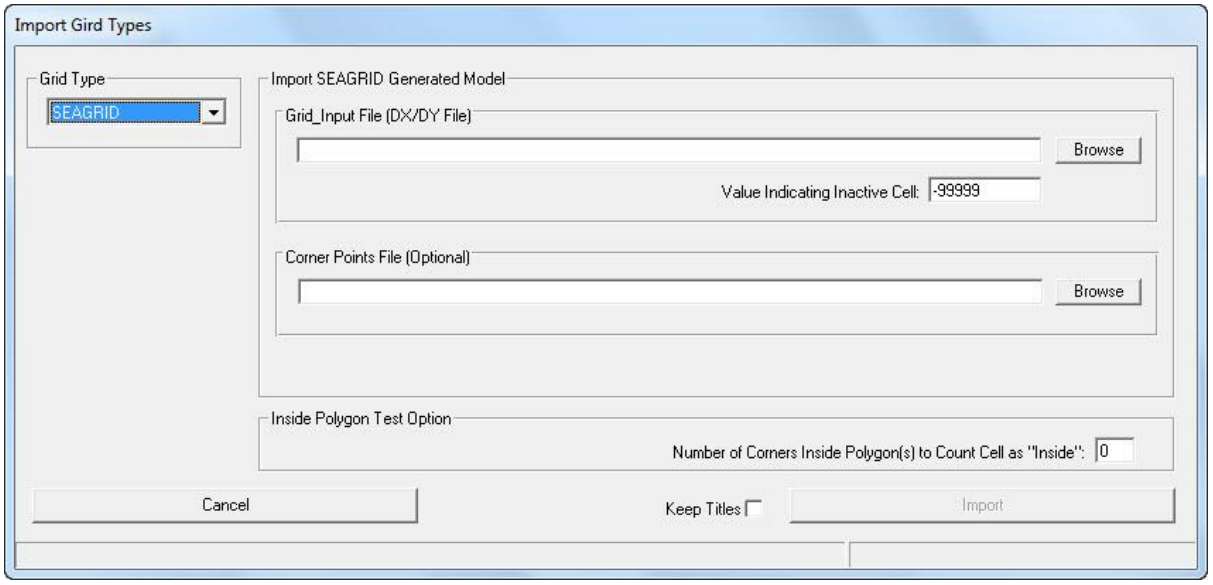


Figure 4-11 Import SEAGRID browsing window.

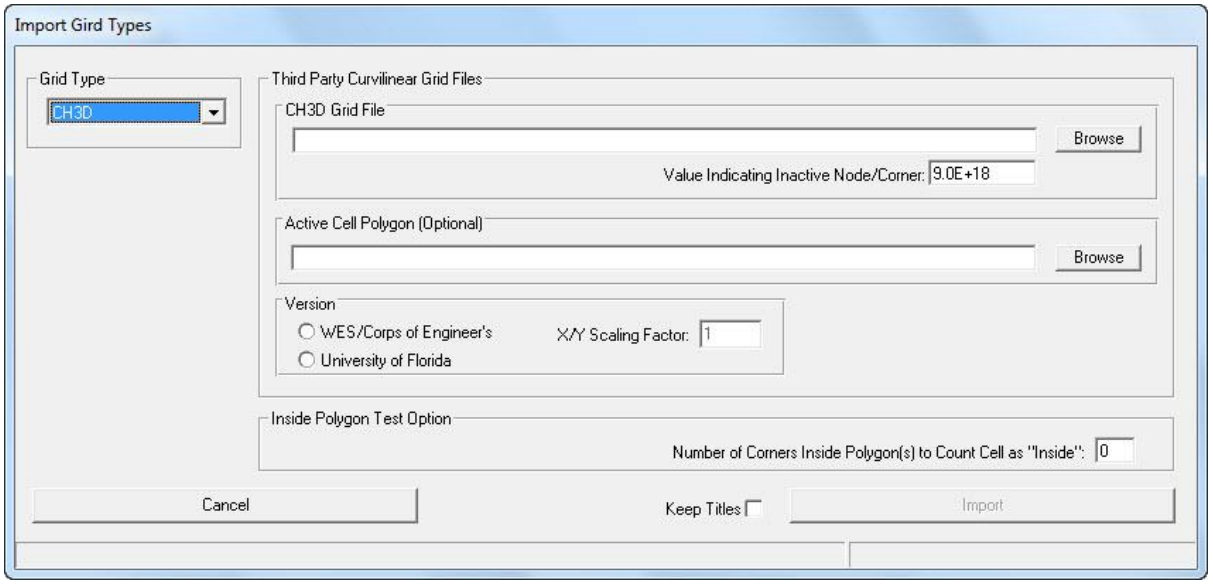


Figure 4-12 Import CH3D browsing window.

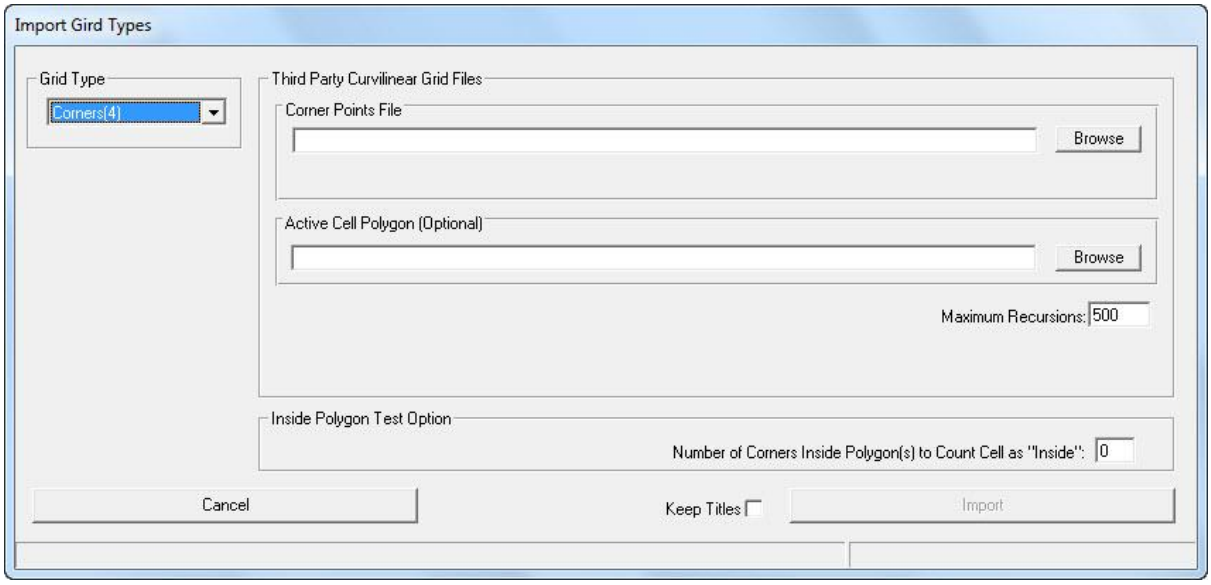


Figure 4-13 Import Corners(4) browsing window.

4.1.7 Export Current Layer

The *Export Current Layer* option allows the user to export the current layer when many layers are loaded and can be seen in *Layer Control*. To do this, the user selects a layer so the selected layer will become the current layer then choose *Export Current Layer*, a separate window appears, shown in Figure 4-14 which allows the user to save the layer in the desired folder.

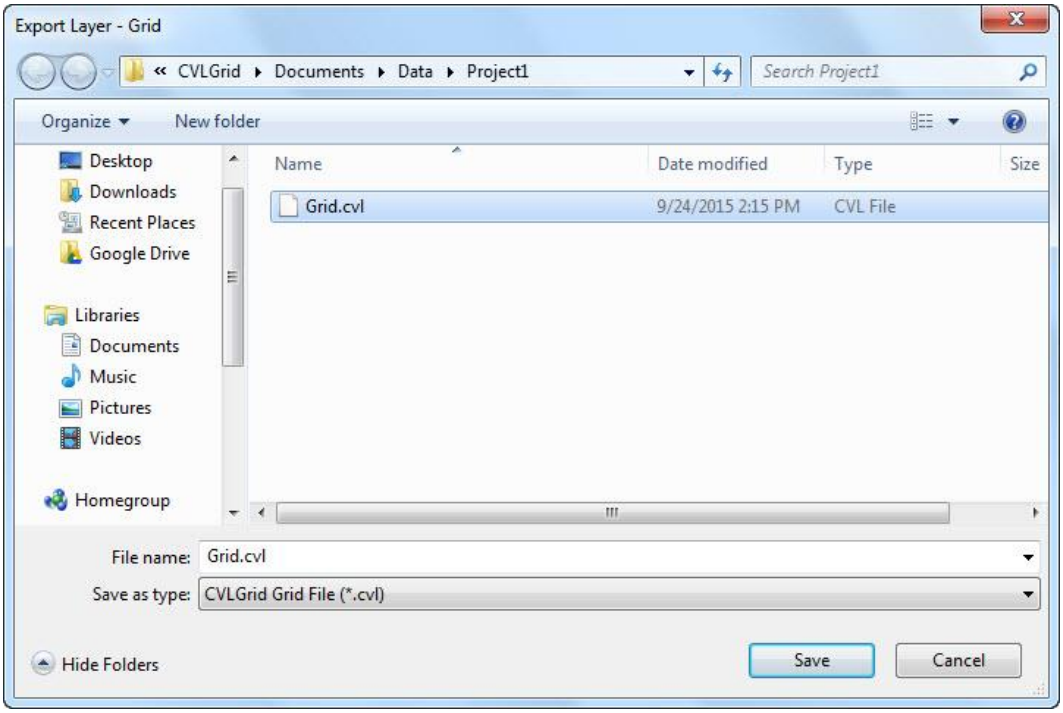


Figure 4-14 Export layer browsing window.

4.1.8 Export All Layer(s)

The *Export All Layers* option allows the user to export all layers at the same time. Each separate window corresponding to each layer appears in turn to allow the user to save layers in desired folder.

For example, there can be three layers including grid, splines, and overlay layers, so if the user exports all layers there are three separate windows to set the name and location of the exported file. Figure 4-15 shows exporting of a grid layer. Figure 4-16 shows exporting of an overlay layer and Figure 4-17 shows exporting of a spline layer.

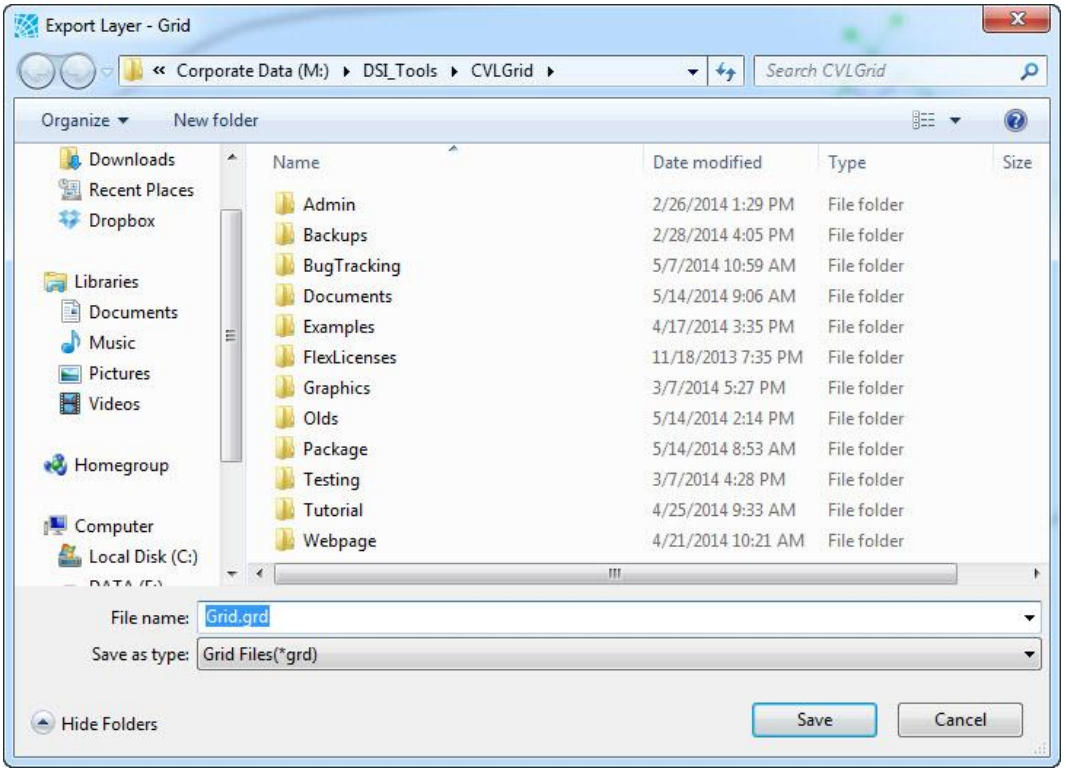


Figure 4-15 Export layer - grid browsing window.

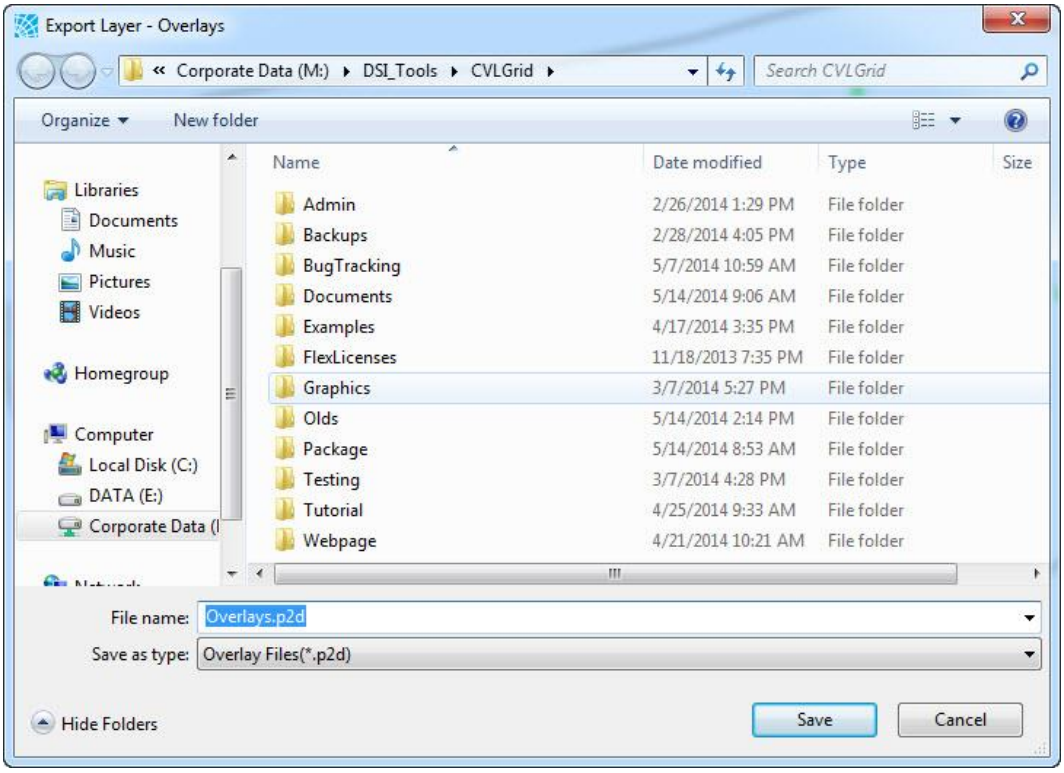


Figure 4-16 Export layer - overlays browsing window.

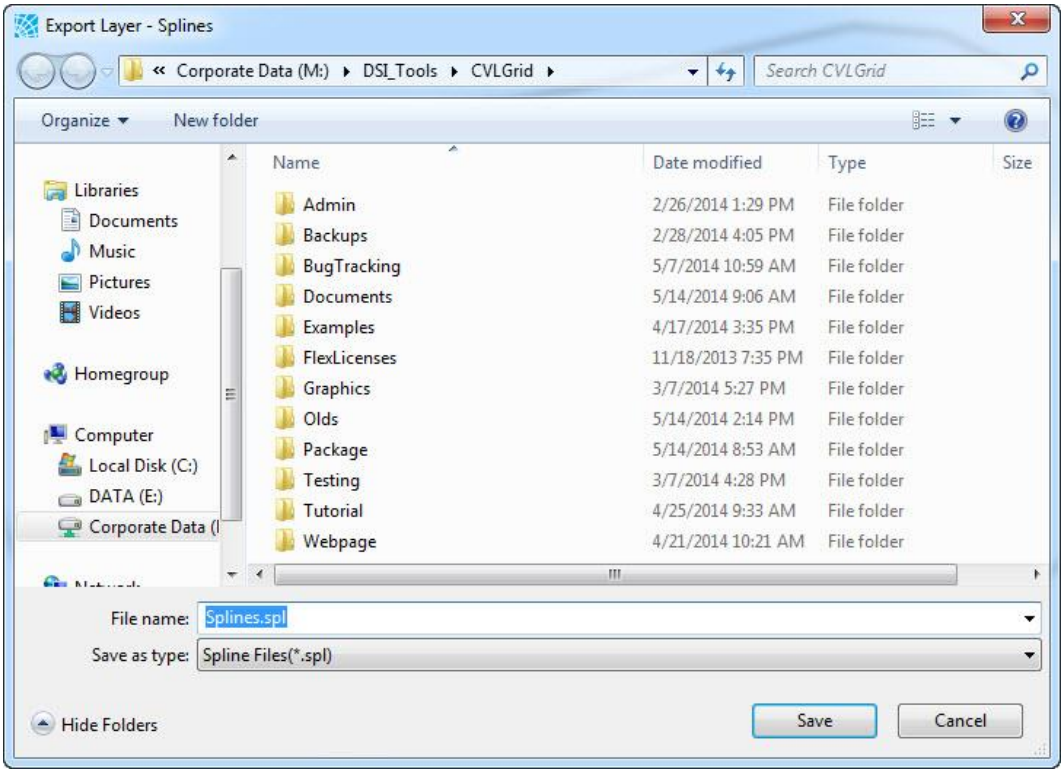


Figure 4-17 Export layer - splines browsing window.

4.1.9 Exit

This option allows the user to exit the program. The confirmation message box will be displayed as Figure 4-18 when the option is chosen.

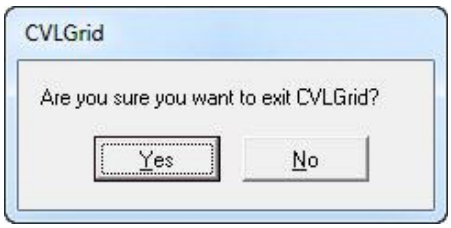


Figure 4-18 Exiting confirmation.

4.2 DTM Tools Menu

CVLGrid allows the user to load a Digital Terrain Model file (DTM) into the workspace. A DTM is a simple X, Y, Z ASCII file that contains information of coordinates and terrain elevation. It may be accessed with the *DTM Menu* button as shown in Figure 4-19 .

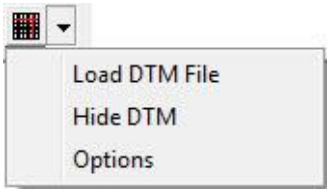


Figure 4-19 DTM dropdown menu.

4.2.1 Load DTM File

Selecting the *Load DTM File* allows the user to load a DTM file. This browsing window is shown in Figure 4-20. After clicking on *Open* button, the DTM File is loaded as shown in Figure 4-21.

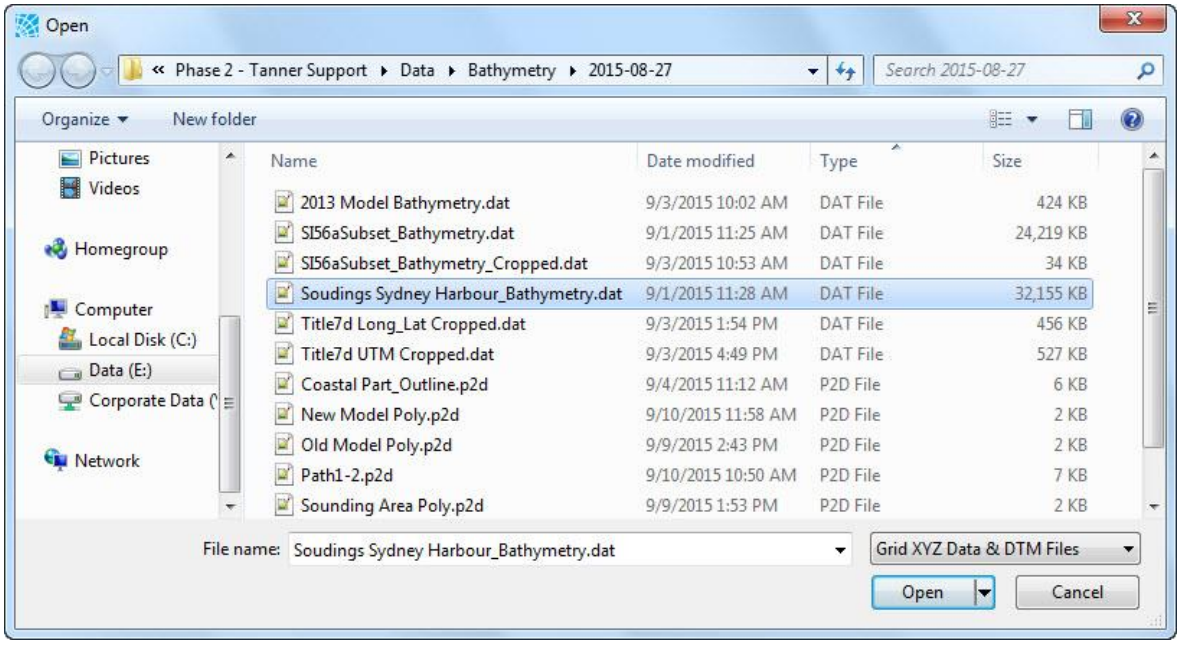


Figure 4-20 DTM browsing window.

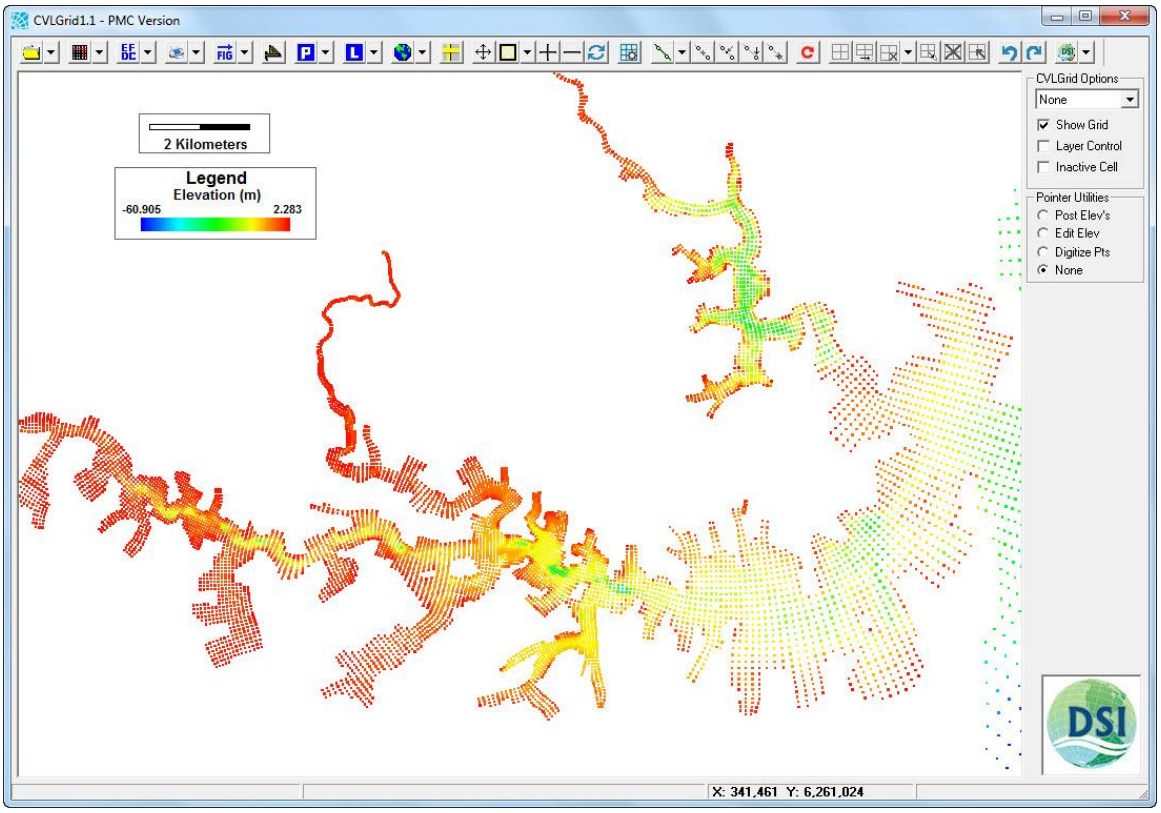


Figure 4-21 DTM file loaded.

When a DTM file loaded, a tool set of *Pointer Utilities* appears on side menu as shown in Figure 4-22. These are described in the following sections.



Figure 4-22 Pointer utilities.

4.2.1.1 Post Elevations

Selecting the ***Post Elev's*** pointer utility displays the coordinates and elevation value (Z) of a specific point on workspace in the yellow text box by RMC as shown in Figure 4-23.

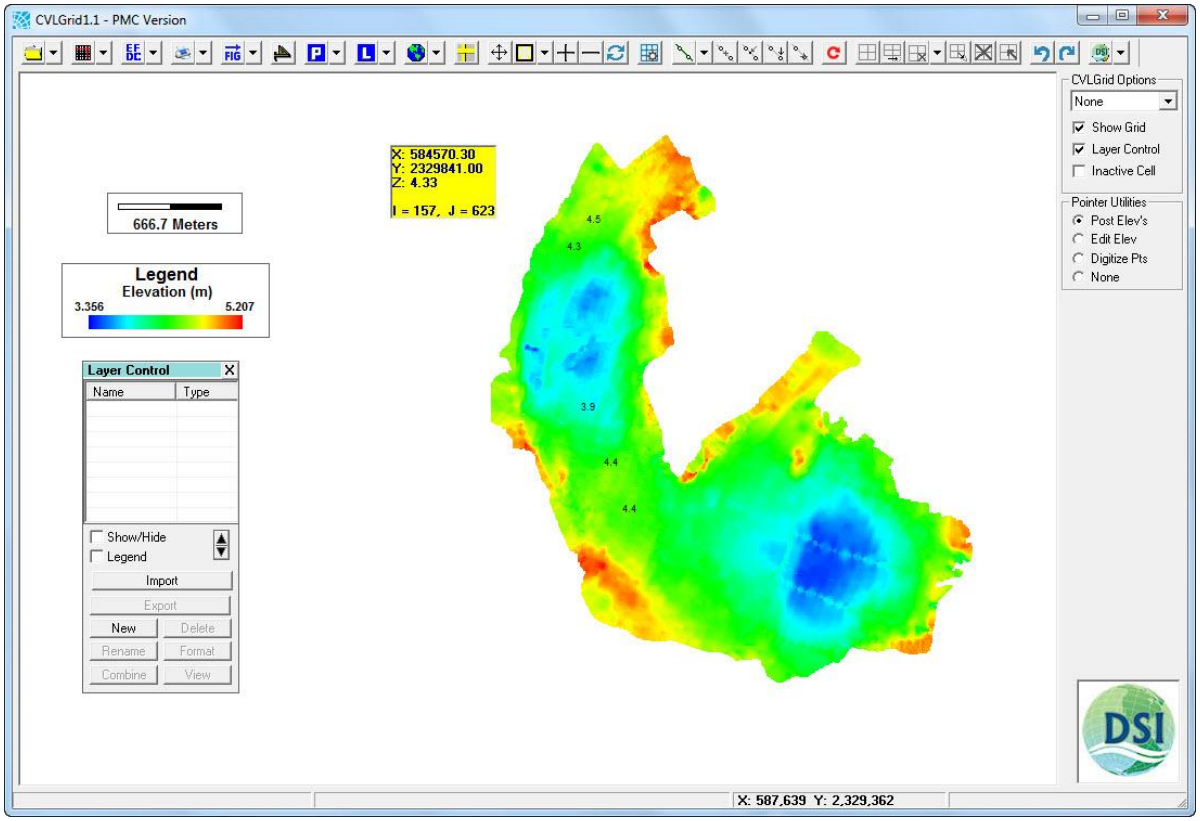


Figure 4-23 Point elevation utility.

4.2.1.2 Edit Elevation

When the *Edit Elevation* function is selected a frame appears which allow the user modify elevation value (Z) of a specific point on workspace by entering a number or using operators on the white box as shown in Figure 4-24.



Figure 4-24 Elevation adjustment factor.

4.2.1.3 Digitize Points

The *Digitize points* function allows the user digitize a series of points by RMC continuously. The X and Y values are recorded and can be pasted to a text file as shown in Figure 4-25 and Figure 4-26.

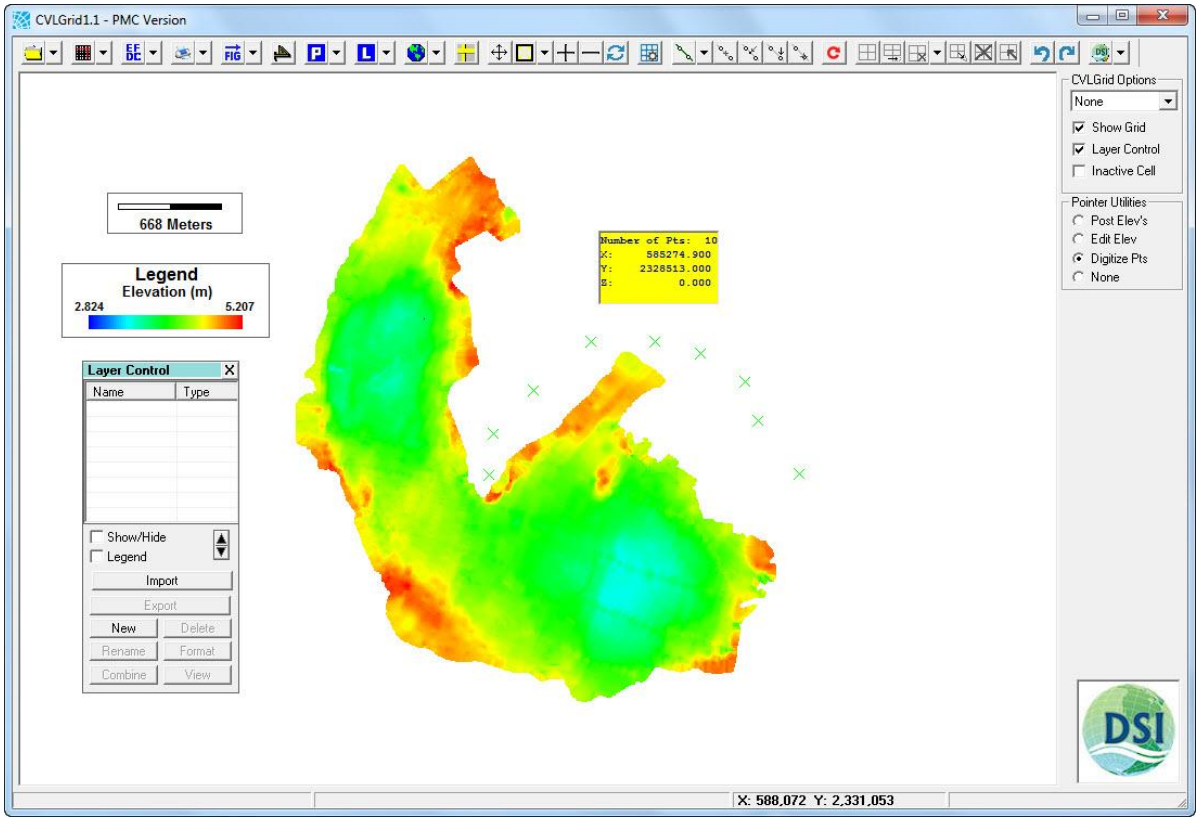


Figure 4-25 Digitize points utility.

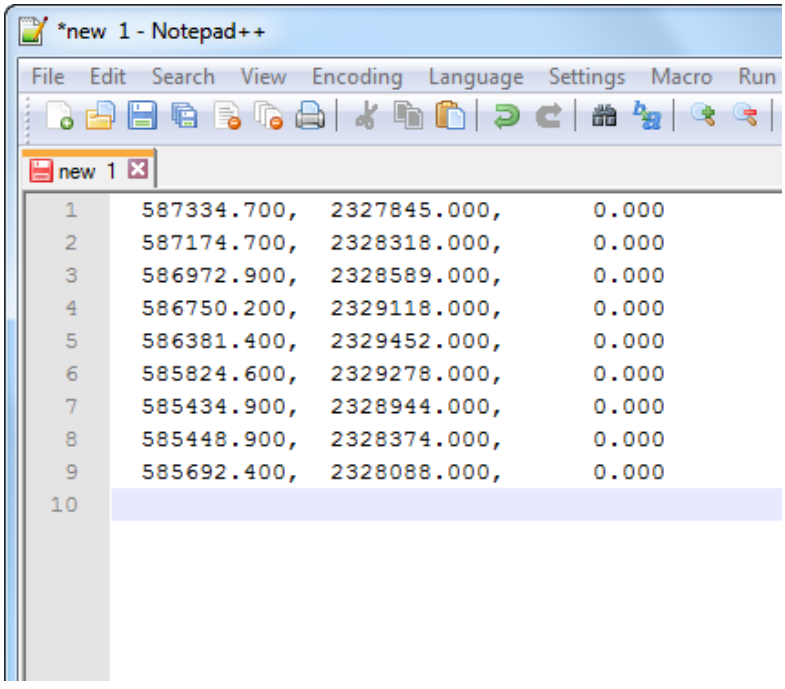


Figure 4-26 Digitized points.

4.2.1.4 None

The *None* option allows the user to end all previous pointer utilities.

4.2.2 Hide/Show DTM

This option allows the DTM file to be displayed or hidden.

4.2.3 DTM Options

When the *Options* is selected a window in appears allowing the user browse to the DTM file and load it as shown in Figure 4-27. When a file has been selected, click the *Load* button to load the information.

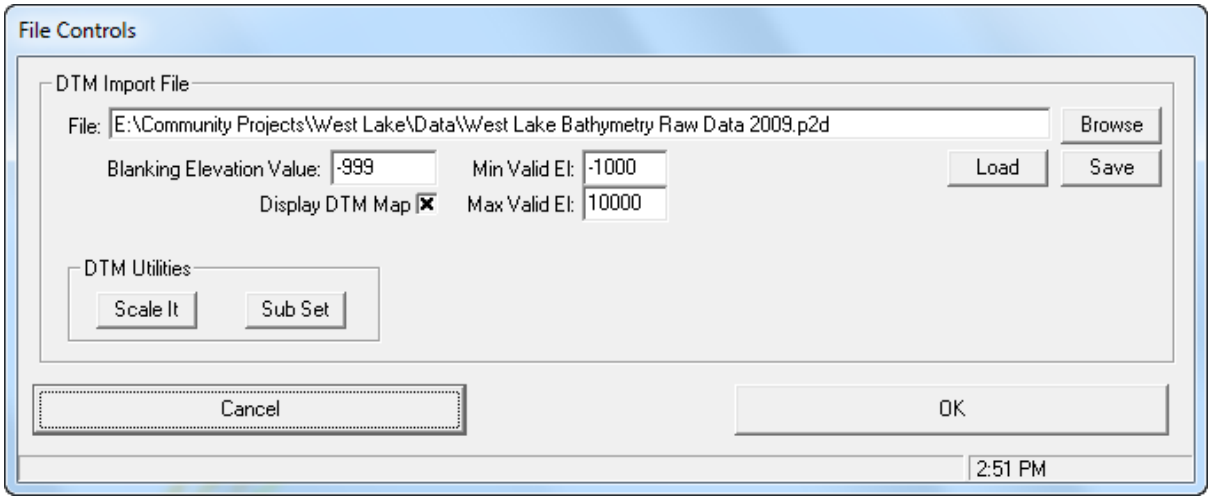


Figure 4-27 File controls window.

Blanking Elevation Value

This option selects what value to fill in missing elevation data points. -999 is the standard input for missing values.

Display DTM Map

This option displays the data file in the workspace when checked.

Min and Max Valid Elevation

These two input boxes let the user adjust the range of elevation values allowed by DTM Tool. By default, this range is usually set to -1,000 m minimum and 10,000 m maximum. The user can use this range to isolate unwanted elevations or to encompass all the elevation points in a very wide range.

Load Options

This frame allows the user to attach data to the data already loaded by clicking the *Append Data* box. The user can also adjust the size of the data points in the workspace by changing the value in the *Diameter of DTM Points* input box.

DTM Utilities

This frame offers two options to the user. The *Scale It* button allows the user to input a conversion factor that will convert the units from meters to another unit such as feet. The input window is shown in Figure 4-28.

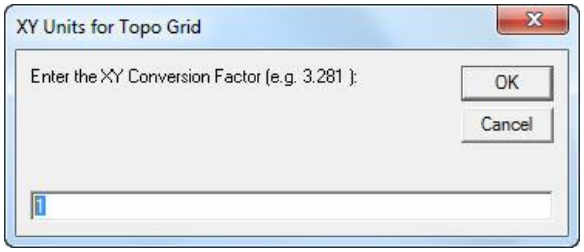


Figure 4-28 Unit conversion input window.

The second button in this frame is *Sub Set*. Selecting this button displays the window shown in Figure 4-29. The various options available here are described in detail below.

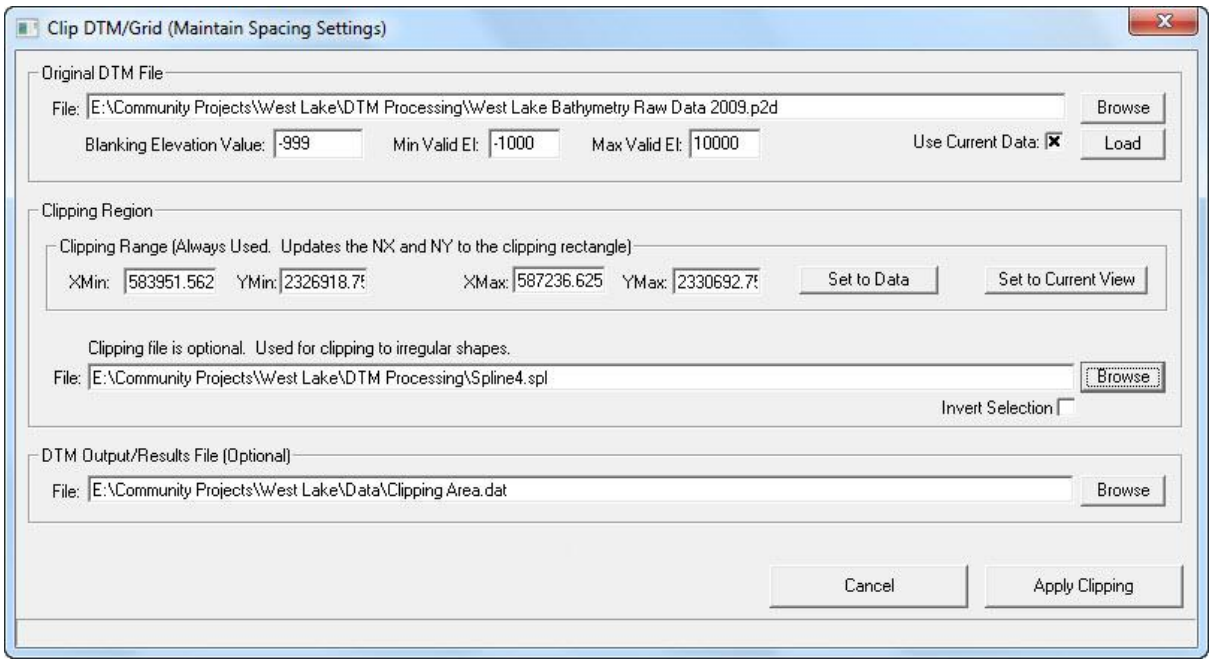


Figure 4-29 DTM clipping window.

Clipping Region

The user is able to enter Min and Max values for X and Y of two defined points for clipping rectangle. If the *Set to Data* button is selected, the fields for X/YMin and X/Y Max values will fill in automatically based on DTM data file loaded. If the *Set to Current View* button is selected, the clipping region will be generated base on the screen view set by the user.

The user can also define the clipping region by browsing to the desired file then loading. The region outside of the clipping line will be cropped. In case the *Invert Selection* check box is selected, the region outside of the clipping line will be retained.

The user can save the region after clipping by browsing to the desired directory and naming the output file. Then click the *Apply Clipping* button for the changes to take effect.

Figure 4-30 and Figure 4-31 show visualizations of DTM data before and after clipping by a polyline.

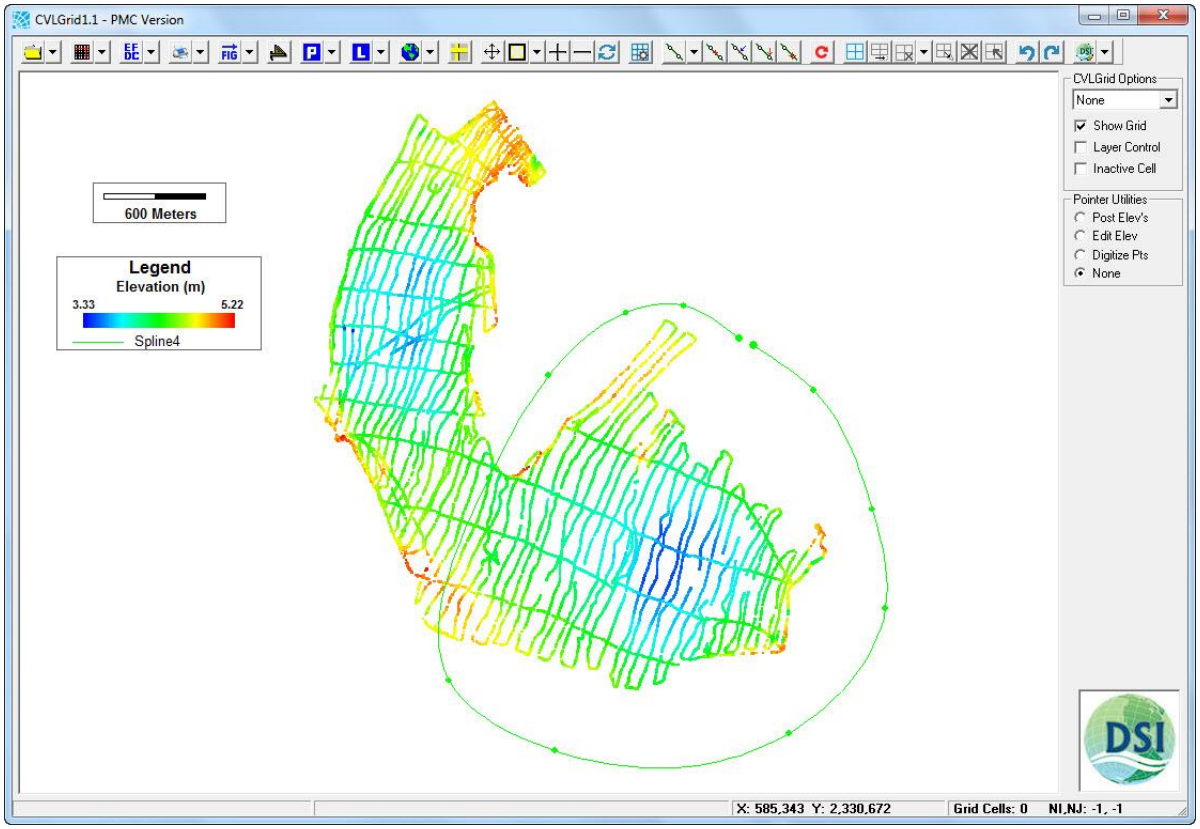


Figure 4-30 DTM data with clipping line before clipping.

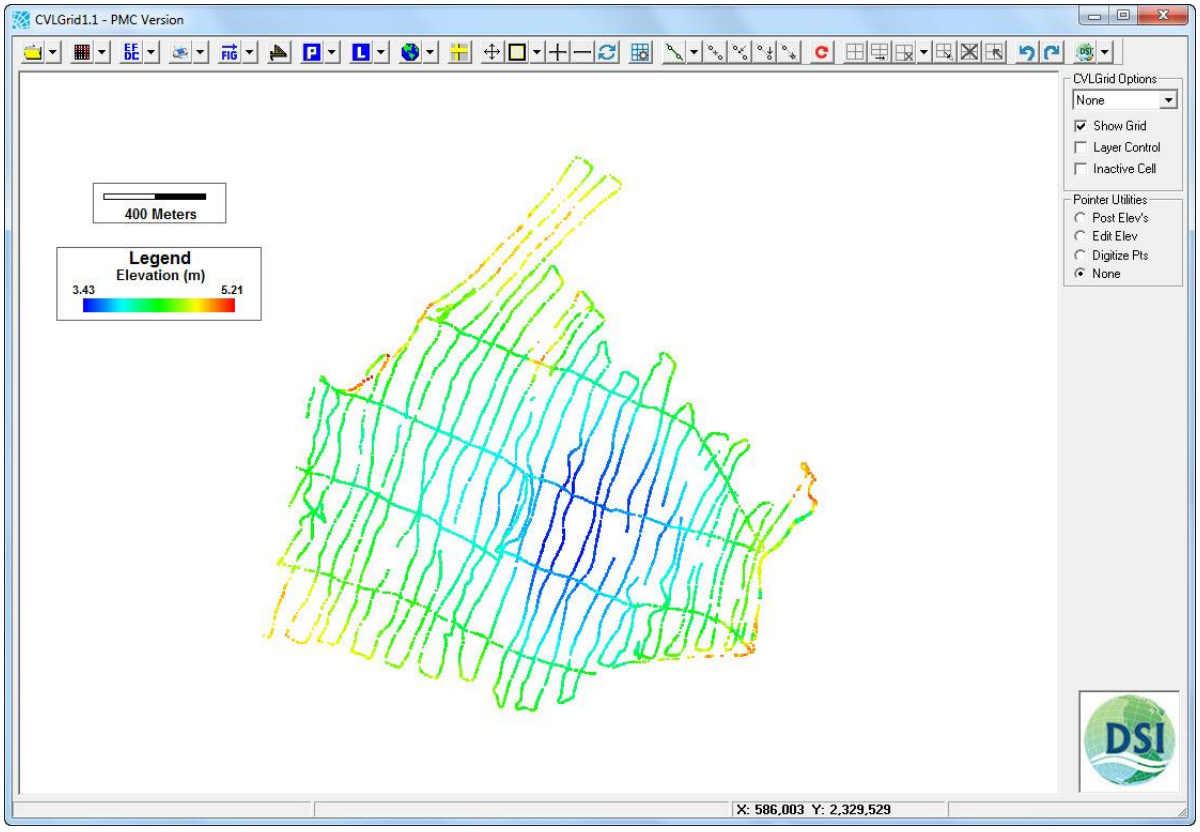


Figure 4-31 Retained region after clipping.

4.3 EFDC Tools Menu

CVLGrid allows the user to load an EFDC grid from an existing EFDC model directly into the workspace. This is done using the *EFDC Tools* button. The options in this menu include loading, saving and displaying EFDC Grid tools as shown in Figure 4-32. Sub-options shown in the dropdown menu are described below.

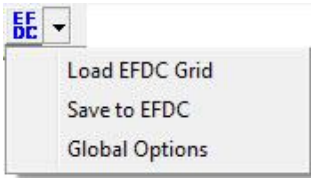


Figure 4-32 EFDC dropdown menu.

4.3.1 Load EFDC Grid

This *Load EFDC Grid* option allows the user to load a grid of an existing EFDC model. This browsing window is shown in Figure 4-33. After clicking the *OK* button the EFDC Grid is loaded in Figure 4-34.

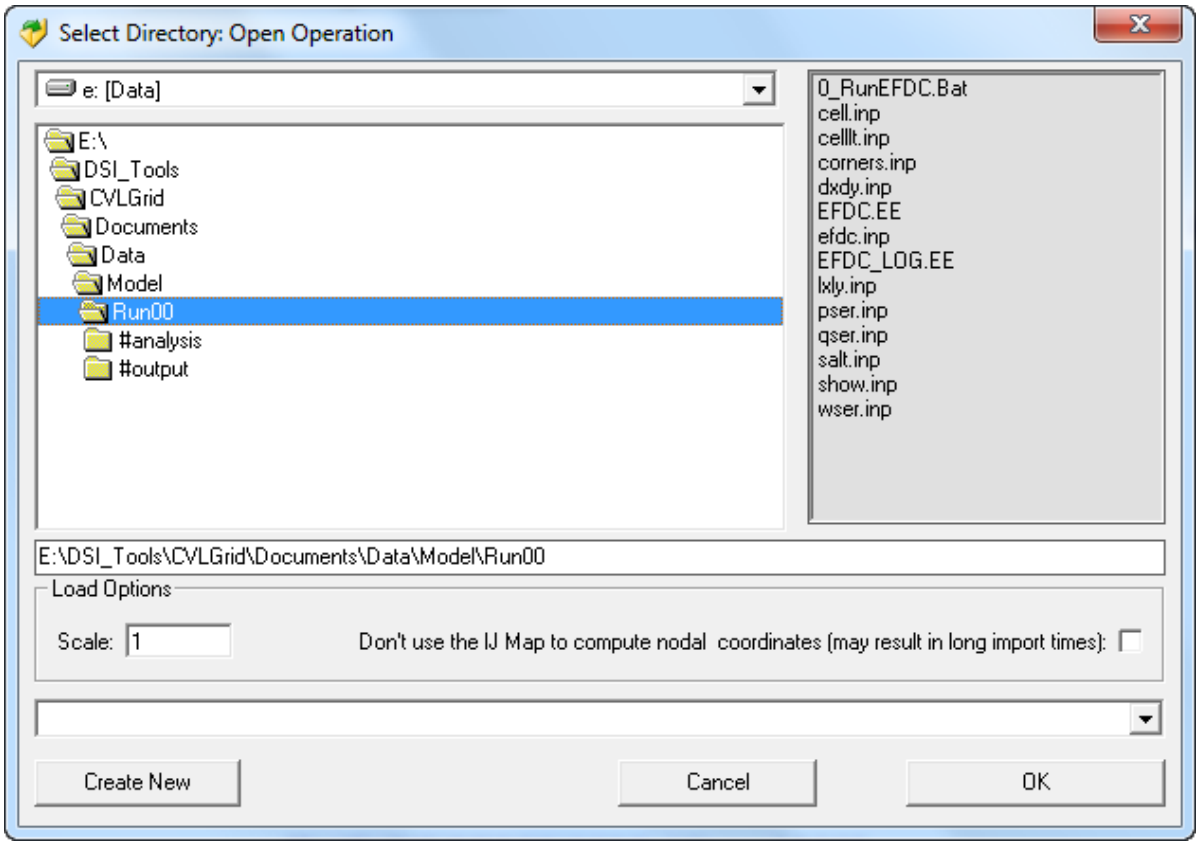


Figure 4-33 EFDC model browsing window.

EFDC Grid Loading Options

The "*Scale*" input box allows the user to apply a conversion factor to the centroid units used in the legacy LXLY file. The EFDC_Explorer default X, Y unit is meters. Many applications use kilometers or miles as the units for the cell centroids provided in the LXLY file. To correctly display the model these cell centroid coordinates must be converted to meters.

CVLGrid can perform this function by entering the conversion factor in the *Scale* box when loading the model for the first time. Note: When a model is loaded and then viewed but looks like a cluster of large cells stacked on top of one another, it is likely to be a LXLY units conversion issue. Try reloading the model with an appropriate scale factor.

The *"Don't use the IJ Map to compute nodal coordinates"* box will force CVLGrid to search through all the cell corners to build the nodes.

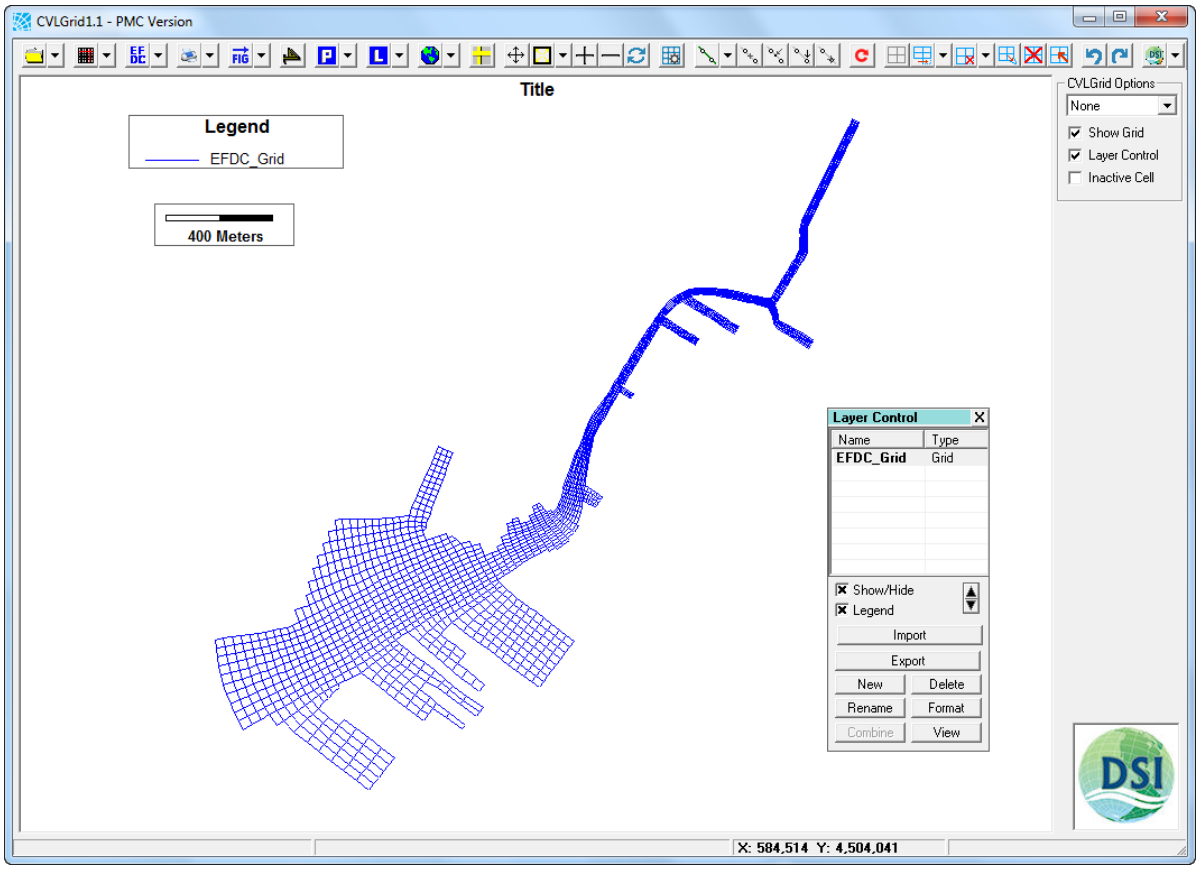


Figure 4-34 EDFC grid loaded.

4.3.2 Save EFDC

When an EFDC Grid is loaded, as shown in Figure 4-34, the user may save the grid after modifying it. If the I, J map has not been altered then the grid can be loaded straight back into EE without needing to change the boundary condition assignments. In other cases when a grid has been generated the user will want to build an EFDC model with the grid. Use the *Create New* button to create a new directory so as not to save over the original grid as shown in Figure 4-35.

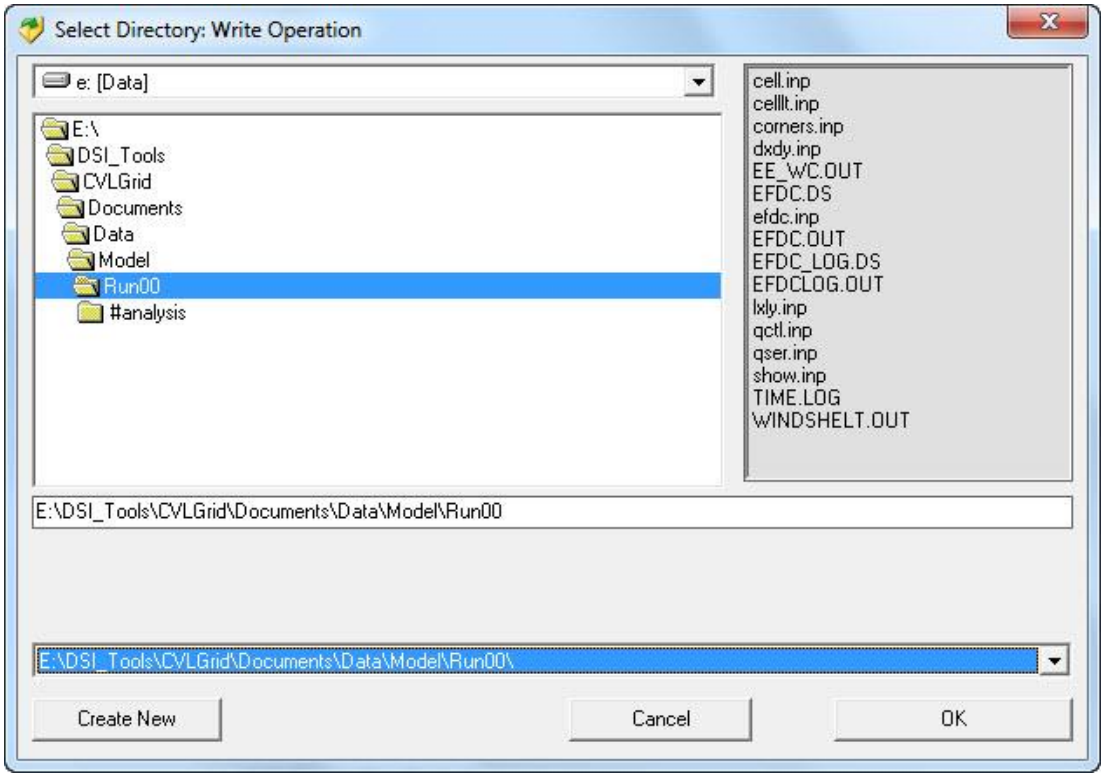


Figure 4-35 EFDC grid save browsing window.

An option window shown in Figure 4-36 appears after the user clicks the *OK* button shown in Figure 4-35. The allows the user to either save the model with existing settings of the model by the use before or default settings of the model by CVLGrid.

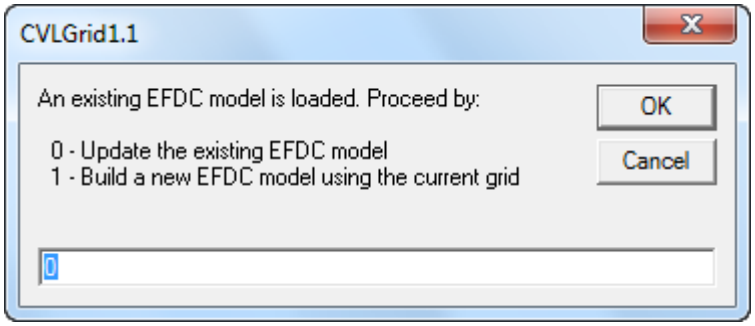


Figure 4-36 Save EFDC model option window.

4.3.3 Global Options

This option allows the user set attributes of the model as roughness, water surface elevation, and number of water layers as shown in Figure 4-37.

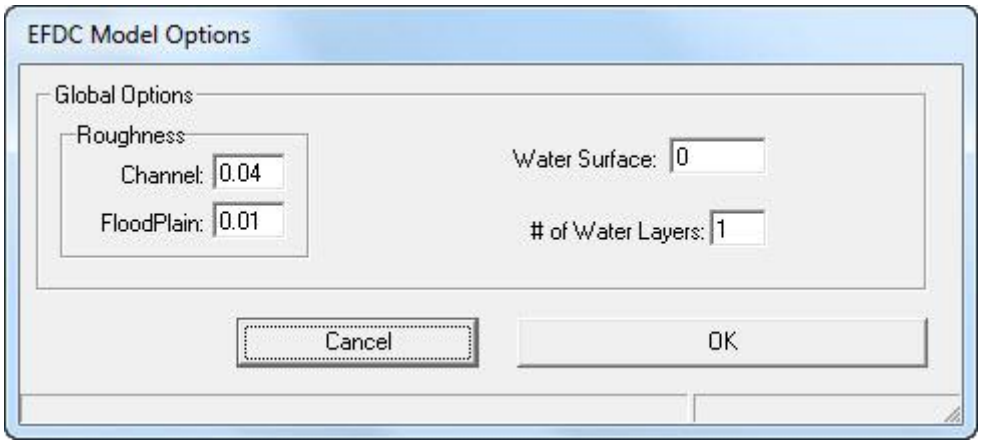


Figure 4-37 EFDC model options.

4.4 Printer Menu

When a grid has been created the user may print an image of the current workspace using the *Printer Options* button shown in Figure 4-39.

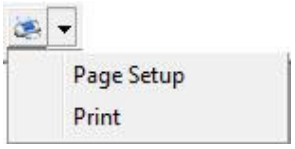


Figure 4-38 Printer menu options.

4.4.1 Page Setup

In the *Page Setup* option a separate window appears allowing the user to make adjustments before printing as shown in Figure 4-39. The top of the window shows the name of the printer that the document will be sent to. This can be changed by selecting the *Show Printer's* Icon.

Page Orientation allows the user to decide whether the page will be printed in landscape or portrait mode.

Margins allows the user to set the size of the page margins. One inch is the default.

Printable area defines the portion of the workspace that will be printed.

Background selects whether or not the background or boarder will be printed. The white box on the right represents a preview of the page and margins.

Selecting *Ok* will save the settings and apply them when the user selects *Print*. *Cancel* closes the print setup window.

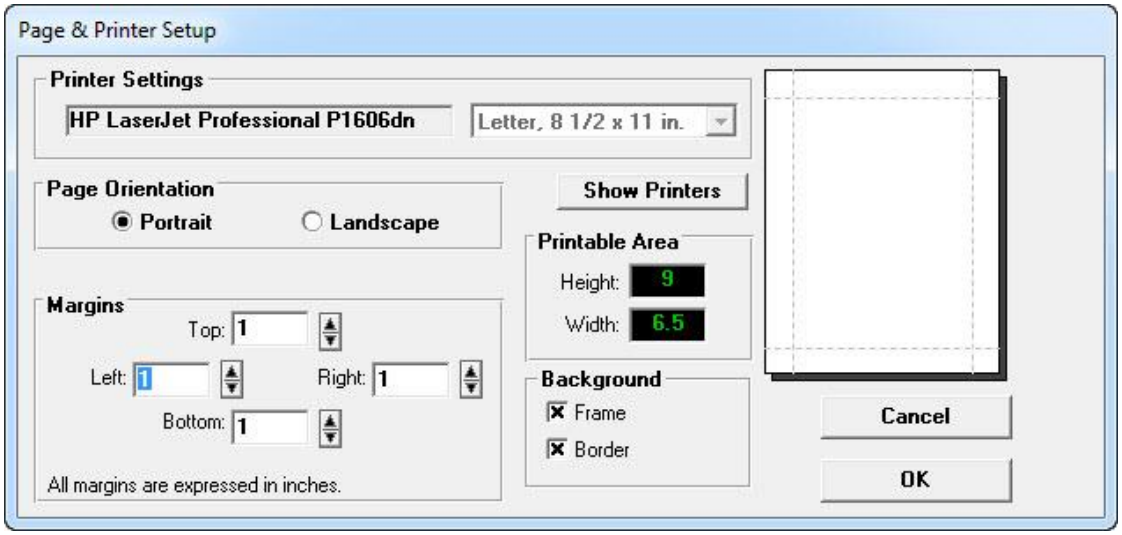


Figure 4-39 Page and printer setup window.

4.4.2 Print

This option applies the settings from the *Page Setup* option and prints the document.

4.5 Exporting Images

CVLGrid allows the user to export images as either Metafiles or Bitmaps using the *Export Image* button, which has the  icon.

4.5.1 Export to a Metafile

To export a metafile the user can select *Export to Metafile* button from the *Export Images* button in the main toolbar. This option opens a window, shown in Figure 4-40, that allows the user to name the file and browse to a location on the hard drive. The file exported has an EMF extension that can be opened in various graphic utilities (e.g. DSI's Picture Viewer) and will display whatever is on the CVLGrid workspace.

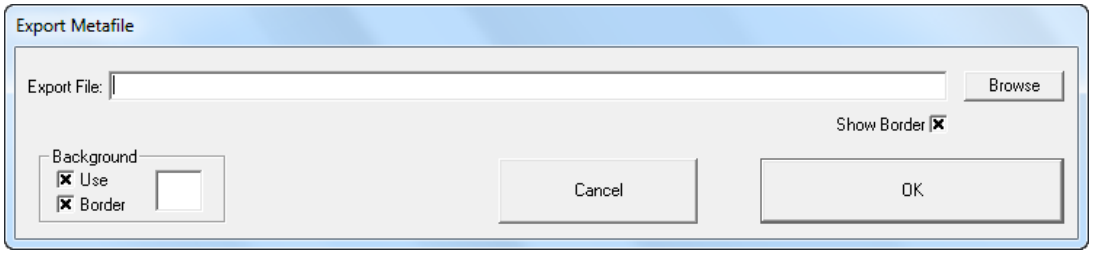


Figure 4-40 Export to metafile.

4.5.2 Export to Bitmap

The user can also export everything on the CVLGrid workspace to a bitmap file by selecting *Export to Bitmap* option from the menu of *Export Images* button as shown in Figure 4-41. There are a variety images formats that can be saved including JPG, BMP, PNG, GIF and TIF.

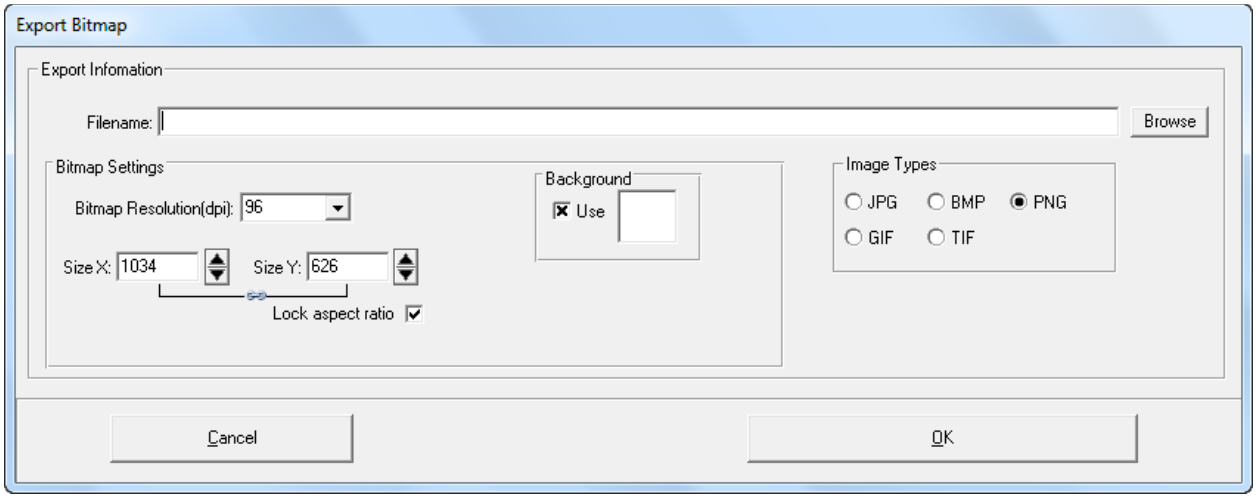


Figure 4-41 Export to bitmap.

4.6 Display Options Menu

The *Display Options* menu can be accessed by clicking on the  icon in the main toolbar. This icon opens a window, shown in Figure 4-42, that presents the user with a number of display options described more fully below.

4.6.1 Main Title

The first option when you open the *Display Options* menu is the *Main Title* frame. Here the user can input the name of the project and it will be displayed in the workspace. The *Format* button next to this frame allows for formatting changes such as font, size, color, position, and angle (shown in Figure 4-43). The *Hide Title* text box will hide the title from the workspace

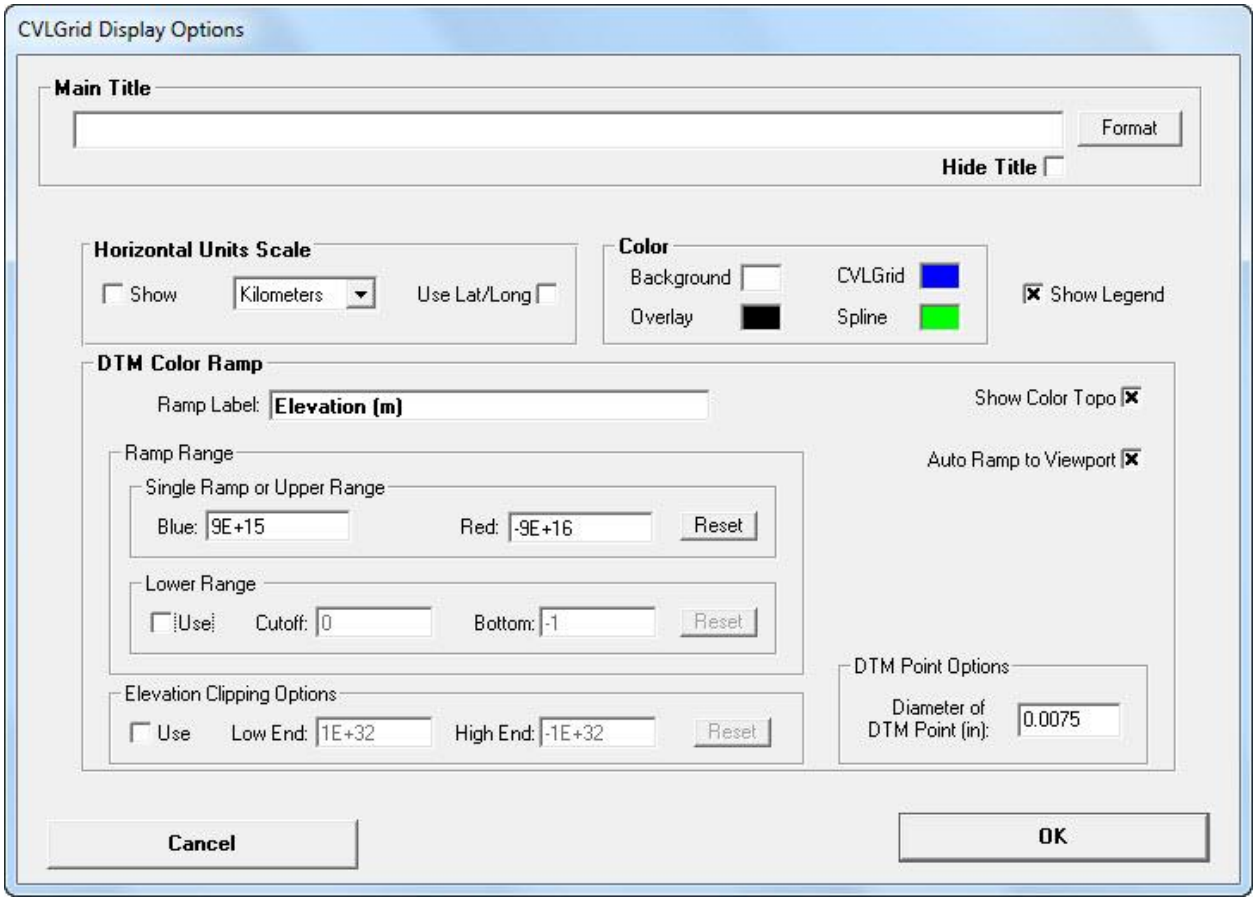


Figure 4-42 Display options window.

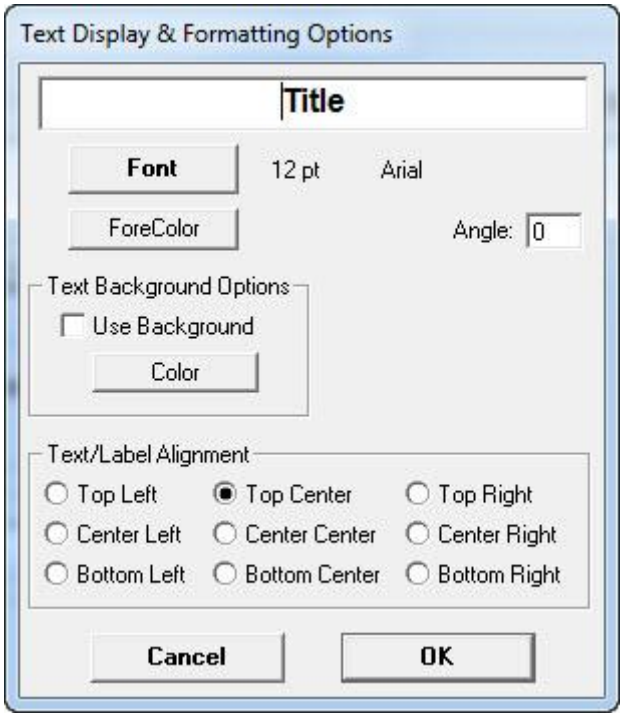


Figure 4-43 Main title display formatting.

4.6.2 Horizontal Units Scale

The *Horizontal Units Scale* frame provides the user the option of adding a scale legend to the workspace. This legend gives a visual representation of distance and is based on the size of the model domain and its UTM coordinates, shown in Figure 4-44. The units of this legend can be changed to show meters, kilometers, feet, miles and nautical miles, however, the different units are simply converted from the base unit, meters.

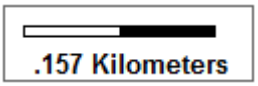


Figure 4-44 Horizontal units scale legend.

4.6.3 Background Color

The *Background Color* frame allows the user to change the color of the workspace background from its default color white. This option opens a window with many color options to choose from.

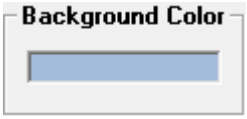


Figure 4-45 Background color setting.

4.6.4 DTM Color Ramp

The *DTM Color Ramp* frame deals with elevation and the color gradients to represent different elevations. If the user is only using a 2D polygon then this section is irrelevant. Figure 4-46 shows this section for reference.



Figure 4-46 DTM color ramp sub-sections.

Ramp Label	Give the color ramp a name
Single Ramp or Upper Range	Specifies the range of color from blue to red
Lower Range	Specifies the lower range of the color ramp
Show Color Topo	Shows the elevation topography in color based on the color ramp.
Auto Ramp to Viewport	Change Min, Max, and Average number of View Options “Ortho”, “Dx”, and “Dy” on Legend when view window changed.
Elevation Clipping Options	Specifies the elevation range for clipping
DTM Point Options	Allow the user set the diameter of DTM Points in inches

4.7 Data Posting Options

Posting files contain coordinate data that link a specific structure or geographic feature to a particular location. The *Date Posting Option* button and menu options are shown in Figure 4-47. This tool enables the user to load a posting file and display the locations of multiple structures or features in relation to each other.

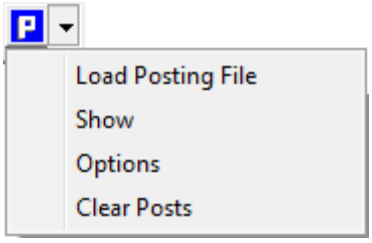


Figure 4-47 Data posting menu options.

4.7.1 Load Posting File

This option loads a browser window displayed in Figure 4-48 from which the user can locate and select the desired posting file. The file types that can be loaded are explained in Section 1.6 When a posting file is selected, the settings window in Figure 4-50 is displayed before anything is loaded to the workspace. This settings window is explained in Section 4.7.3 below. Figure 4-51 shows the display window with data from a posting file loaded.

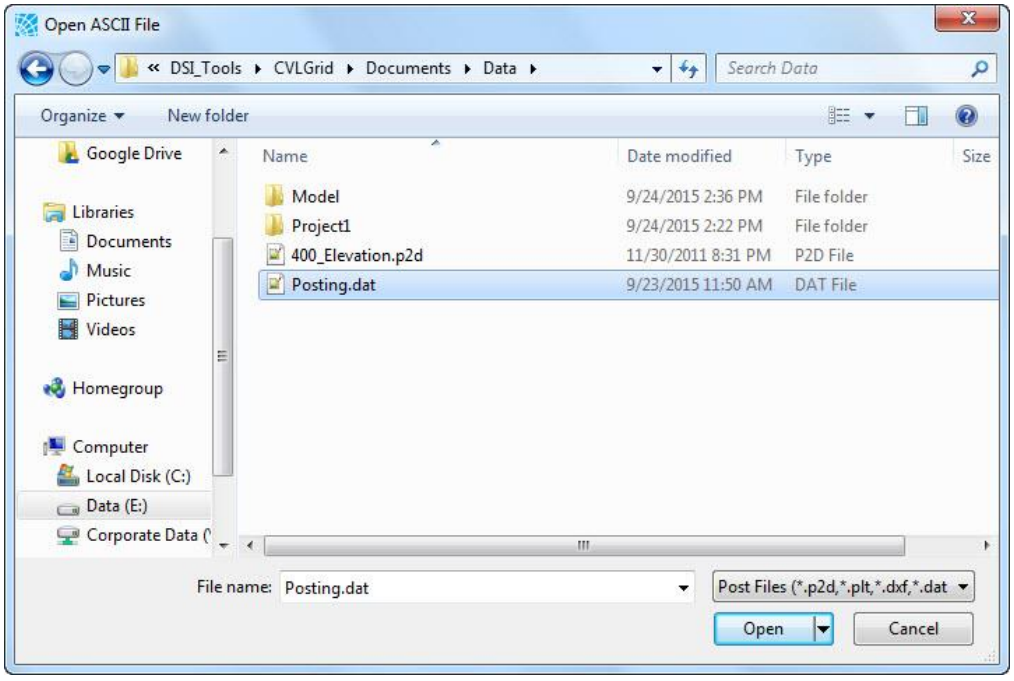


Figure 4-48 Loading posting file.

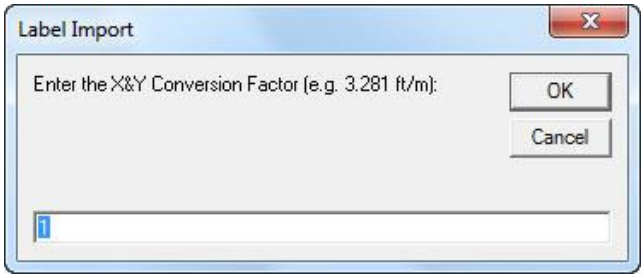


Figure 4-49 Loading posting file: label import.

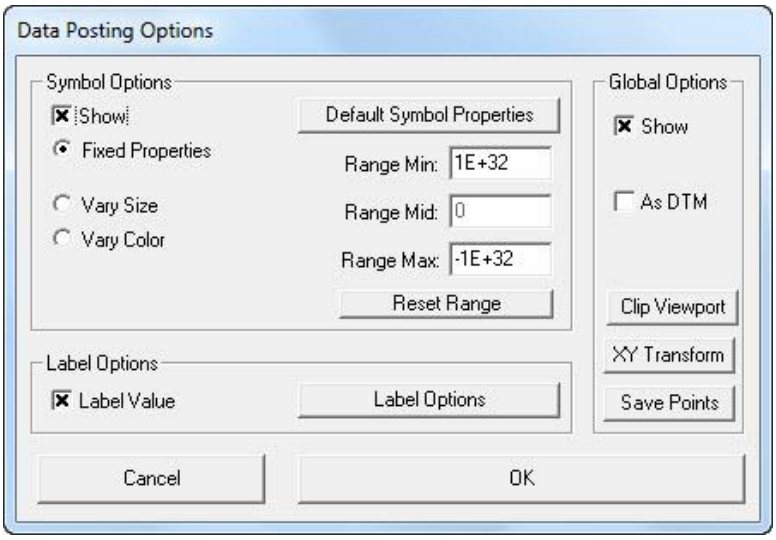


Figure 4-50 Data posting options window.

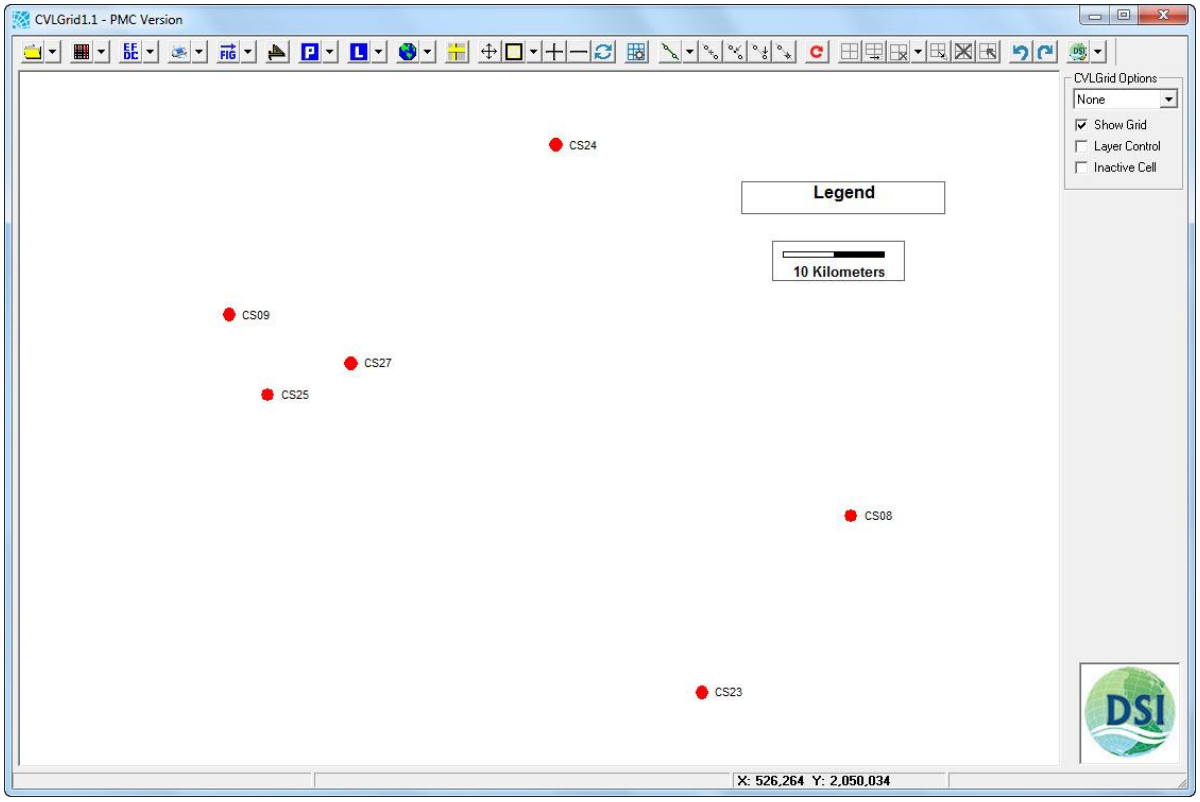


Figure 4-51 Posting file data displayed in the workspace.

4.7.2 Show/Hide

Shows or hides the data from the posting file.

4.7.3 Options

The *Data Posting Options* form changes what will be displayed in the workspace as shown in Figure 4-50. The following list describes some of the adjustments that can be made.

Default Symbol	Allows the user to change the symbol that indicates each structure or feature.
Properties	Changes can be made to the type of symbol, color, and fill
Clip Viewport	Clips all the data that is outside the current view in the workspace
XY Transform	Allows the user to shift or rotate the coordinates of the data loaded
Save Points	Saves all the data points currently in the workspace
Label Options	Allows the user to change the text that labels each point. The text, font, color, and positioning can all be adjusted from this option

4.7.4 Clear Posts

This option clears the workspace of all the posts data that was previously loaded.

4.8 Label Options

Similar to posting files, label files contain coordinate data that link a text label to a specific location. This tool can be used to label stations (posts) in the workspace that do not have labels or to label geographic features important to the project. This tool can be used by selecting the *Label Options* button shown in Figure 4-52.

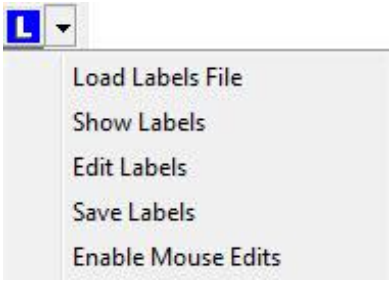


Figure 4-52 Label menu options.

4.8.1 Load Labels File

This option displays a browser window from which the user is able to select the desired labels file. Once the file is selected it is displayed in the workspace. If a posting file was previously loaded, the labels with the same coordinates will show up right next to the posts.

4.8.2 Hide/Show

Hides or reveals the label files in the workspace.

4.8.3 Edit Labels

This option allows the user to make changes to multiple labels currently loaded in the workspace. When selected the window in Figure 4-53 appears.

In this window, the list of labels currently in the workspace is displayed in a sub-window on the left. At the bottom of this labels list the user can add or delete labels using the *Insert* and *Delete* buttons.

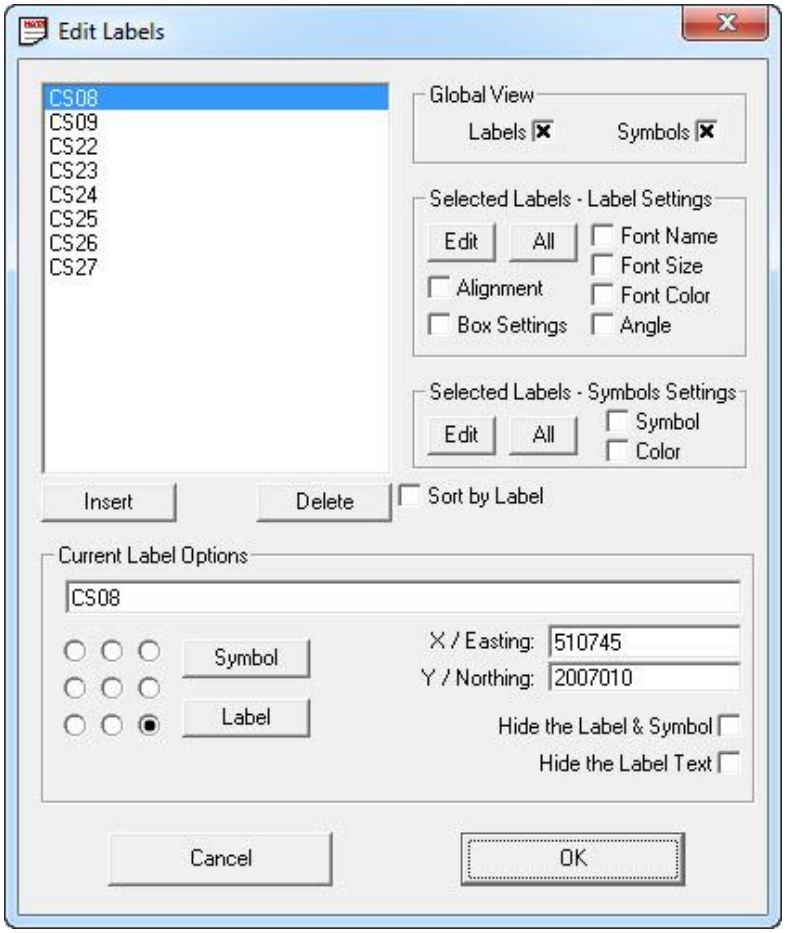


Figure 4-53 Edit Labels window.

The following list describes the sub frames in the *Edit Labels* window.

Global View	The two check boxes select what is visible in the workspace. The user can make both labels and symbols visible or just one or the other, or none.
Label Setting	This sub frame applies changes only to the label selected from the list on the left.
Symbol Settings	This sub frame applies changes to the symbol of the label selected from the list on the left.
Sort by Label	Sorts the list alphabetically.
Current Label Options	Displays the information from the label currently selected in the list and allows the user to change the settings of that particular label.

4.8.4 Save Labels

This option saves the current labels in the workspace including the adjusted settings.

4.8.5 Enable/Disable Mouse Edits

This tool allows the user to move manually labels by LMC on the label and dragging it to another location. When enabled a black square appears next to the label indicating that the label can be moved.

4.9 Geo-Reference Background

CVLGrid allows the user to load a geo-reference image to display in the workspace along with the grid and overlays files. This function aids the user in understanding the model domain and locate points of interest where the grid resolution should be increased. The options in this menu are shown in Figure 4-54.

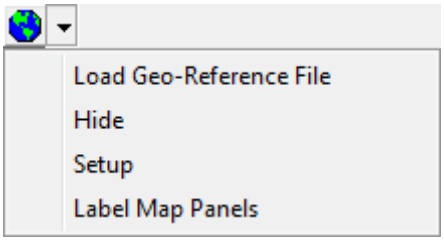


Figure 4-54 Geo-Reference dropdown menu.

4.9.1 Load Geo-Reference File

From the dropdown menu, when *Load Geo-Reference File* is selected, a browsing window appears. The user is then able to navigate to the location of the geo-reference file on the hard drive and load it into the workspace. This browsing window is shown in Figure 4-55, and a loaded geo-referenced file with grid is shown in Figure 4-56. Note that the geo-reference files use the .GEO extension.

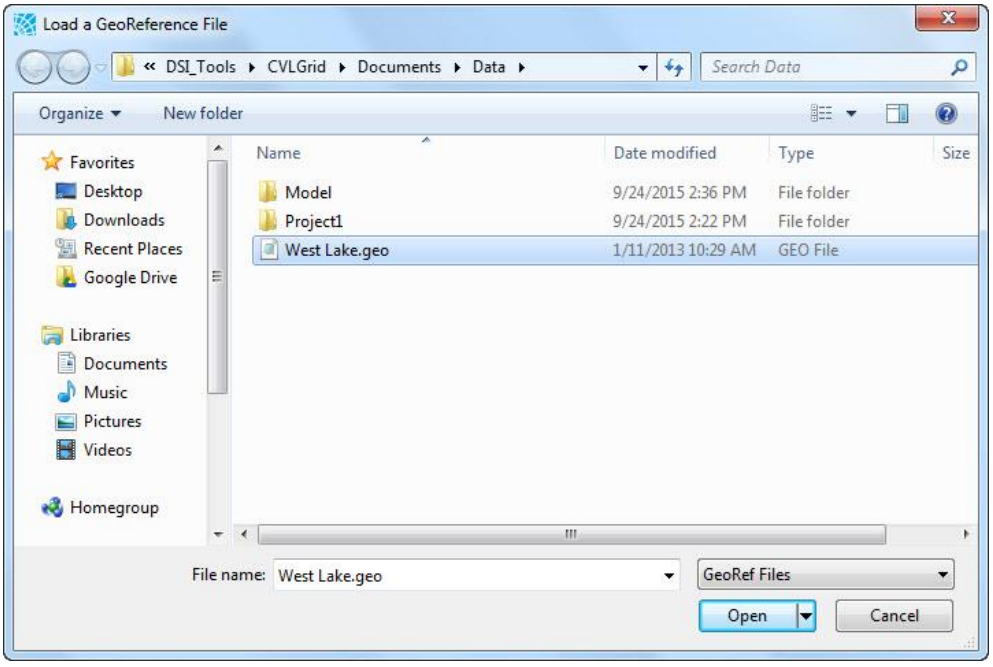


Figure 4-55 Geo-Reference browsing window.

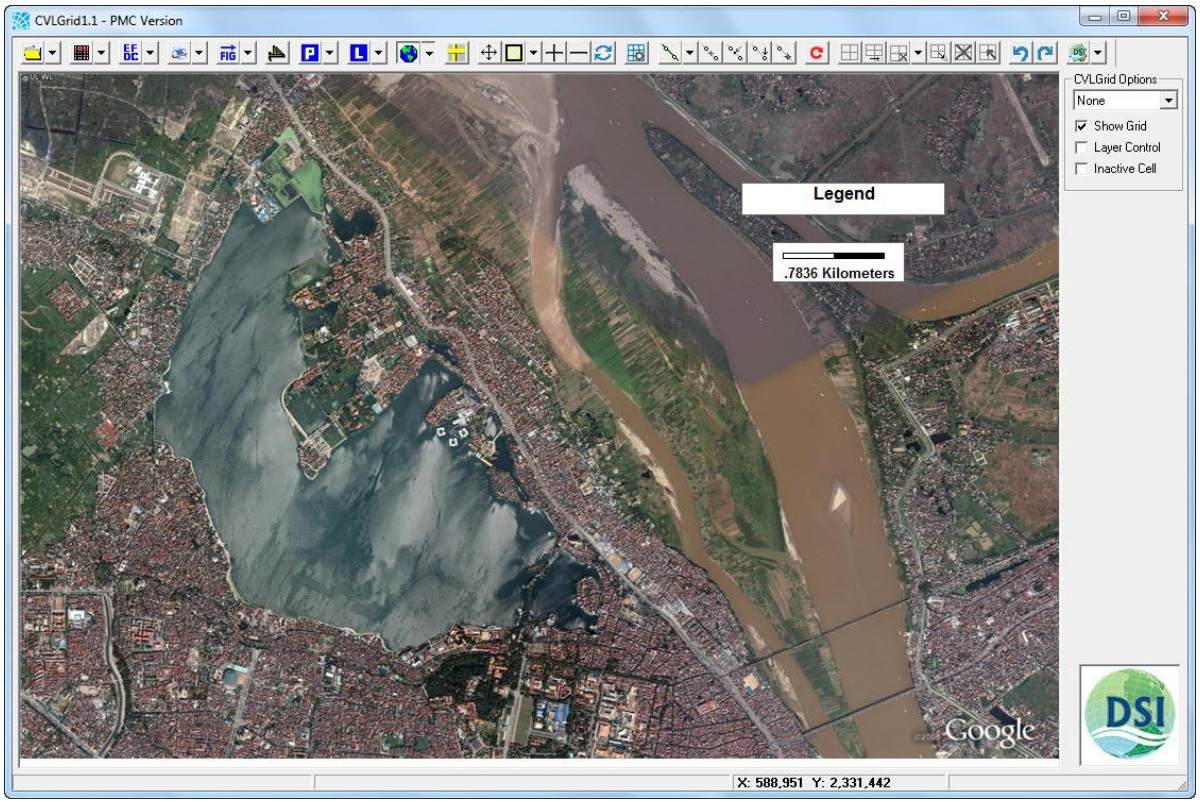


Figure 4-56 Geo-Referenced file loaded in CVLGrid.

4.9.2 Setup

The Setup option opens a separate window that allows the user to create a geo-referenced image. The user is able to load a bitmap image exported from a third party mapping tool such Google Earth (the bitmap image here is JPEG image), for example, and specify the coordinates of two points on the image. Points are specified by RMC on the point location and selecting point 1 or 2 from a drop down menu. The corresponding point coordinates must then be input to the appropriate box in the left toolbar. This window is shown in Figure 4-57 and the file menu is shown in Figure 4-58. Detailed explanation of how to use this tool is provided in a separated document available from DSI.



Figure 4-57 Geo-Reference setup window.

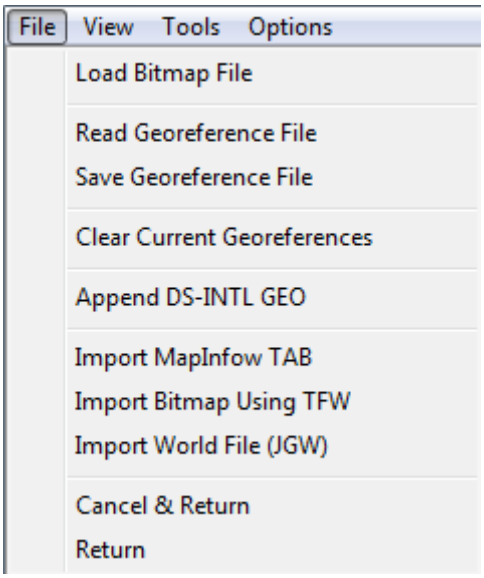


Figure 4-58 Geo-Reference setup window file menu.

4.9.3 Label Map Panels

This option simply allows the user to label the geo-referenced map.

4.10 Distance Measuring Tool

The measurement tool is a simple tool that allows the user to click and create lines in the workspace. At the bottom of the workspace the total distance, current line length, Dx and Dy coordinates, and the current line angle are shown. Figure 4-59 and Figure 4-60 show this measurement tool. The measurement tool can be activated by the button shown in Figure 4-59.

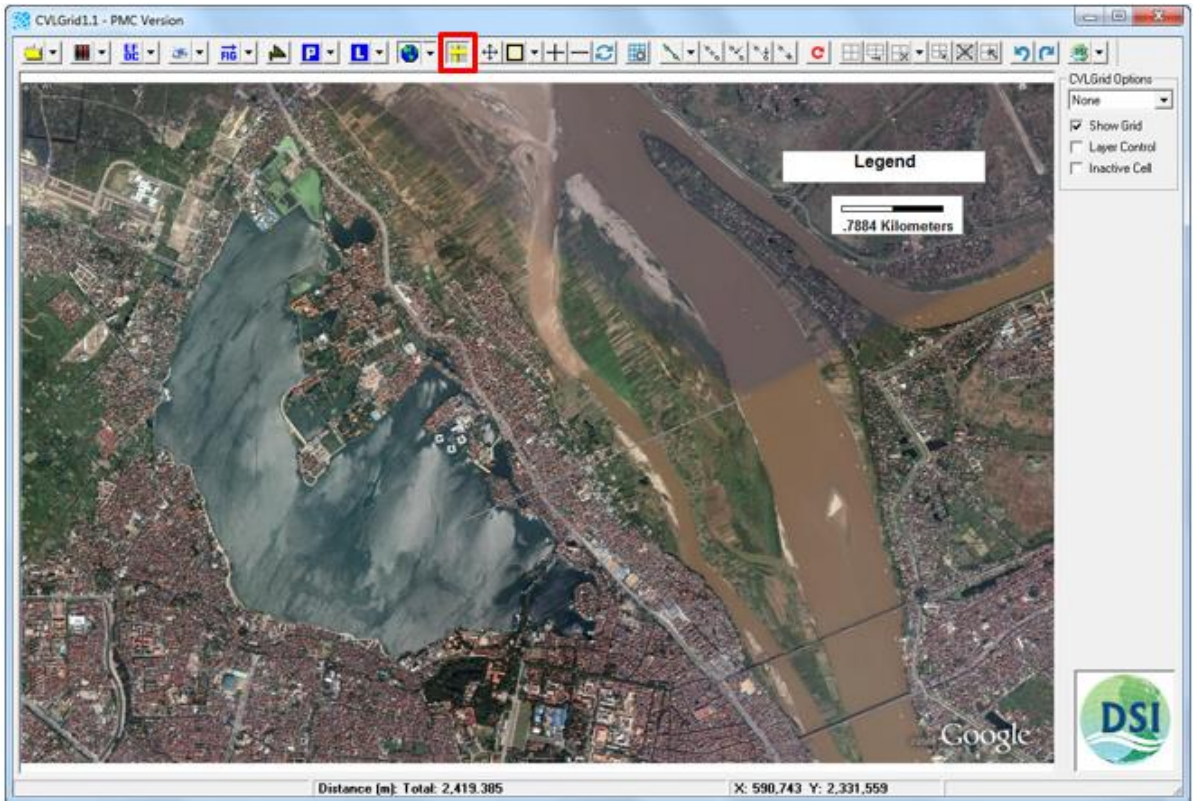


Figure 4-59 Measurement tool



Figure 4-60 Measurement tool information display.

4.11 Navigation Menu

The navigation is fairly straightforward but its various functions are explained below.

4.11.1 Toggle Pan

Toggle pan allows the user to move to different parts of the workspace by clicking in the workspace and dragging to the desired location. This tool can be activated by selecting the icon in the main toolbar. This function can also be activated with the SPACEBAR as a keyboard shortcut.

4.11.2 Zoom to Extents

Zoom extents automatically zooms out so that everything in the workspace is shown. This function can be activated by pressing “Ctrl + E” on the keyboard or by selecting the icon in the main toolbar.

4.11.3 Zoom In/Zoom Out

The user is able to zoom in and out in the workspace by selecting the  icons respectively. The keyboard shortcuts for this are the + and – keys.

4.11.4 Refresh

This button refreshes the workspace and shows the most current spline changes by selecting the  icon. The keyboard shortcut is the F5 key.

4.12 CVLGrid Settings

This menu allows the user to adjust default setting for grid generation. These include various genera settings as well as orthogonalization parameters, smoothing parameter, and scale settings. This menu can be accessed by clicking on the *CVLGrid Settings* button represented by the  icon in the main toolbar. This settings window is shown in Figure 4-61 and Figure 4-62.

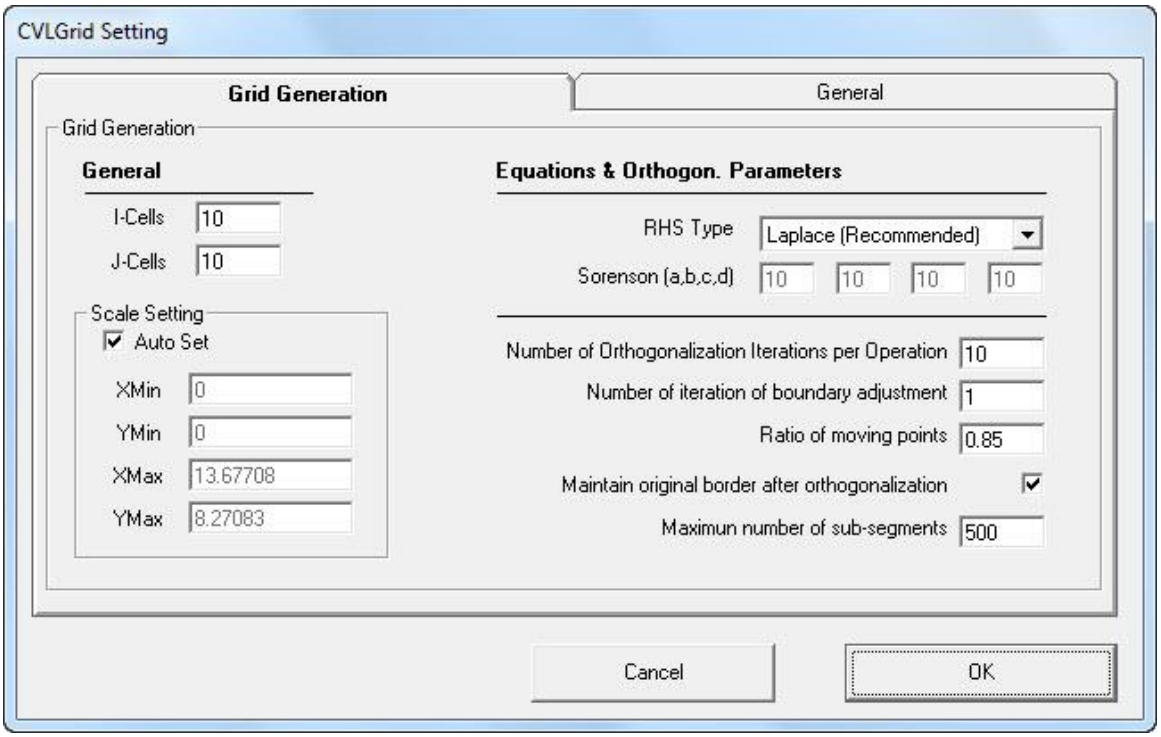


Figure 4-61 Grid generation settings.

I-Cells	Specifies number of cells in the I (X) direction.
J-Cells	Specifies number of cells in the J (Y) direction.
Scale Settings	*Auto Set* is set as default for viewing. The X and Y values can be manually adjusted by unchecking the *Auto Set* box and typing in the desired values in the input boxes. When *Apply* is selected the workspace is refreshed and zoomed in or out based on the X and Y coordinate limitations.
RHS Type	There are two optional choices for grid generation: Laplace and Sorenson.
Sorenson (a, b, c, d)	These are Sorenson coefficients. The user is able to put the coefficients when Sorenson option is chosen for the equation solving option. For *RHS Type*. *a* and *c* for P; *b* and *d* for Q. When a, b, c, d is greater than 100 the Sorenson tends to Laplace methodology in generating grid.
Number of Orthogonalization Iterations per Operation	Specifies the number of iterations to perform during calculation.
Number of Iterations of Boundary Adjustment	Specifies the number of times to move the boundary points. CVLGrid automatically sets these to 10 and 1 respectively. This is recommended if the grid is very large as it can reduce calculation time.
Ratio of Moving Points	Ratio of moving node grid on a segment on the border. 0< ratio <=1
Maintain Original Border after Orthogonalization	Keep grid nodes on the border close to the border as possible when orthogonalize is used
Maximum Number of Sub-segments	Specifies the maximum number of sub-segments

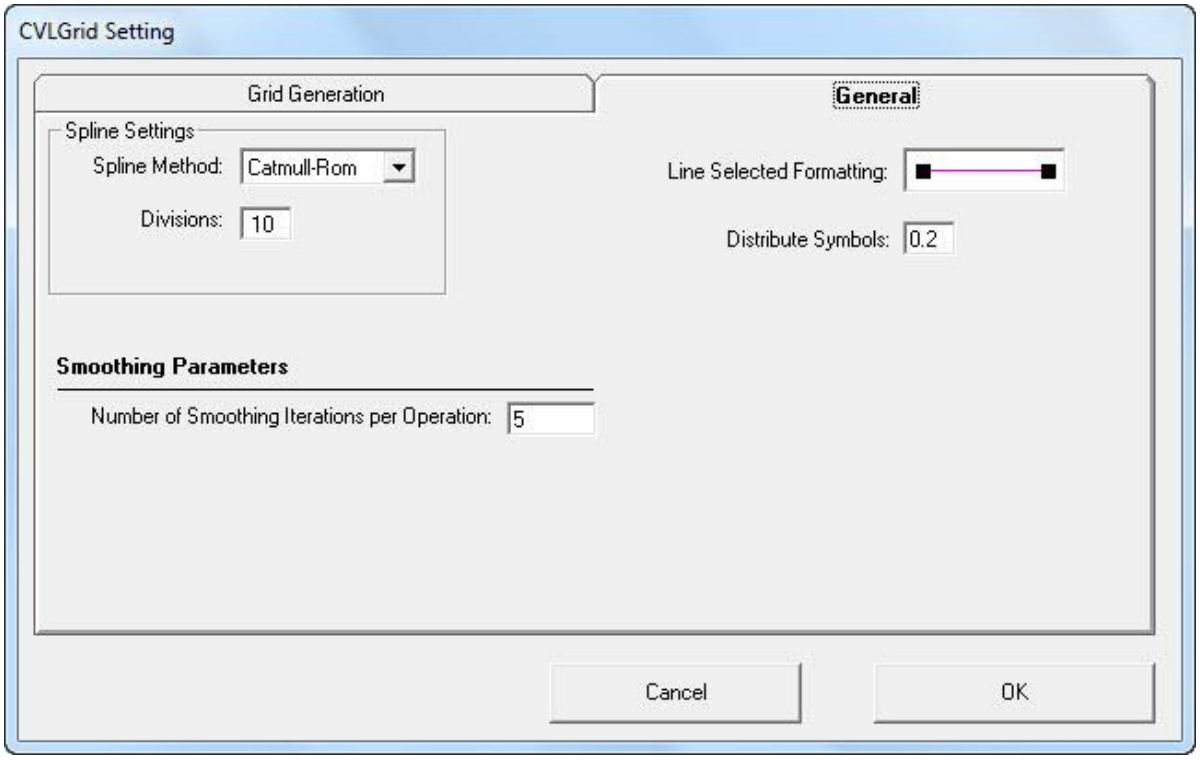


Figure 4-62 General settings.

Spline Method	There are two optional choices for setting the spline type: *B-Spline* and *Catmull-Rom*.
Divisions	Number of point division between two spline points when draw a spline. 1<= Division <=100
Line Selected Formatting	Allows the user to format attributes of line and symbol when clicking on this image , a *Line Formatting* frame appears as Figure 4-63.
Distribute Symbols	Delta value to display space between point symbols in polyline. Default value is 0.2 inch.
Number of Smoothing Iterations per Operation	Specifies the number of iterations for smoothing

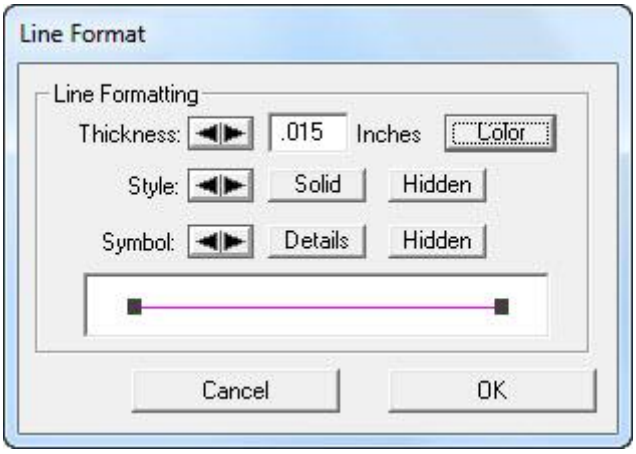


Figure 4-63 Line formatting.

4.13 Spline Menu

As mentioned in Section 3.2 , splines are the most important part of generating grids. A spline can be either a straight line or a curved line and must contain at least two points. The Laplace equations are based on these splines, thus in order to create correct grids the user must define the splines correctly. In the main toolbar there are five icons that make up the "Spline Menu", shown in Figure 4-64.



Figure 4-64 Spline menu icons.

Table 4-1 Spline menu button options.

	This button allows the user to create a new spline/boundary line. Clicking in the workspace creates a point. Moving the mouse and clicking in another location creates another point with a line joining the two points. A point is indicated by a circle (if a spline is drawn). The spline/boundary line continues as the user continues to create new points. RMC to end drawing a spline.
	This button allows the user to delete previously created splines. When this button is selected, clicking on a spline will delete it.
	This button allows the user to insert points on an already created spline. Once a point is inserted it can be moved to reshape the spline.
	This button allows the user to move points on a spline. This enables a spline to be adjusted to better shape the model domain.
	This button allows the user to delete points on a spline.

4.14 Repeat last CVLGrid operation

The user may repeat the last CVLGrid operation which includes *Refine*, *Orthogonalize* and *Smooth* grid in global and local aspects with the *Repeat* button  on the main toolbar.

4.15 Grid Menu

Once the splines have been created and edited the user is able to generate a grid. At this stage the *Create Grid* button is available to be selected as shown in Figure 4-65.



Figure 4-65 Grid menu before creating a grid.

The number of cells must be specified by the user after clicking the *Create Grid* button. After a grid has been created the whole grid menu is available to the user and is made up of eight buttons as shown in Figure 4-66.



Figure 4-66 Grid menu.

This menu is used to create and edit grids in the workspace. The user has various options when deleting nodes as in the drop down menu as shown in Figure 4-67 and Figure 4-68. Table 4-2 provides further detail on each of the Grid menu buttons.

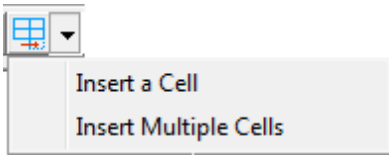


Figure 4-67 Insert cell icon options.

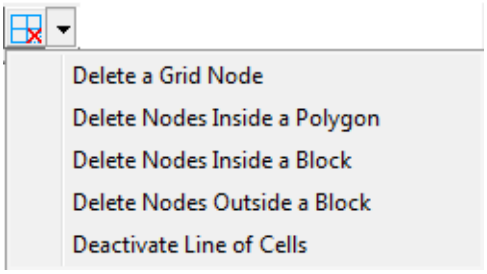


Figure 4-68 Delete cells icon options.

Hint: *before creating a grid make sure that the splines are drawn correctly and are in the best position to create a very orthogonal grid.*

Table 4-2 Grid menu button details.

	This button will generate a grid within the area enclosed by the splines. The number of cells in the I and J direction will depend on the user's input in the *Operations* toolbox.
	This button allows the user to insert one or multiple cells within the generated grid. There two options in the drop down menu as shown in Figure 4-67. *Insert a Cell* will add one cell at a time to the generated grid. To add a cell, click just outside the grid where the extra cell is desired. The new cell will be added and connected to the nearest cell(s). *Insert a Multiple Cells* will add a number of cells to the generated grid by selecting two nodes on same marginal grid line. The new cells will be added and connected to the nearest cell(s).
	This button allows the user to delete one or multiple cells within the generated grid. There are a number of options in the drop down menu as shown in Figure 4-68. *Delete a Grid Node* will immediately delete the node of a cell selected by the hand icon. *Delete Nodes Inside a Polygon* allows the user to create a polygon with LMC. Closing the polygon with a RMC will delete multiple cells inside the polygon. *Delete Nodes Inside a Block* is used by clicking two nodes to indicate the corners of the grid block that you want to delete. A block is selected which just contains the selected nodes. Clicking the second node will delete all cells of the block area. *Delete Nodes Outside a Block* is used by clicking four nodes to indicate the corners of the grid block. A block is selected which just contains the selected nodes. Clicking the fourth node will delete all cells in the area outside the selected block. *Deactivate Line of Cells*: Instead of deactivating cell by cell on the grid, select this function then draw a line by LMC in order to intersect multiple cells. RMC to end the line and multiple cells along that line will be deactivated.
	This button will move nodes on a grid. To move a node on the grid, LMC on the desired node then LMC on the position it should be moved to. Any line connected to that node will move also.
	Delete the entire grid selected. Simply click this icon and then select the grid to delete. Note: any grid connected separately to the grid selected will also be deleted.
	Select a grid so that it can be edited. If there are several grids in the workspace that are not connected the user will only be able to edit one at a time. This icon selects a grid and makes it active for editing.
	Go back to a previous operation (Undo).
	Go forward to next operation.

4.16 4.16 DSI Button

The DSI button provides the user access to the *About, Help,* and *License Manager* options. The *About* screen informs the user of the release date for the current version of CVLGrid. The *Help* menu provides context sensitive user guidance. For the most up-to-date guidance the user should refer to the *Knowledge Base*.In some cases the user settings for screen resolution can cause the DSI button to be hidden. In this case use the F1 button to access *Help,* and use the F3 button to display the *License Manager*.

5 Side Menu

5.1 Viewing Options

The user can select what to view on the main form by selection options on the side bar shown in Figure 5-1. The *CVLGrid Options* dropdown menu allows the user to select what to make visible on the grid. Below this there are a number of other options available on the side bar that can be selected with a check box. These include:

Show Grid	Grid can be shown or hidden by selecting this option.
Layer Control	Layer Control tool can be displayed or hidden on workspace by selecting this option.
Inactive Cell	Inactive Cells of the Grid can be viewed by selecting this option.



Figure 5-1 Side menu.

A number of options are available to the user shown in Figure 5-2, including *"None", "L Index", "I & J", "Corners", "Orthogonal Deviation", "Dx", and "Dy"*. These options are described in the sections below.



Figure 5-2 Viewing options for grid.

5.1.1 None

When the option *None* is chosen then only the grid is displayed in the workspace. This is shown in Figure 5-3.

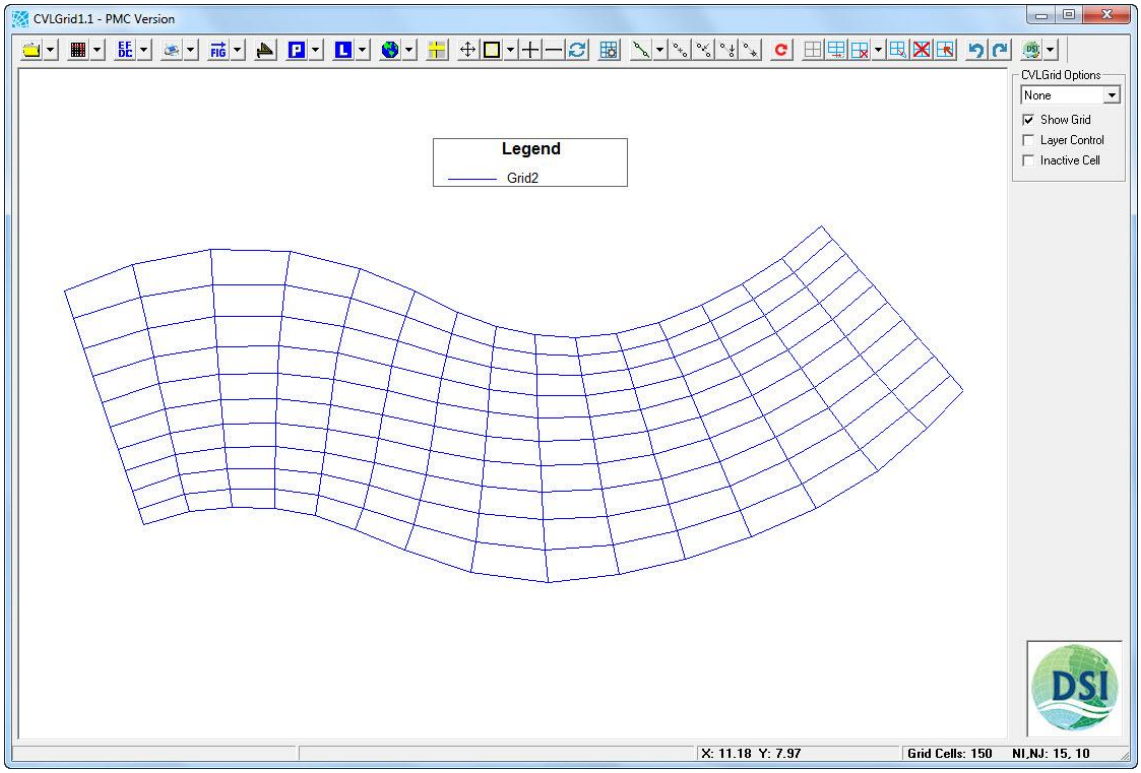


Figure 5-3 Index: "None" viewing option.

5.1.2 L Index

With a grid is loaded, this option allows the user to see the index number of the grid cells as shown in Figure 5-4.

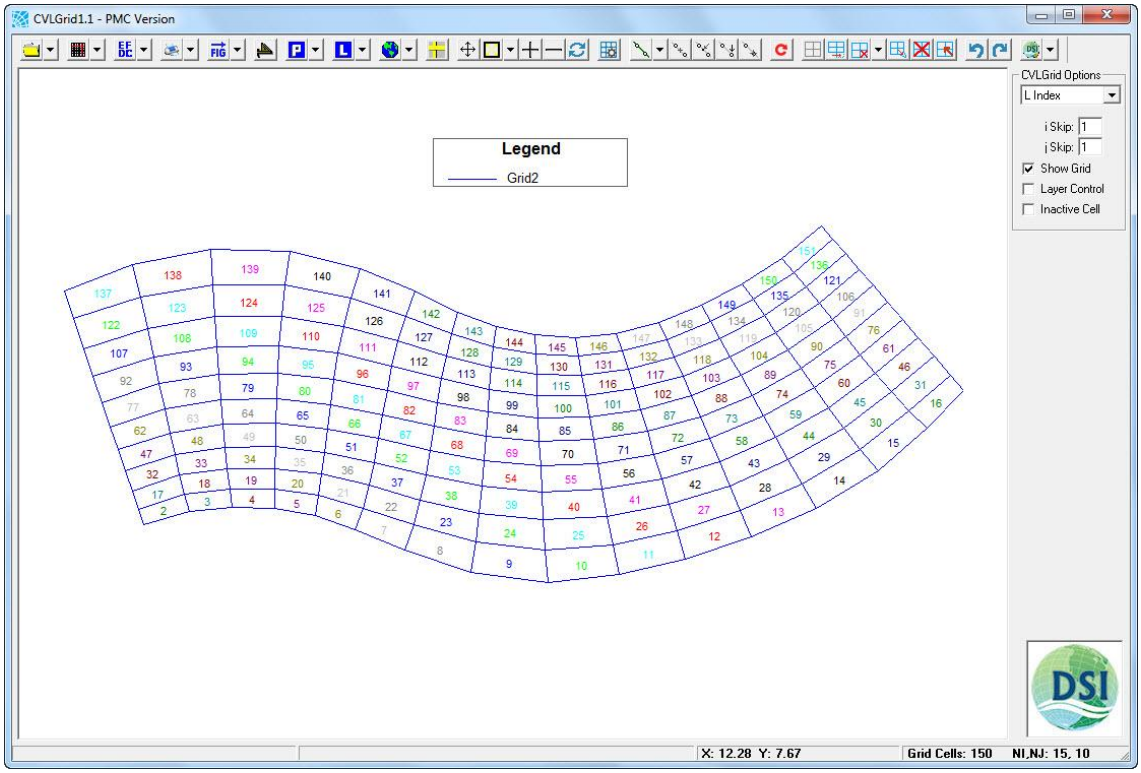


Figure 5-4 "L Index" viewing option.

5.1.3 I & J

This option displays the I and J index for each cell node of a grid as shown in Figure 5-5.

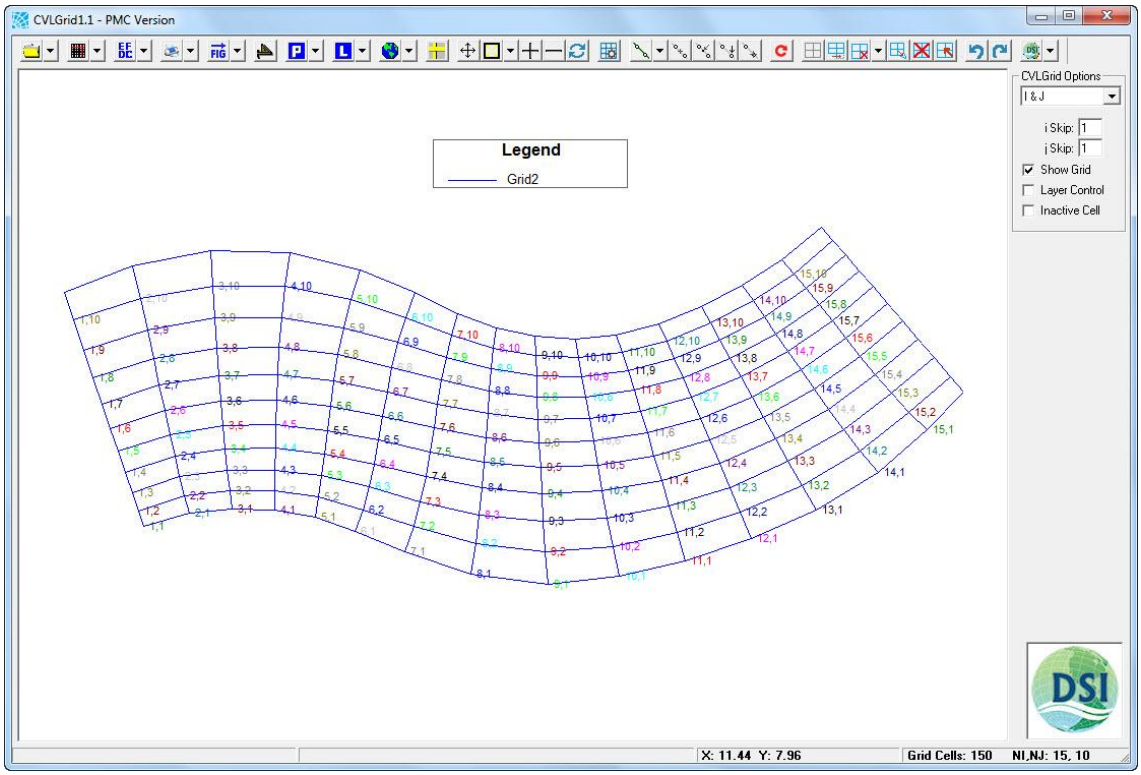


Figure 5-5 "I & J" index viewing option

5.1.4 Corners

The option displays the index order of corners in each node. This is shown in Figure 5-6.

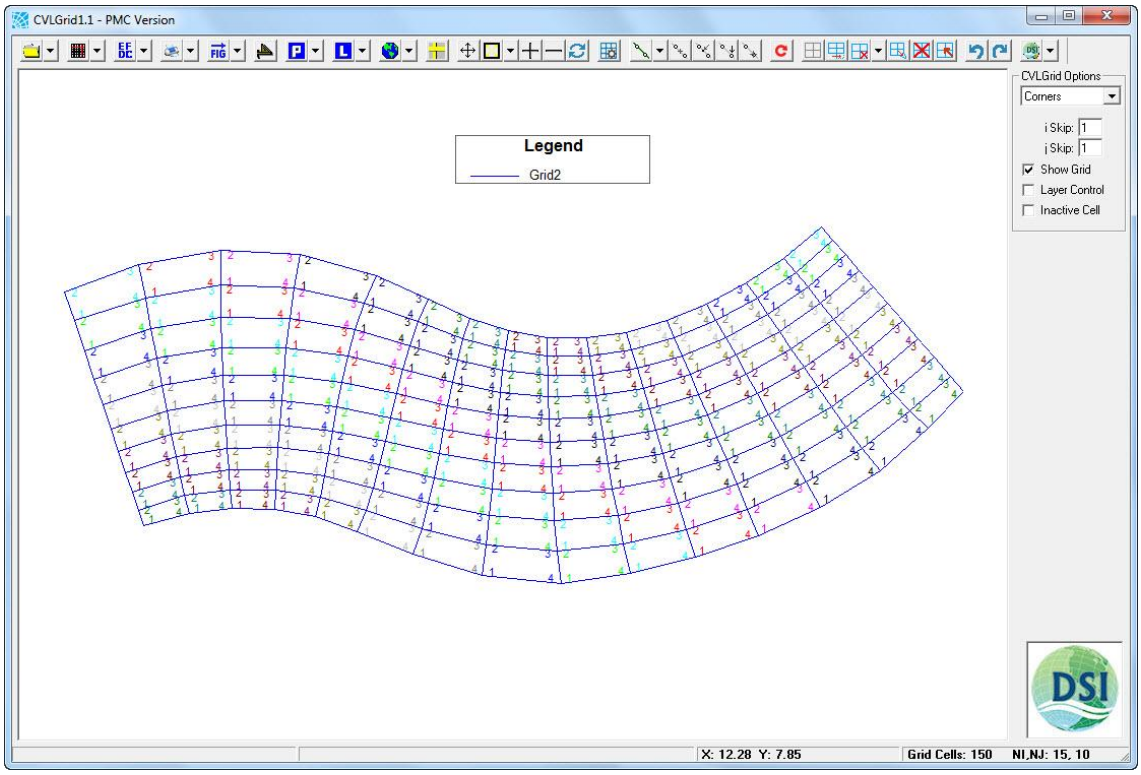


Figure 5-6 "Corners" index viewing option.

5.1.5 Orthogonal Deviation

This option displays the orthogonal deviation of the grid. Each cell is colored based on its orthogonality and a legend is displayed showing the scale as shown in Figure 5-7.

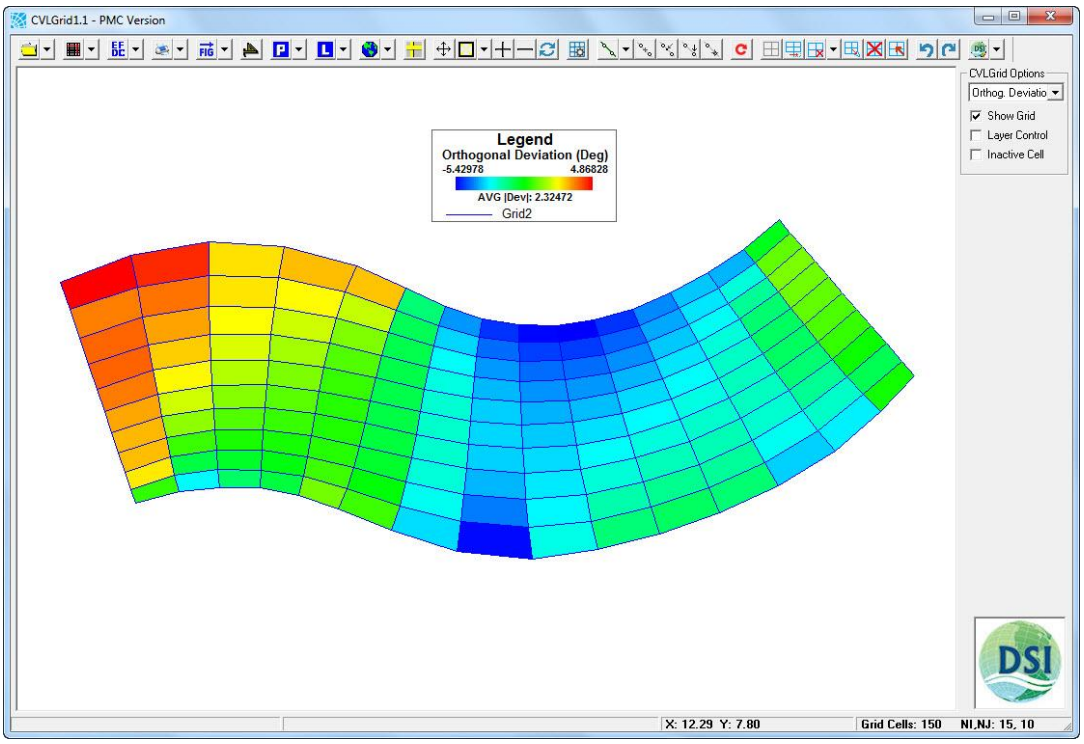


Figure 5-7 "Orthogonal Deviation" viewing option.

5.1.6 Dx

This option displays a map of the cell size in the X axis as shown in Figure 5-8.

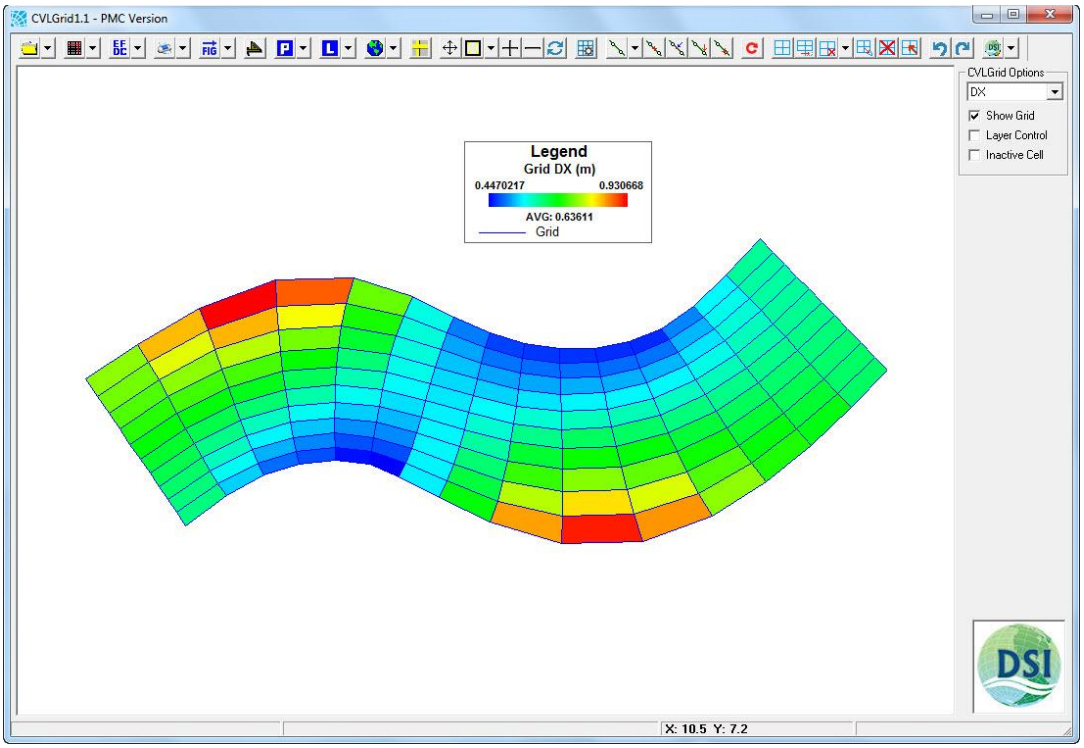


Figure 5-8 "Dx" viewing option.

5.1.7 Dy

This option shows a map of the cell size in the Y axis. It is shown in Figure 5-9.

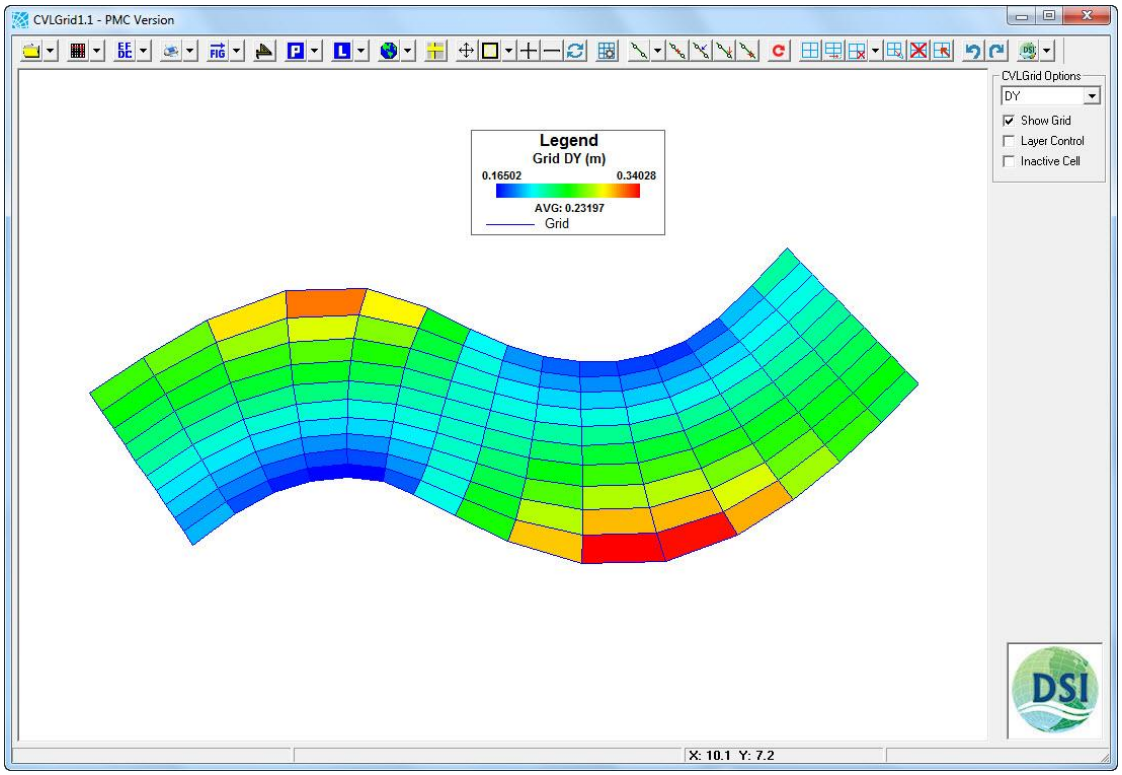


Figure 5-9 "Dy" viewing option.

5.1.8 Cell Map

The Cell Map Viewing Opt displays the cell as stored in the computer memory. Figure 5-10 shows an example cell map using the Uniform Grid option. The I and J axes can be labeled, as they are in this example.

Deactivated cells can be viewed using this view by checking “Inactive Cell” box. The inactivated cells are displayed with gray color.

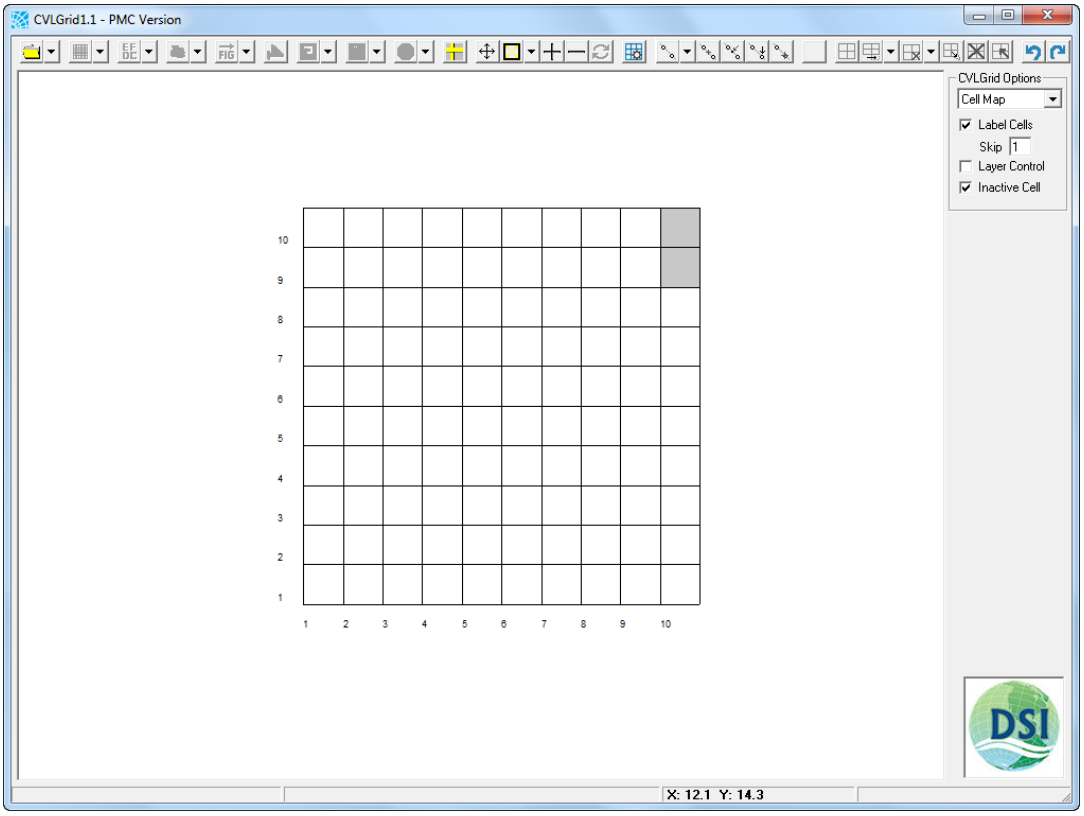


Figure 5-10 "Cell Map" viewing option.

5.2 Layer Control

The *Layer Control* toolbox, shown in Figure 5-11, provides the user with a number of options for selecting and manipulating the layers such as grids, splines and overlays as described below.

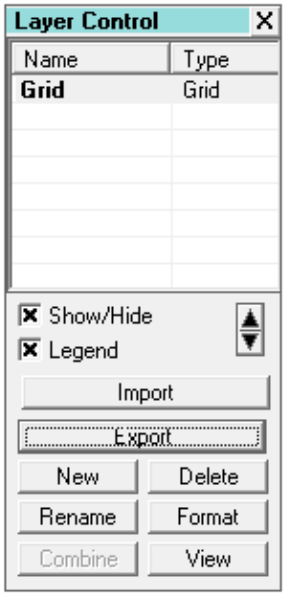


Figure 5-11 Layers toolbox.

Options available to the user from the *Layers Tool* are as follows:

Show/Hide	Makes the layer visible or hidden. When the the box is checked the layer is visible
Legend	For the layer selected this option will show or hide that layer label on the legend. If box is checked then the legend is visible.
Import	Imports a polyline files or GRD file. Also loads CVL grid file.
Export	Opens a browser window to choose a location to save the selected layer. Options include the following extensions CVL, GRD, P2D, KML, LDB, SHP, ECOMSED, SEAGRID, CH3D. This option allows the user to export a grid to a third party applications such as DELFT3D or other modeling tools.
New	Creates a new layer
Delete	Deletes the selected layer
Rename	Renames the selected layer
Format	Changes the color, symbol and line type of the lines on layer
Combine	Combines selected layers into one, this function only applied for splines and overlay layers.
View	View the selected layer

Note that users may also drag and drop a file onto the workspace to open it.

The user can also edit a selected layer by using the list of options available when RMC on the items list, the enabled items list varies by layer type as shown in Figure 5-12, Figure 5-13 and Figure 5-14 for grid, spline and overlay layers respectively.

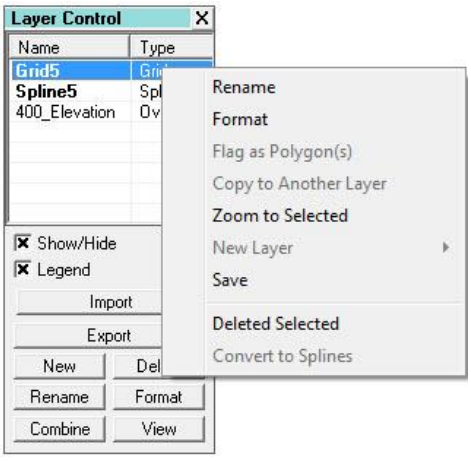


Figure 5-12 Layer items menu for grid when RMC.

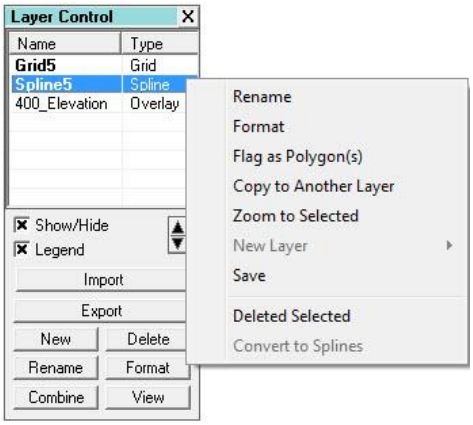


Figure 5-13 Layer items menu for spline when RMC.

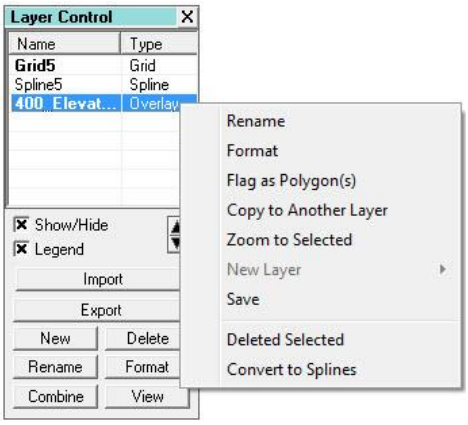


Figure 5-14 Layer items menu for overlay when RMC.

It should be noted that the *Convert to splines* option is available only for overlay layers. The purpose is for the overlay layer to be converted to a spline layer.

In addition, to create a new layer, the user can RMC on blank space of *Layer Control*, shown in Figure 5-15.

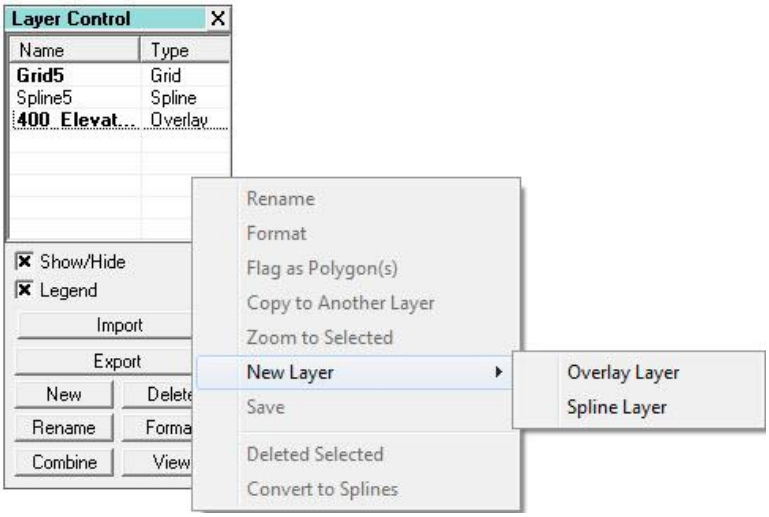


Figure 5-15 Create new layer when RMC.

6 Grid and Cell Operations

When the grid is loaded on the workspace then RMC will display a list of grid and cell operations.

Cell operations such as *Deactivate/Activate Cell* will only be displayed when RMCing on the grid. On the other hand, the list of *Grid Operation* will be displayed either when RMC on the grid or the outside of the grid as shown in Figure 6-1, Figure 6-2 and Figure 6-3. These operations are explained in detail in the following section.

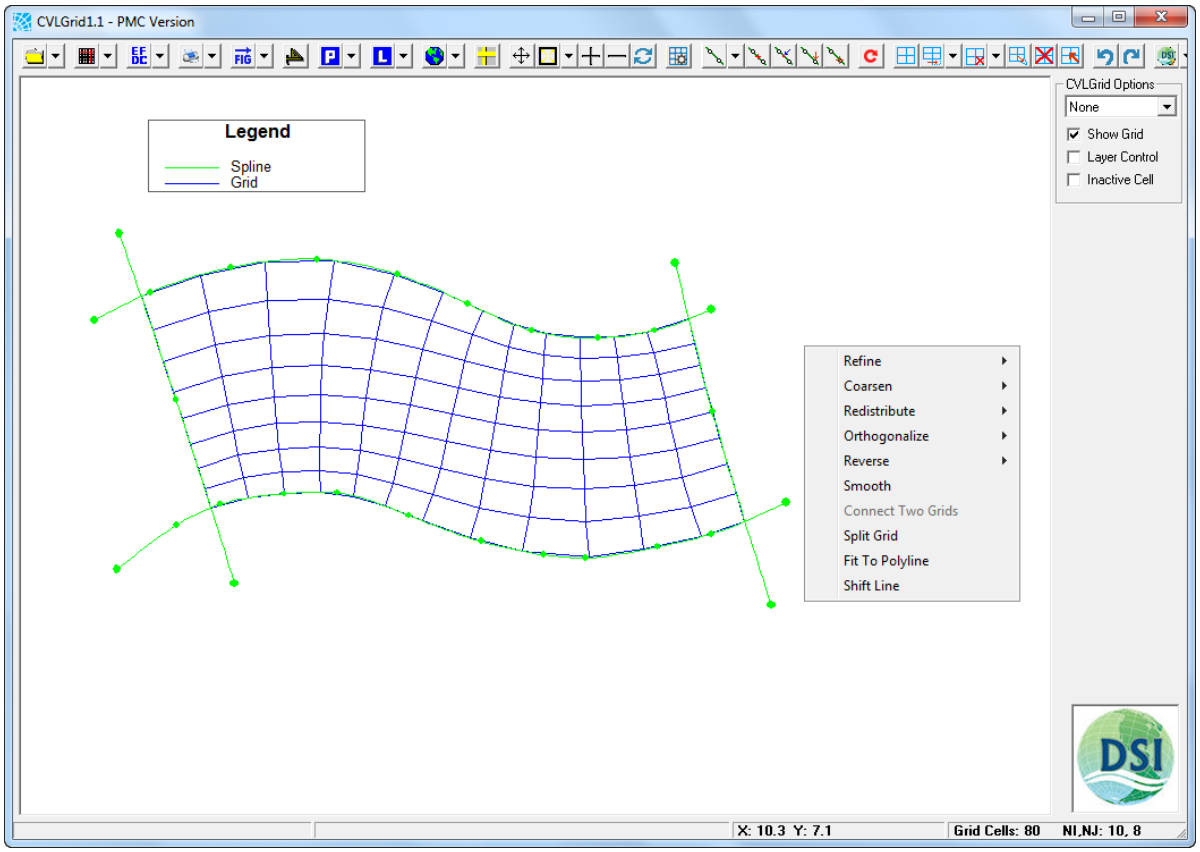


Figure 6-1 Grid operations.

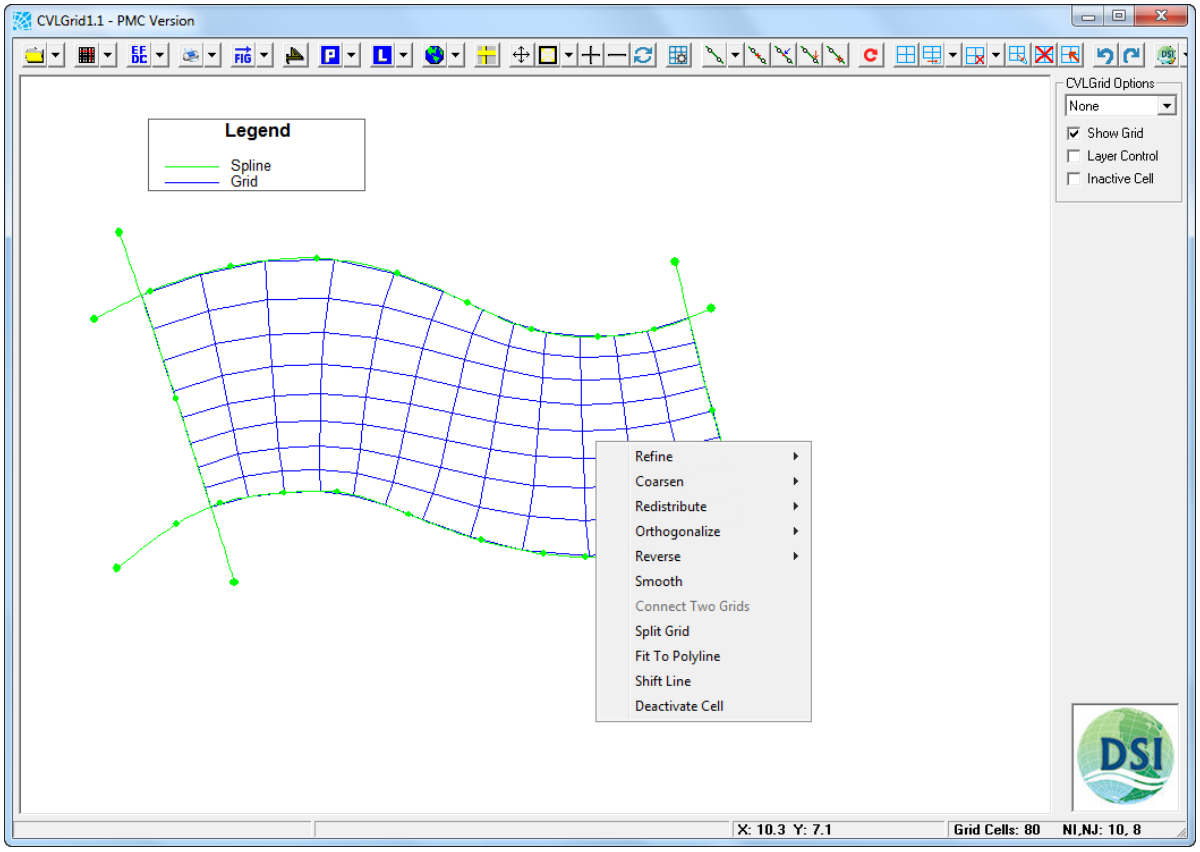


Figure 6-2 Grid and cell operations.

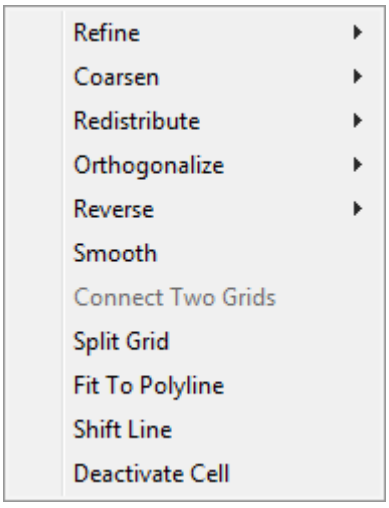


Figure 6-3 Grid and cell operations (detailed).

6.1 Refine Grid

The *Refine* tool increases the resolution of the grid. It operates by multiplying the existing number of cells in the current grid with (IC, JC) independently. There are two refine features: *Refine Global* and *Refine Local* as shown in Figure 6-4. *Refine Global* feature will apply the refinement for the whole grid, whereas *Refine Local* feature will only apply the refinement for the section of the grid selected.

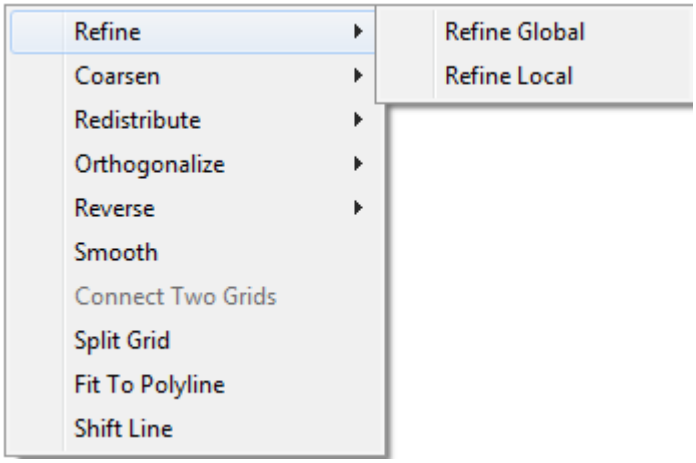


Figure 6-4 Two options for "Refine" grid.

For *Refine Global* feature, the user must define set the multiplier, n I and n J, to enter into the text box. Figure 6-5 and Figure 6-6 show the grid before and after applying *Refine Global* with (2, 2) for the IC and JC indexes. Note that the multiplier must be a whole number.

For the *Refine Local* feature, the user must first define two nodes in the same I or J direction. This will define a "local" zone for which the refinement will be applied. Figure 6-7 and Figure 6-8 show the grid before and after applying *Refine Local* with 2 in the I direction.

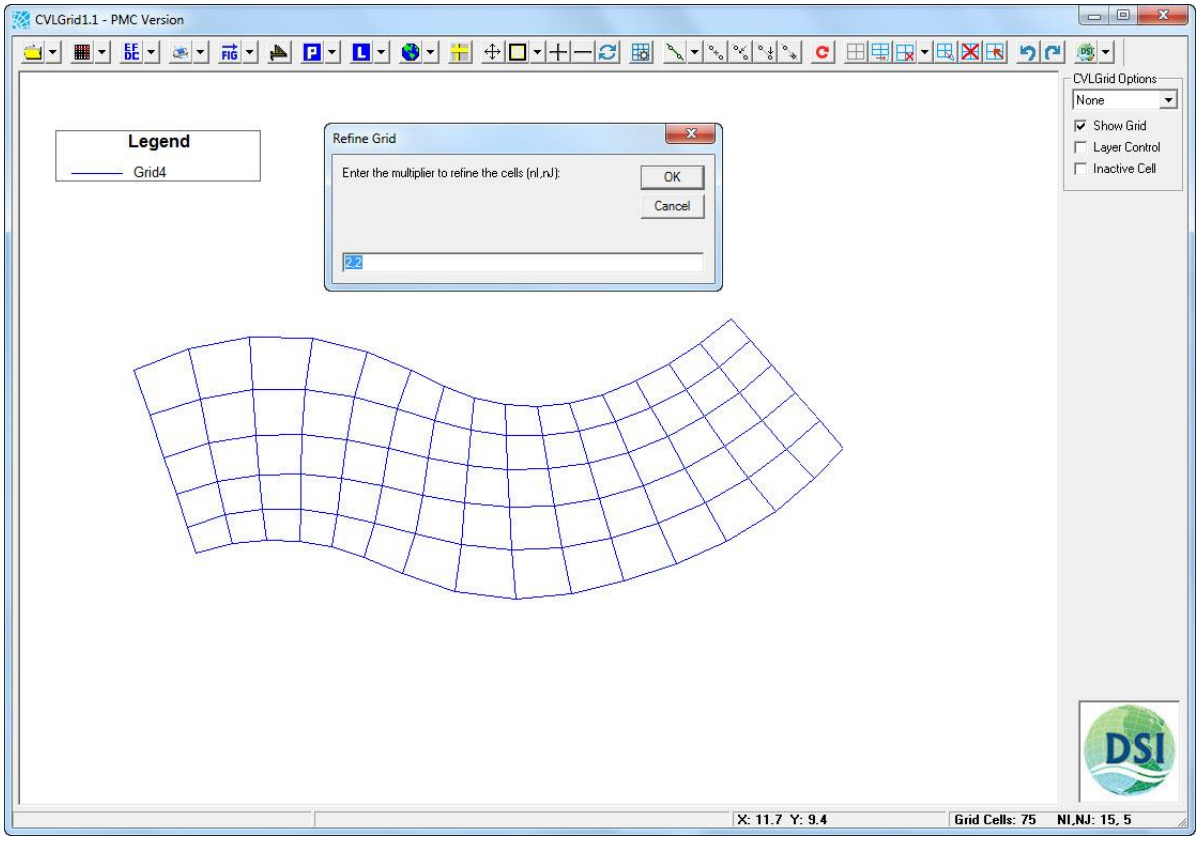


Figure 6-5 Grid Image before applying "Refine Global"

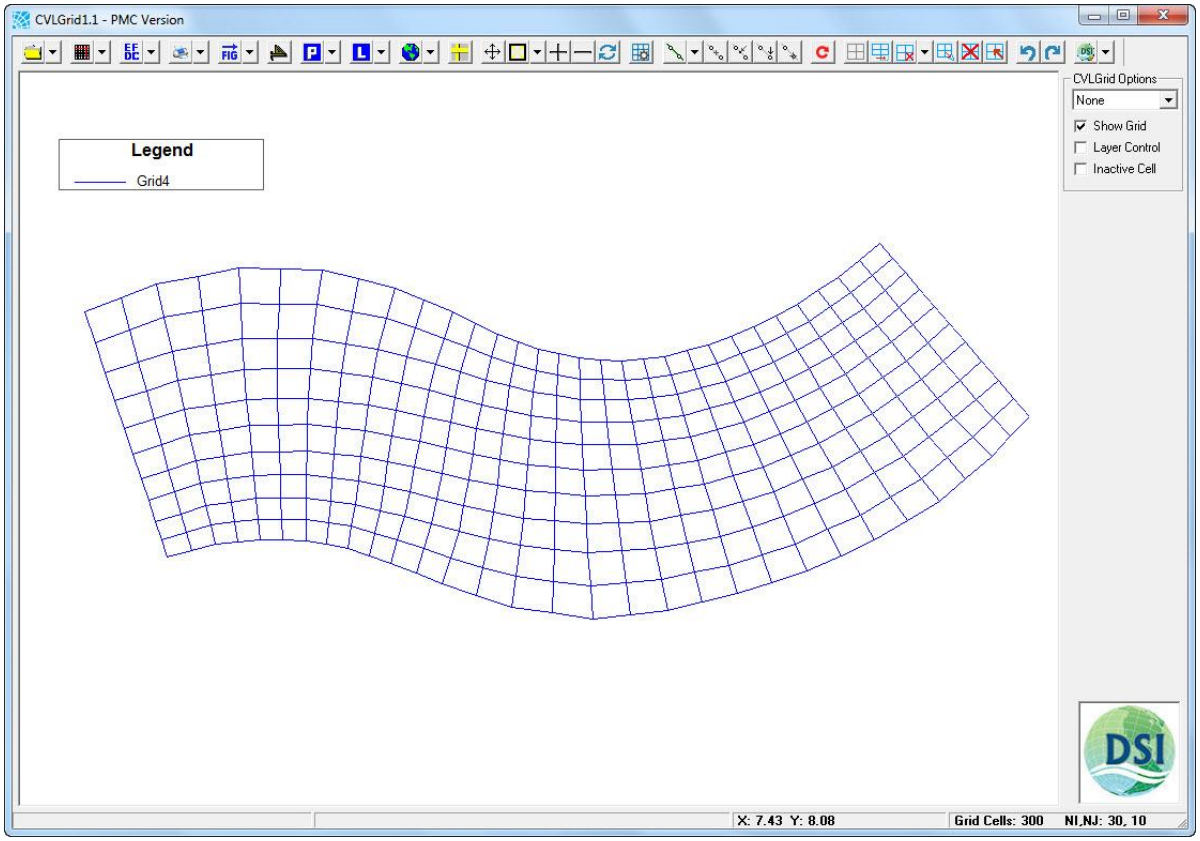


Figure 6-6 Grid image after applying "Refine Global"

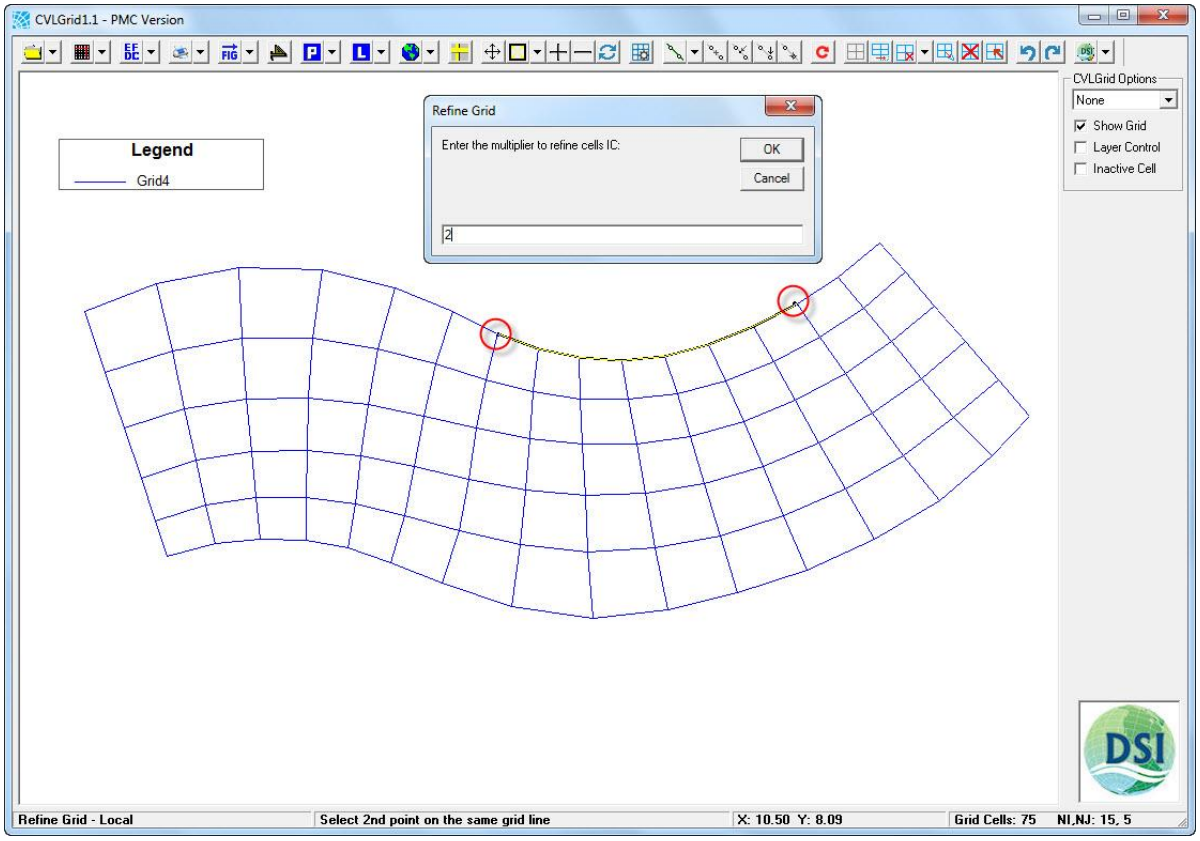


Figure 6-7 Grid image before applying "Refine Local"

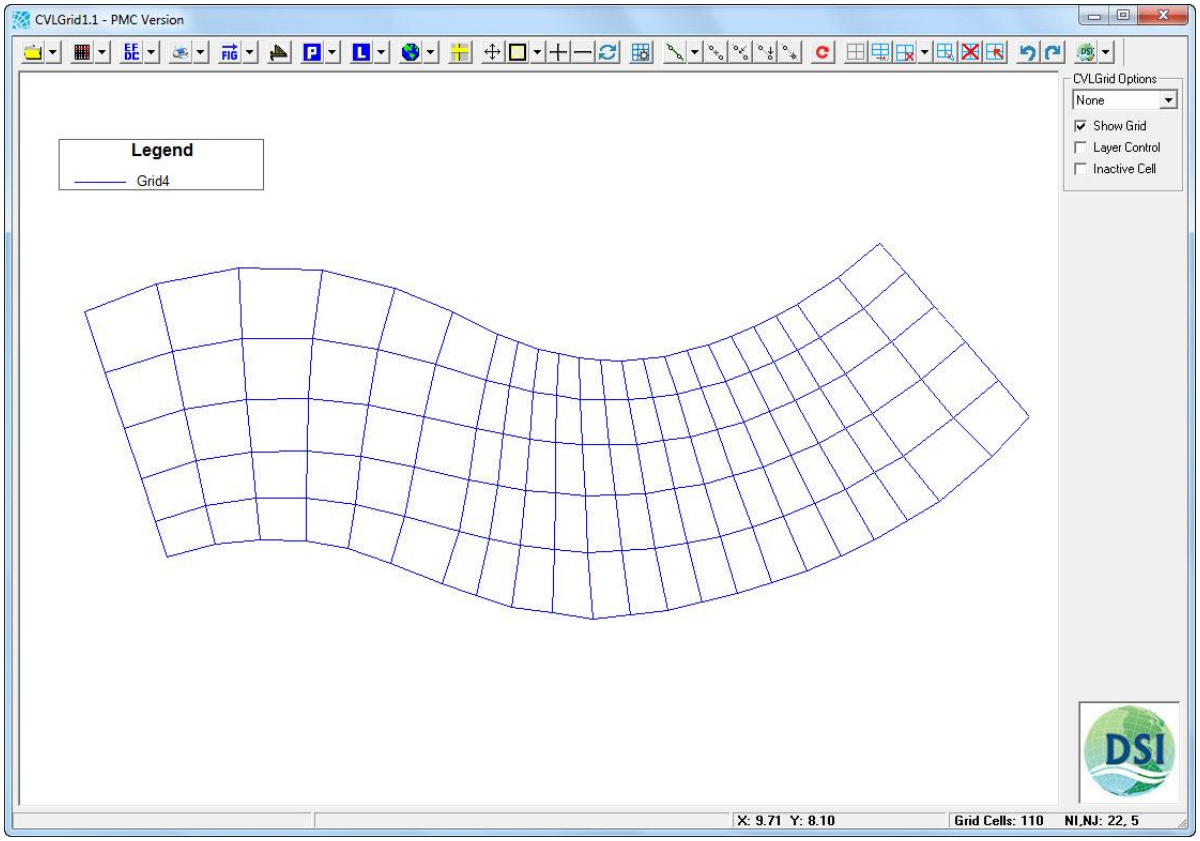


Figure 6-8 Grid image after applying "Refine Local"

6.2 Coarsen Grid

The *Coarsen Grid* tool operates in an opposite way to that for *Refine Grid*. It decreases resolution of the grid.

When a grid is first generated, the grid cells are spaced uniformly across the entire domain. This is not a problem but there may be areas in the grid that are of little importance to the model. In these areas the grid can be coarsened to reduce the grid resolution. By reducing the grid resolution of unimportant areas the calculation time is reduced.

The coarsen tool is dependent on the grid cell the user specified and the settings in CVLGrid Settings. The user must first define the divisor by which cells will be coarsened in the I and/or J direction when prompted.

The tool has *Coarsen Global* and *Coarsen Local* features as shown in Figure 6-9. If a global operation is selected the entire grid will be coarsened by the specified amount. An example is shown in Figure 6-10 and Figure 6-11.

If a local operation is selected, the user must define two nodes on a grid line ofin either I or J axis to be coarsened. An example is shown in Figure 6-12 and Figure 6-13.

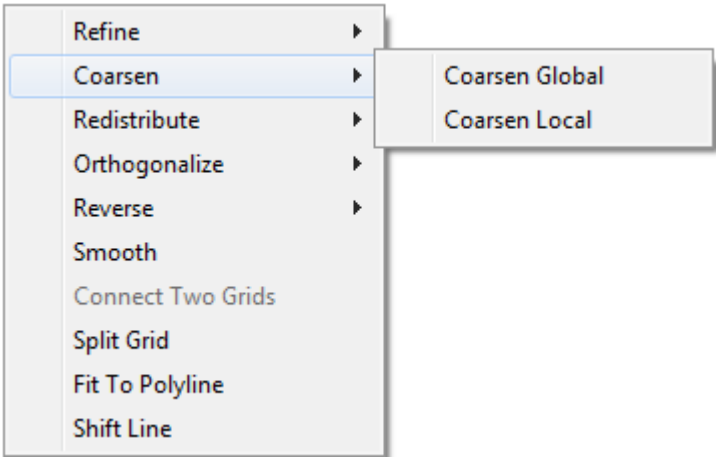


Figure 6-9 Two features of coarsen grid

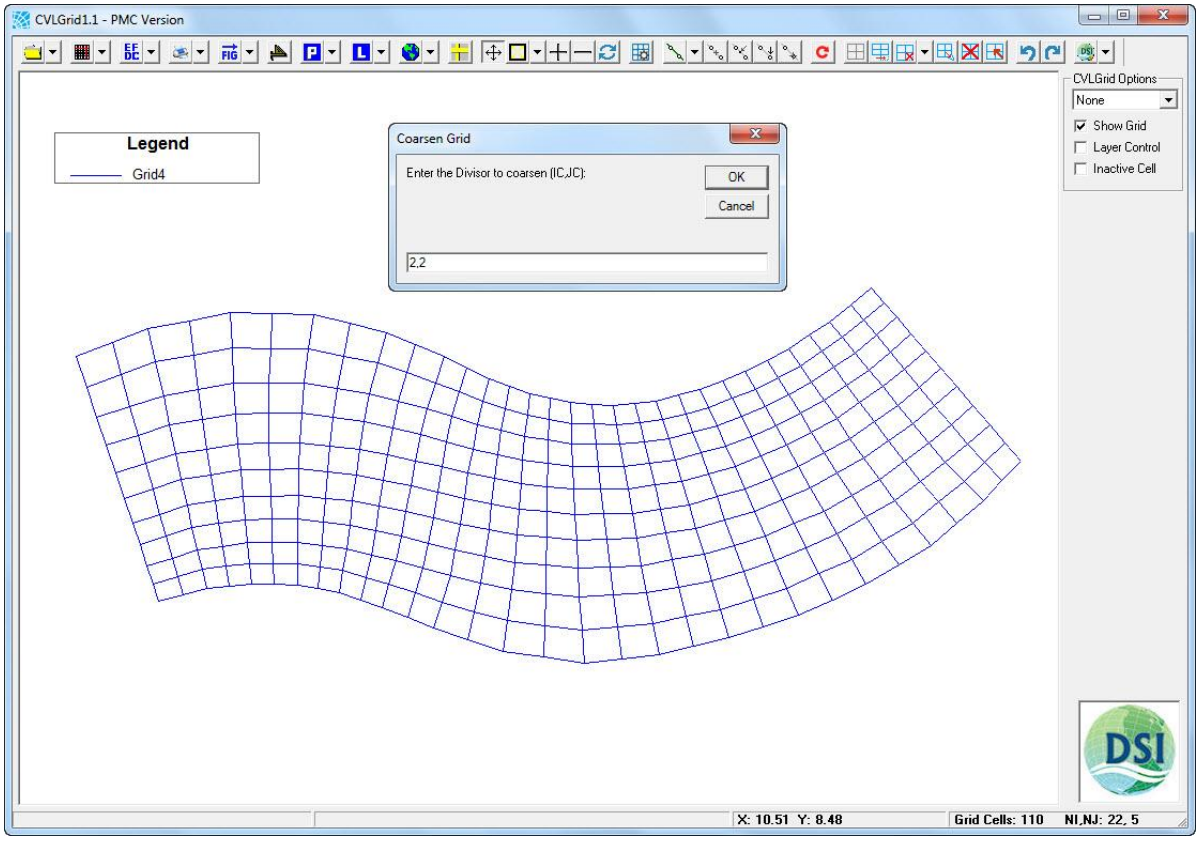


Figure 6-10 Grid image before applying "Coarsen Global."

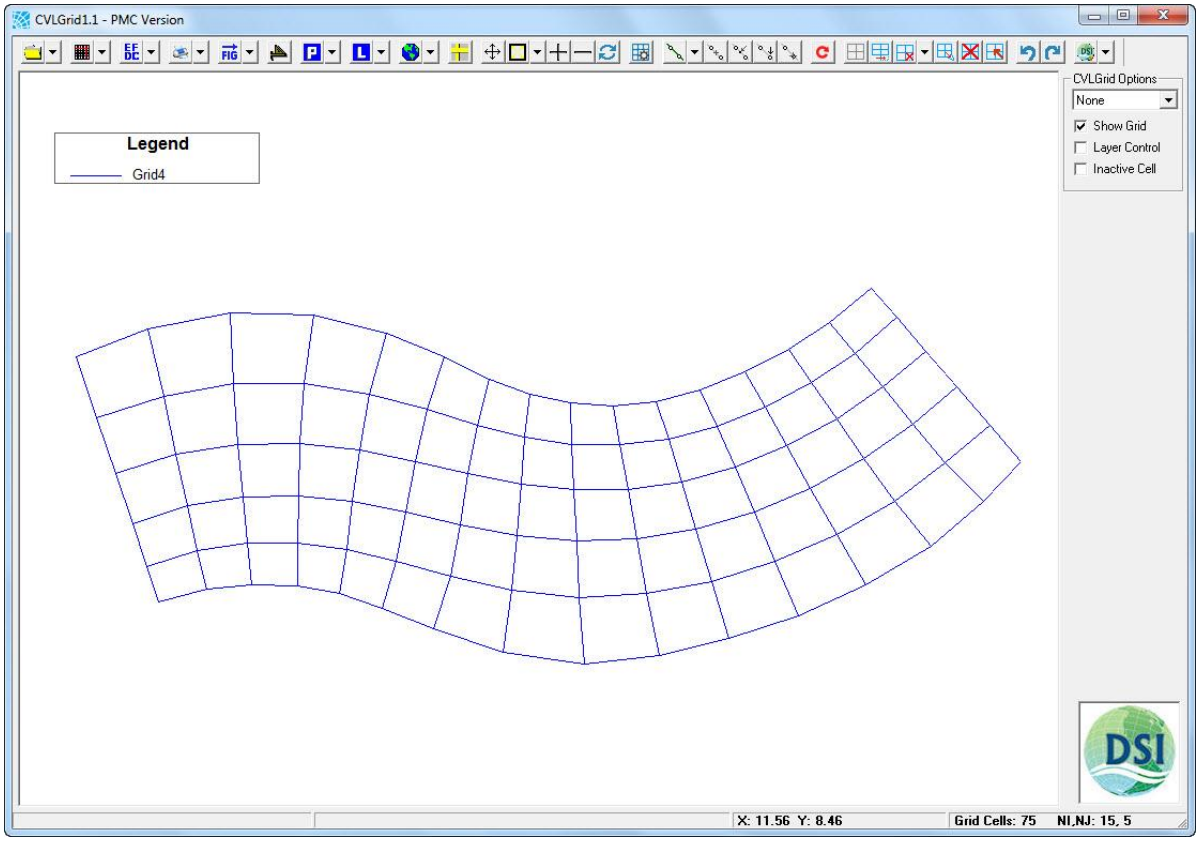


Figure 6-11 Grid image after applying "Coarsen Global."

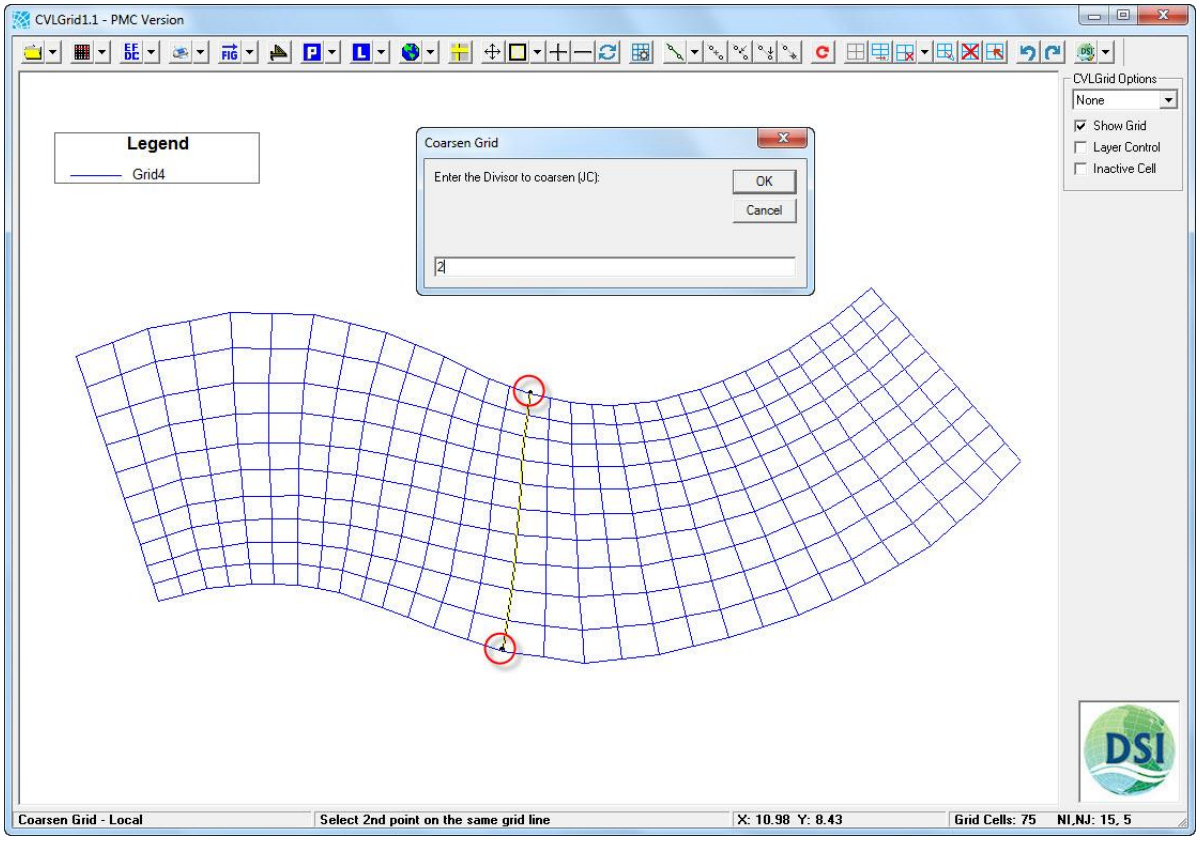


Figure 6-12 Grid image before applying "Coarsen Local."

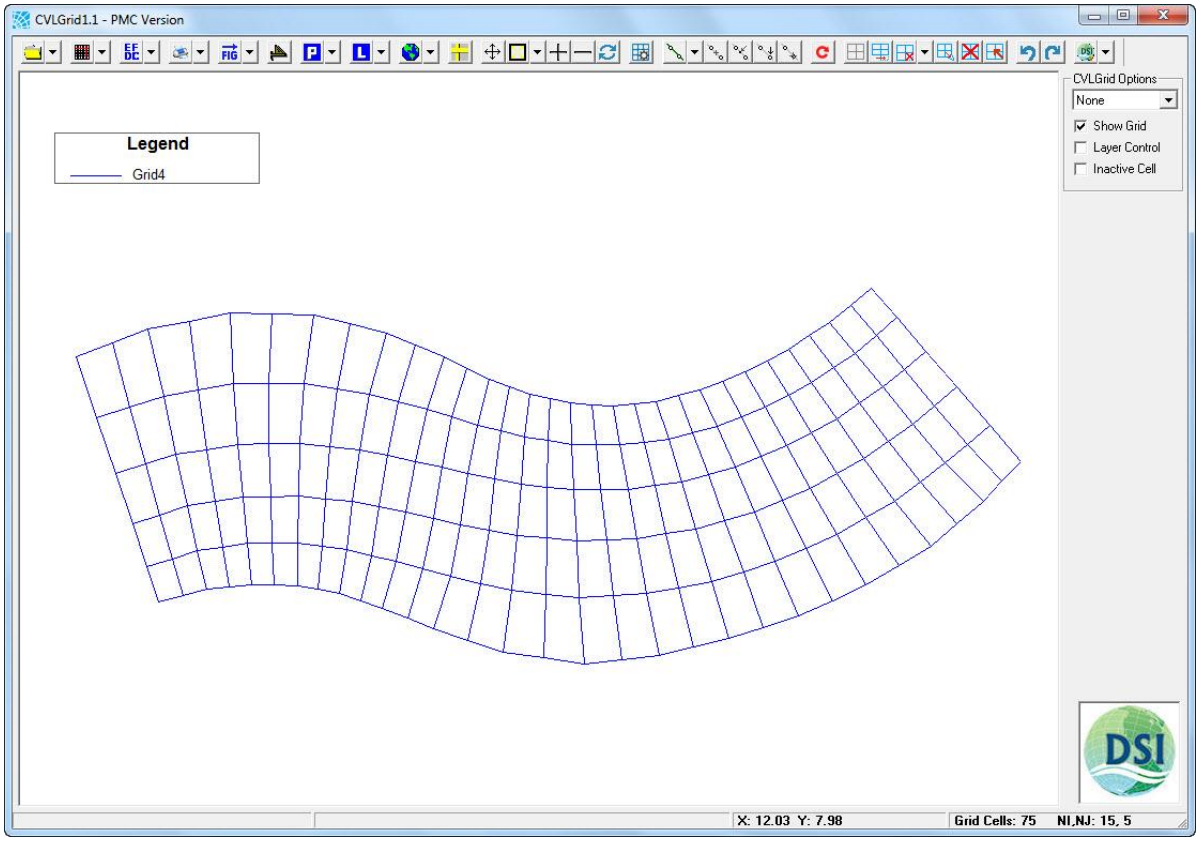


Figure 6-13 Grid image after applying "Coarsen Local."

6.3 Redistribute Grid

The *Redistribute Grid* tool has three options: *Regularize Global* creates even spacing between cell's nodes for entire grid domain. *Regularize Local* creates even spacing between cell's nodes in either the I or the J direction of grid domain by selecting two nodes on a grid line. *Telescoping* creates increasing or decreasing spacing to a side of the grid block specified by three cell nodes and depends on a distance factor value in the *Telescoping* settings.

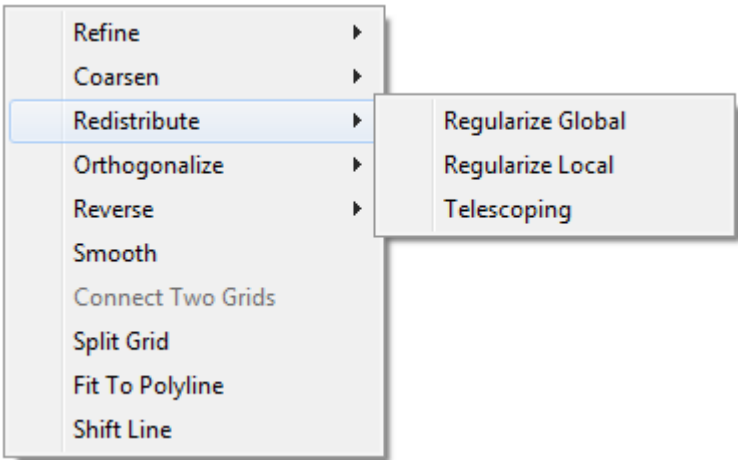


Figure 6-14 Sub-options for redistribute a grid.

Figure 6-15 to Figure 6-17 demonstrates how *Regularize Global* and *Regularize Local* affect the grid.

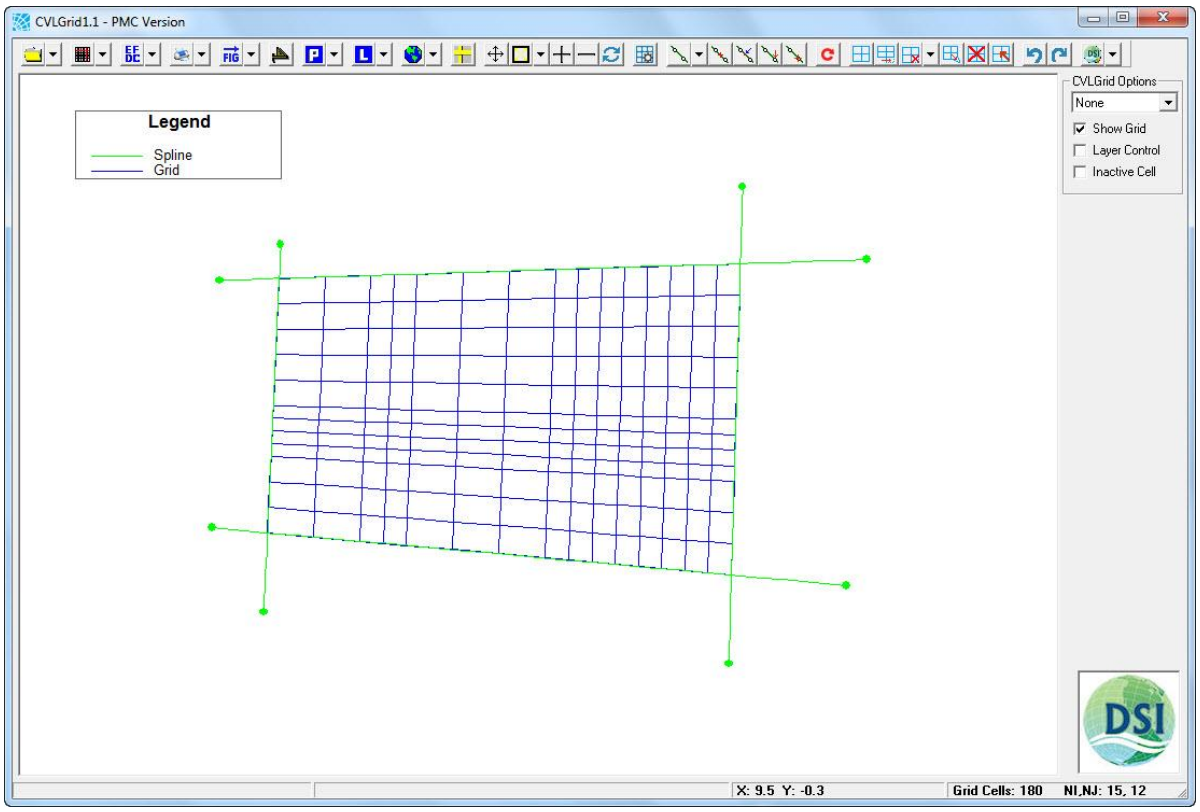


Figure 6-15 Original grid.

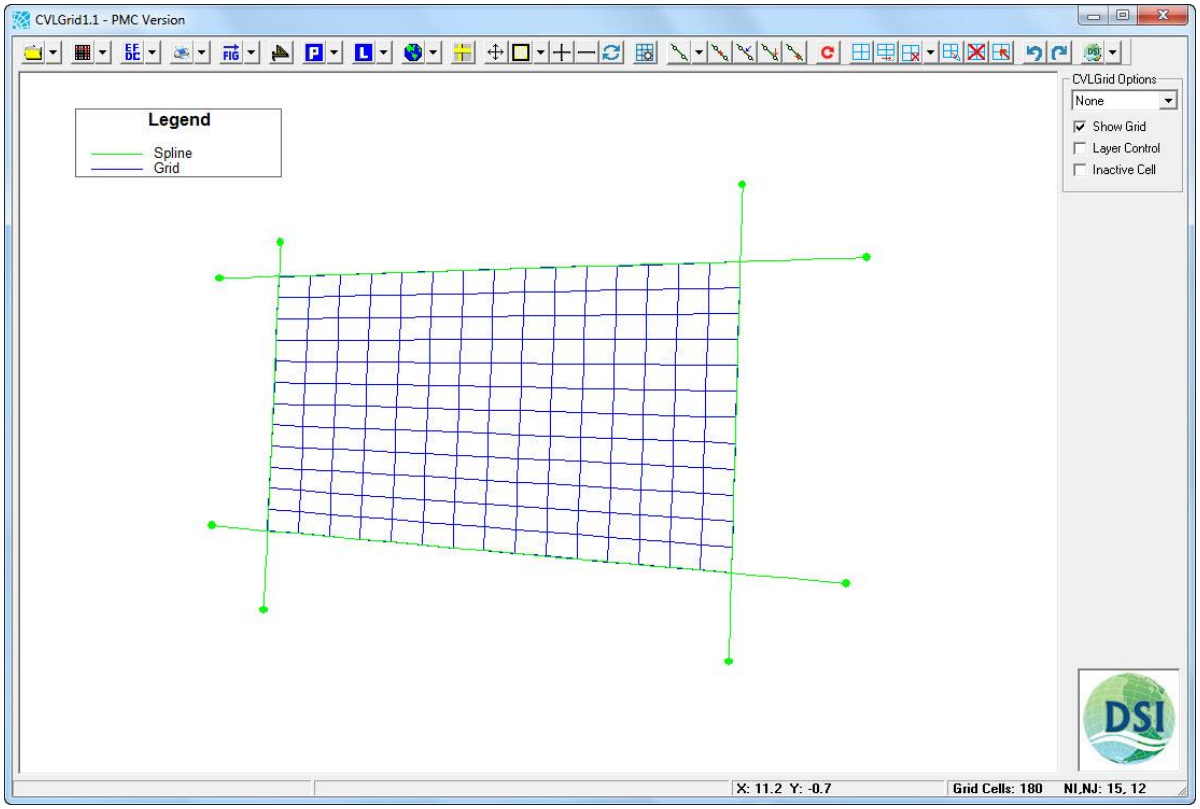


Figure 6-16 Grid after applying *Regularize Global*.

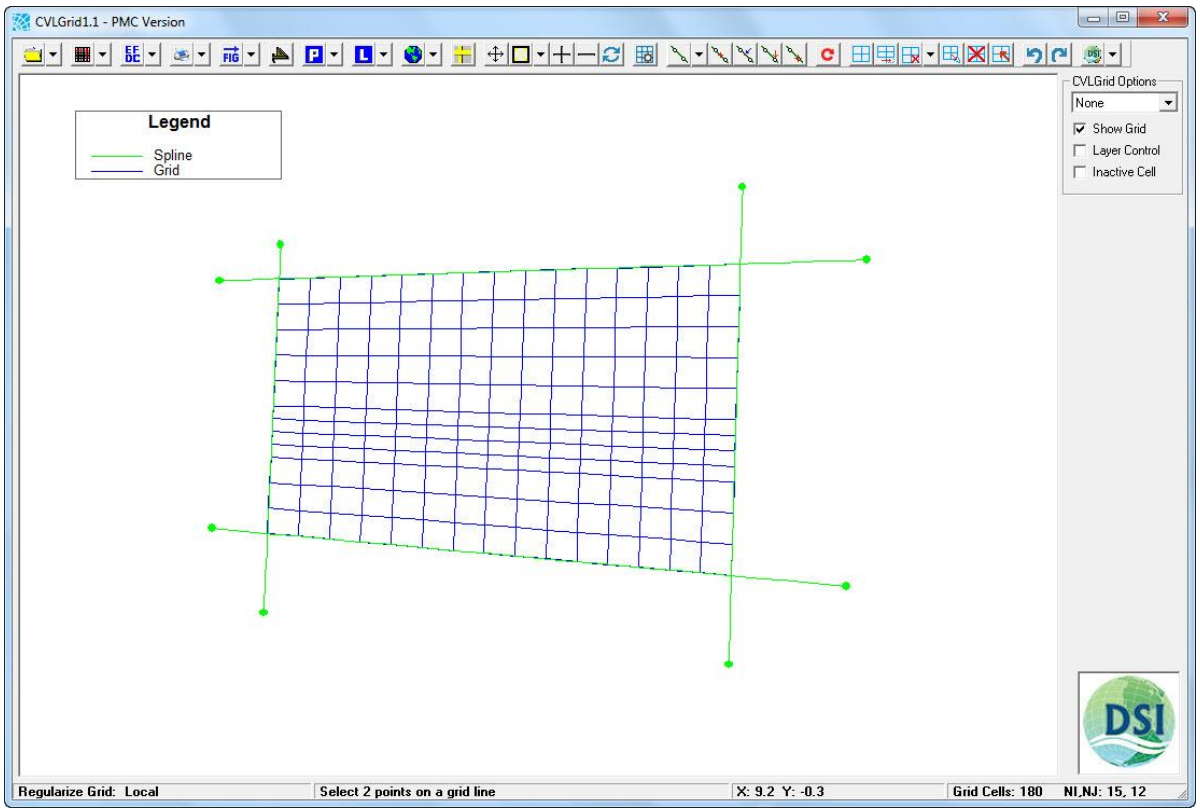


Figure 6-17 Grid after applying *Regularize Local* (in I direction).

When the user selects *Telescoping,* the mouse cursor transforms from an arrow to a cross hair. Three nodes to be selected in the Telescoping option. The user needs to first select two nodes on same grid line which are perpendicular to the direction in which the grid will be telescoped or stretched. The user should then select a third node on grid domain which the end point to which the telescoping will be applied. These three nodes define a grid block which the telescoping will be apply too.

After selecting the third node, a window for the T*elescoping Setting* appears to allow the user to enter the value of a distance factor. The default value of the distance factor is two and the distance factor must be a whole number greater than 1. After entering the telescoping factor click *OK*. The distance between the first two nodes in the grid block increases gradually towards the third node specified.

Figure 6-18 shows the grid before applying telescoping; Figure 6-19 is an example of selecting three grid nodes for telescoping. Figure 6-20 shows the telescoping setting and Figure 6-21 shows the grid after applying telescoping.

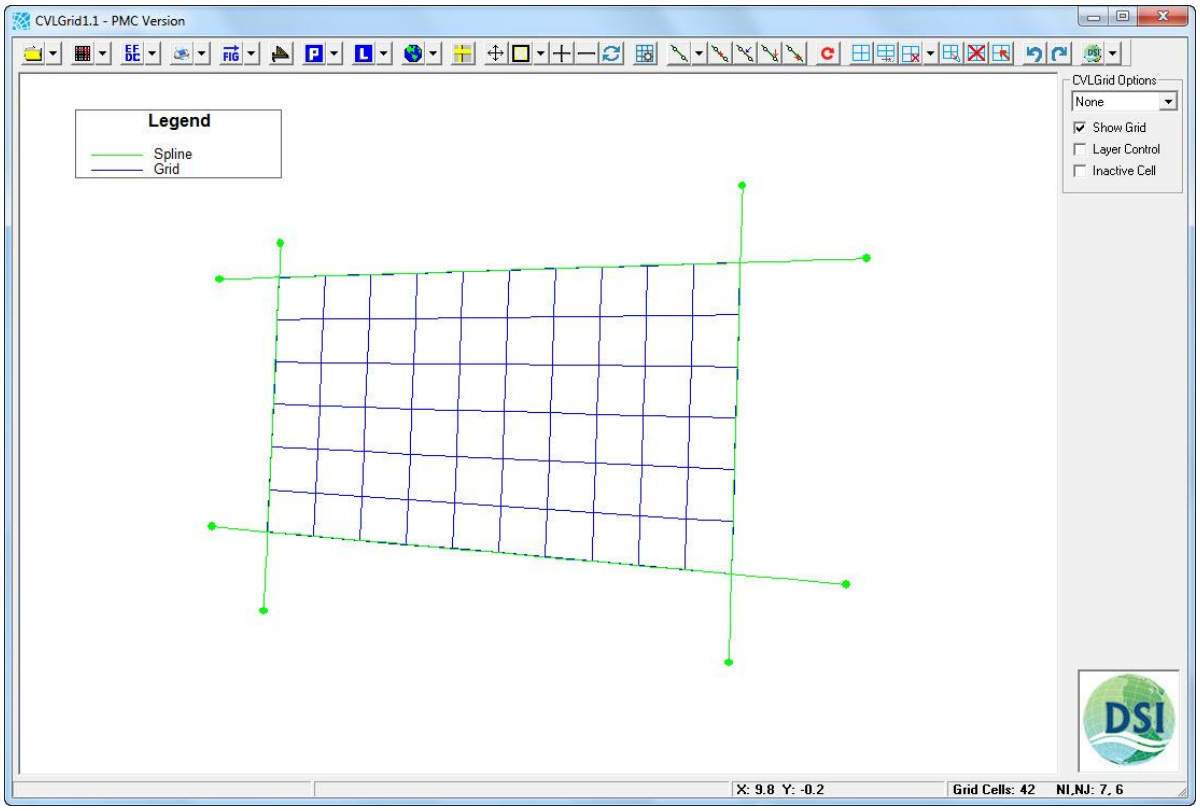


Figure 6-18 Grid before applying *Telescoping*.

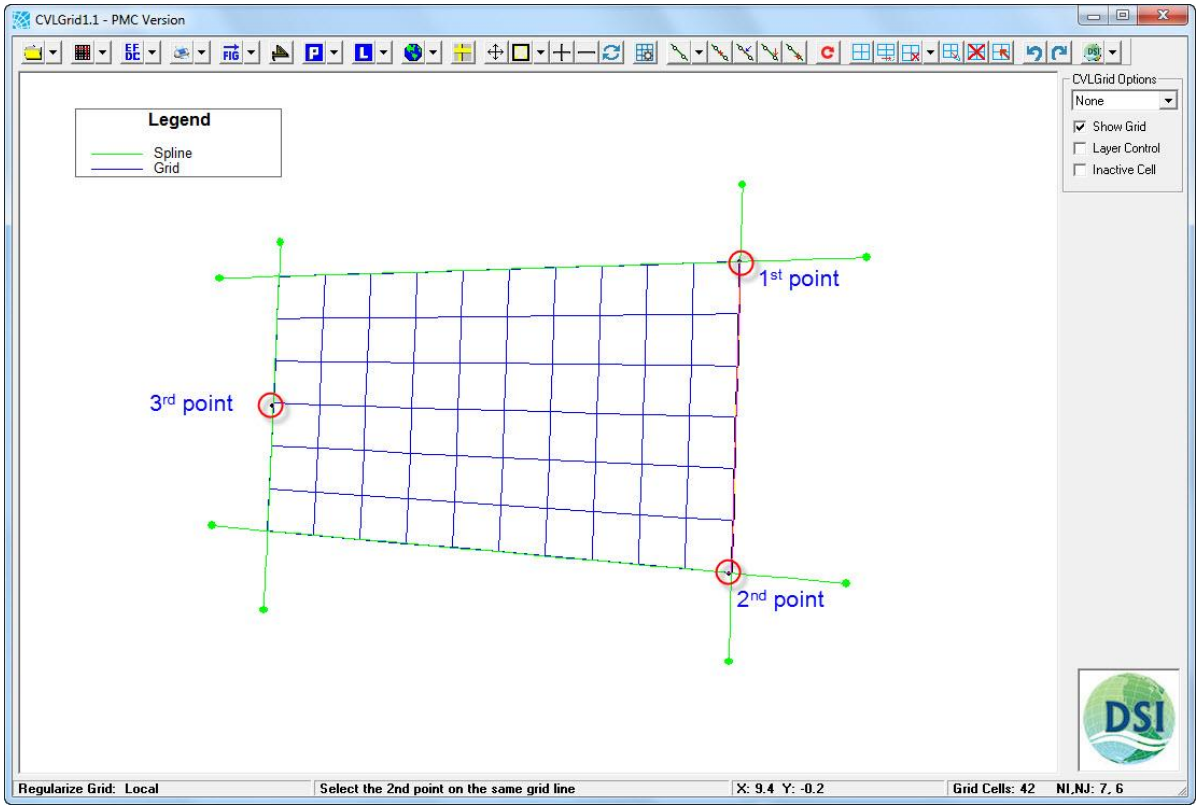


Figure 6-19 Points selection for *Telescoping.*

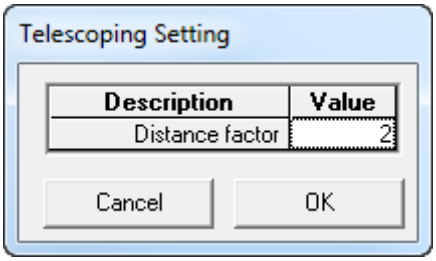


Figure 6-20 Telescoping settings.

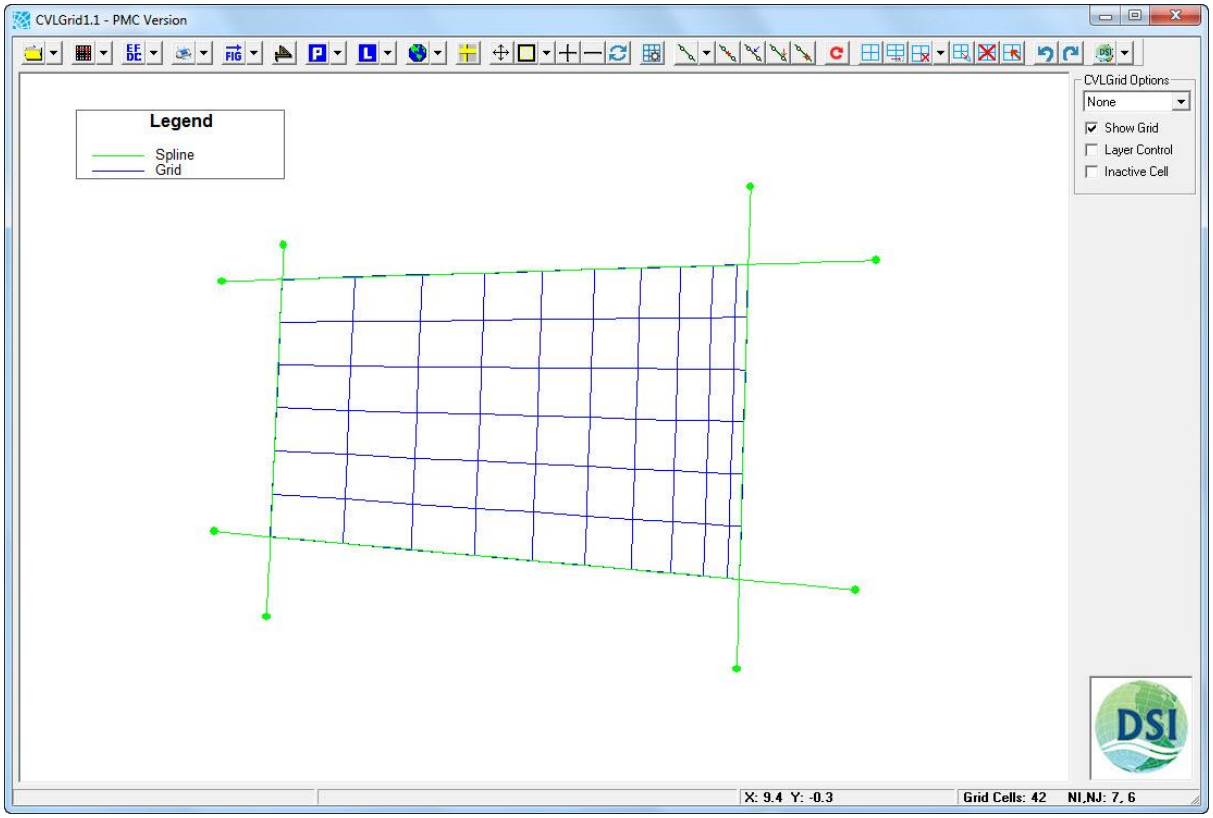


Figure 6-21 Grid after applying telescoping.

6.4 Orthogonalize Grid

The accuracy of a model is increased when the orthogonality of a grid is optimized. A perfect grid would have all cells perpendicular to the boundary lines. This is very difficult to achieve and thus for most EFDC models an orthogonality of under 3° is recommended. The orthogonalization tool is used to reduce the orthogonal deviation of a grid. The parameters of this tool are set in *CVLGrid Setting Options* described earlier in Section 4.12 .

The tool has *Orthogonalize Global, Orthogonalize Local* and *Orthogonalize 1D* features as shown in Figure 6-22.

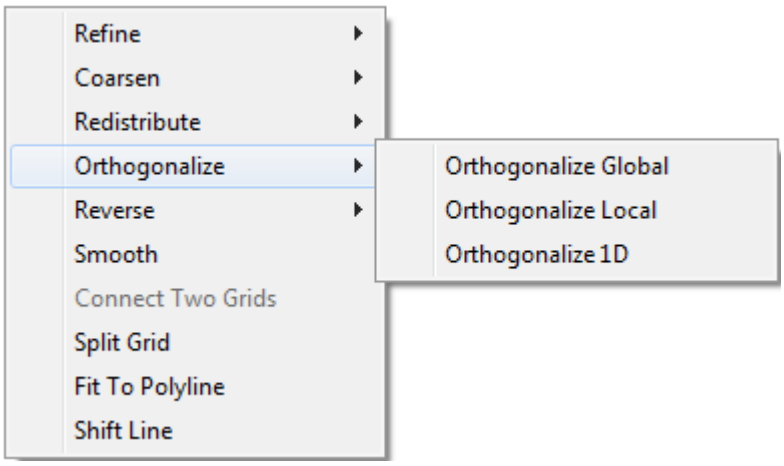


Figure 6-22 Options for orthogonalizing a grid.

If a global operation is selected the entire grid will be orthogonalized immediately. If a local operation is selected, the user must define two grid nodes and the area of the grid limited by the two points will be orthogonalized. *Orthogonalize 1D* feature is similar to the *Orthogonalize Local* option but is only applicable for a 1D grid region. In case the feature is applied for an improper grid region, a warning message will appear.

The change in grid orthogonality before and after applying *Orthogonalize Global* are shown in Figure 6-23 and Figure 6-24. It may be seen in the first image that the average orthogonality is 20.3. In the second image, after CVLGrid has automatically reduced the orthogonalization, it is only 2.6.

The change in grid orthogonality before and after applying *Orthogonalize Local* are shown in Figure 6-25 and Figure 6-26.

The change in grid orthogonality before and after applying *Orthogonalize 1D* are shown in Figure 6-27 and Figure 6-28.

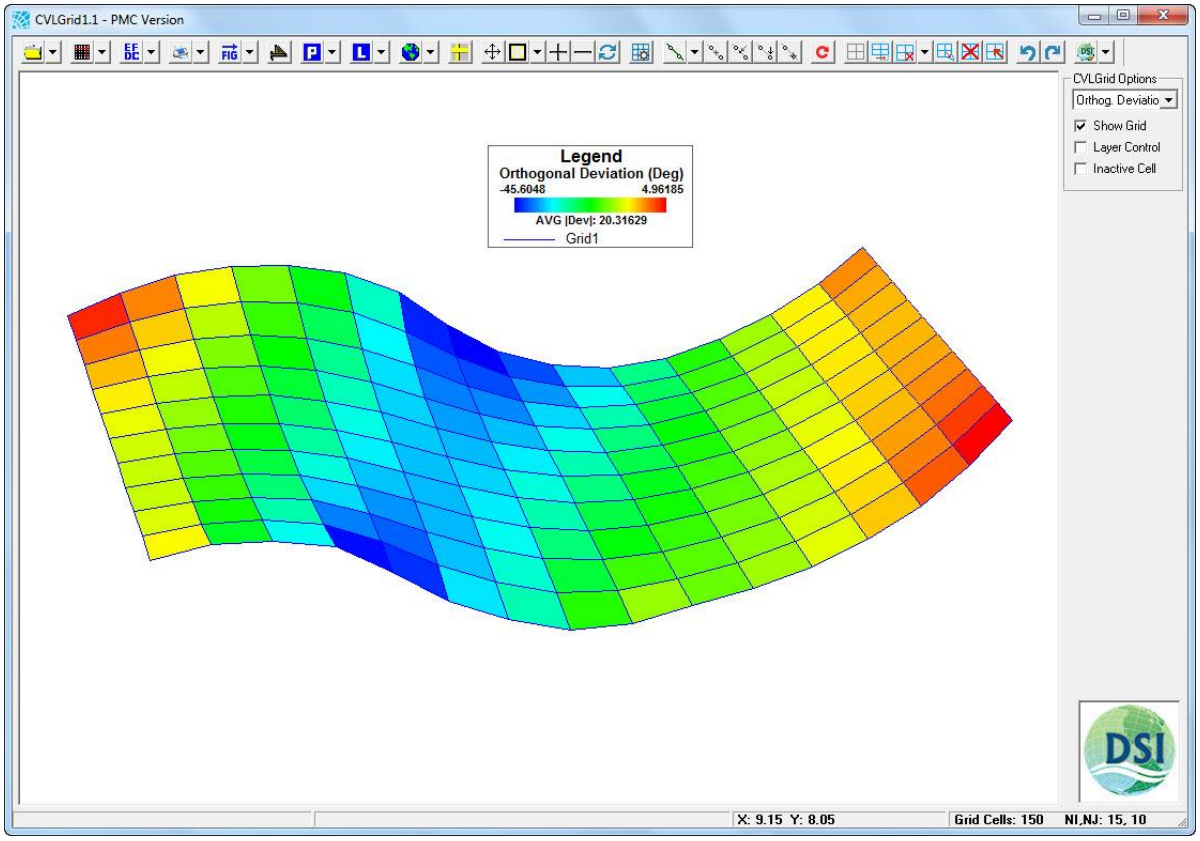


Figure 6-23 Grid image before applying "Orthogonalize Global"

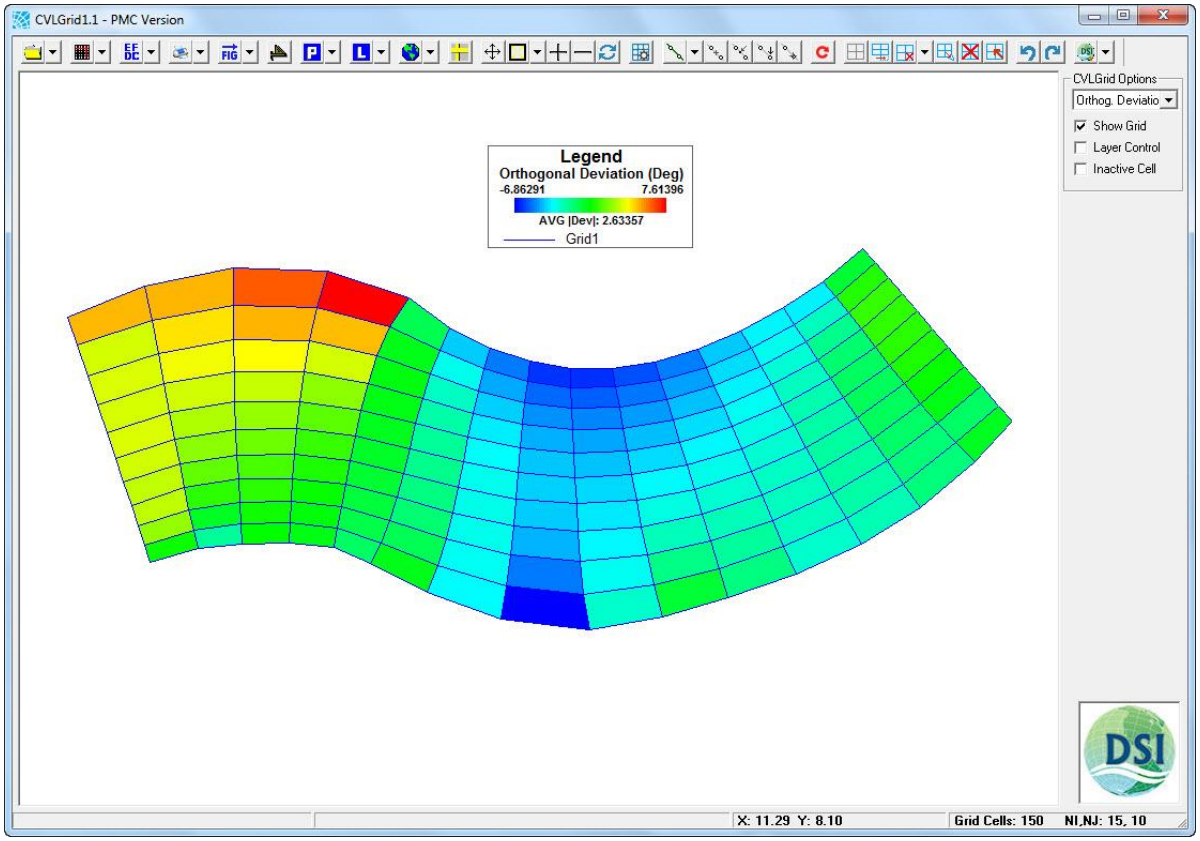


Figure 6-24 Grid image after applying "Orthogonalize Global"

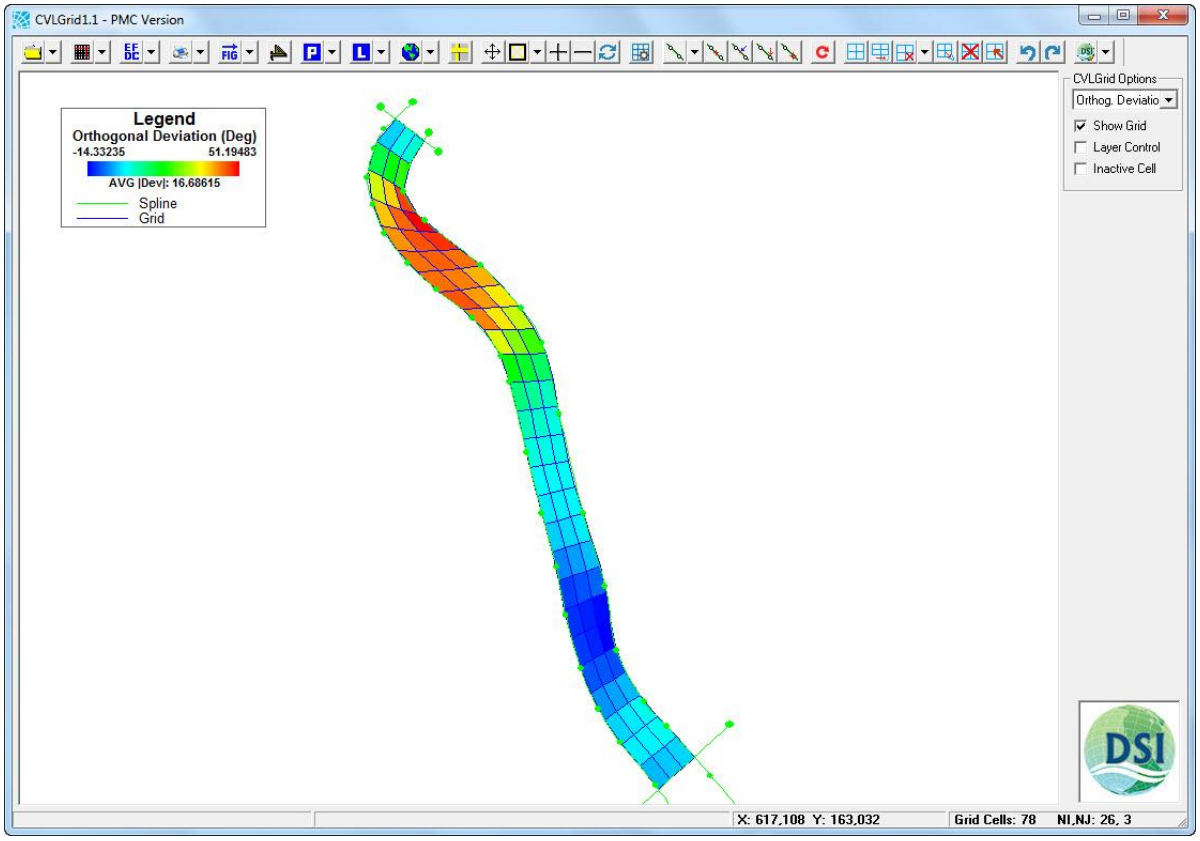


Figure 6-25 Grid image before applying "Orthogonalize Local"

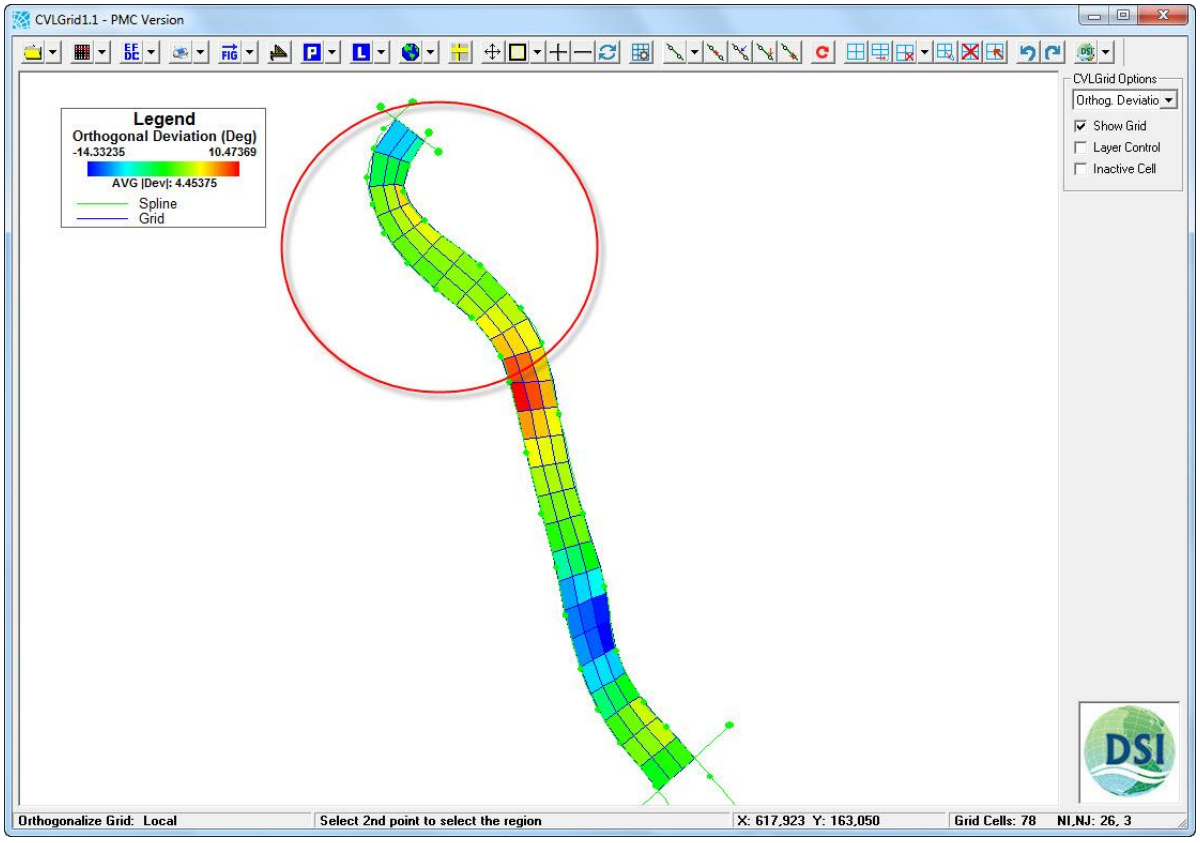


Figure 6-26 Grid image after applying "Orthogonalize Local"

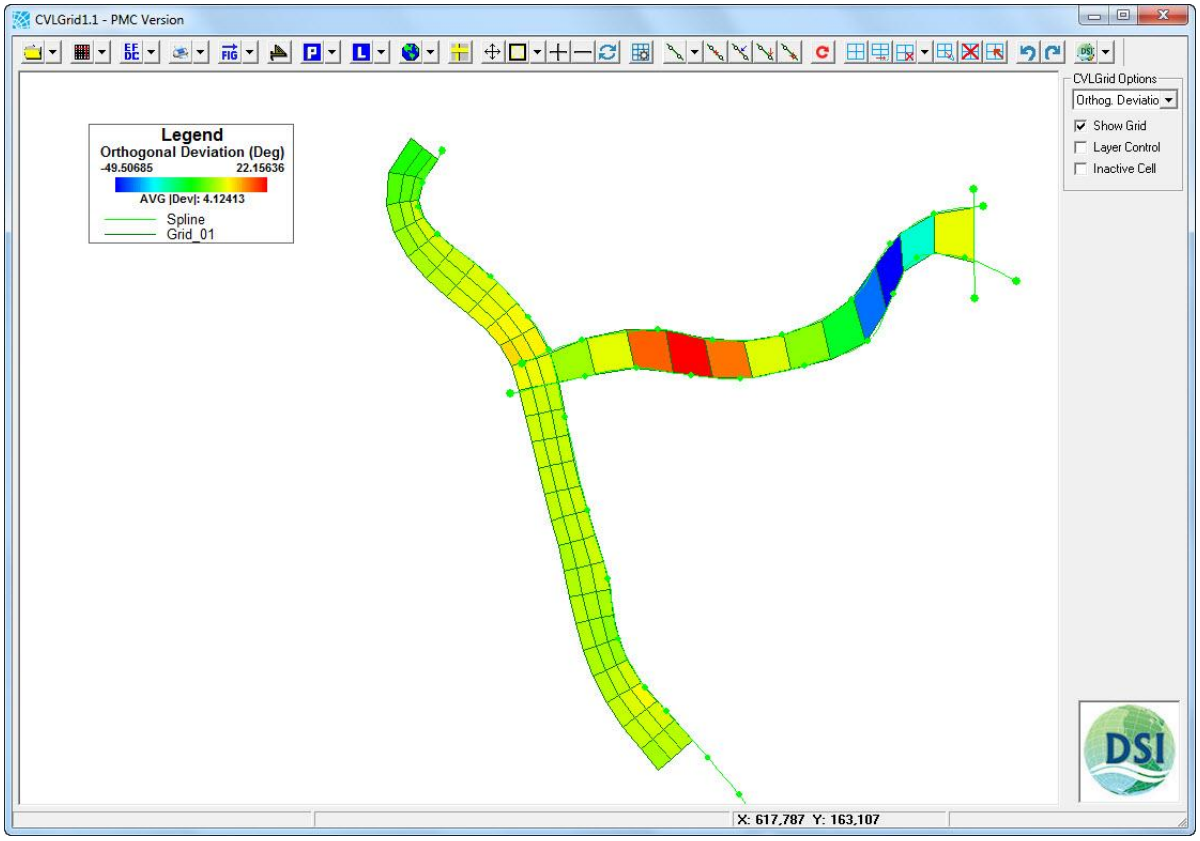


Figure 6-27 Grid image before applying "Orthogonalize 1D"

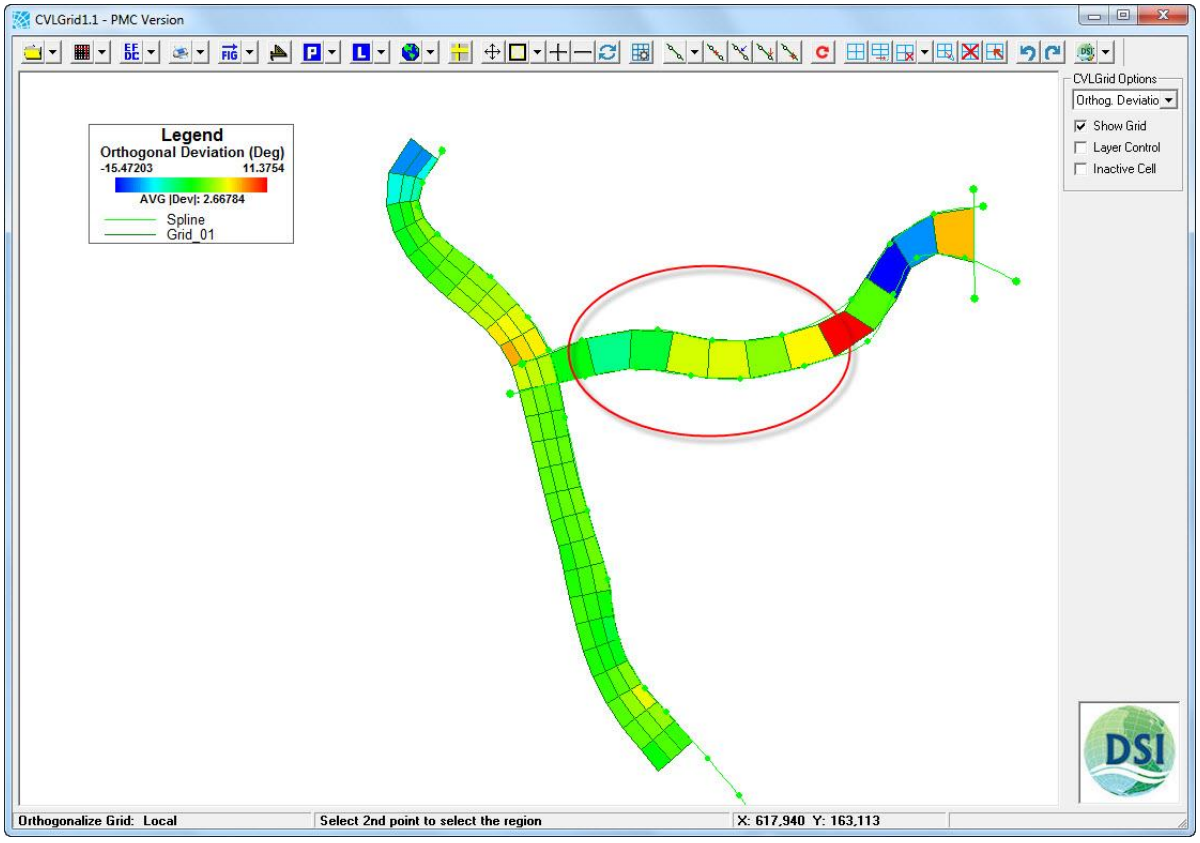


Figure 6-28 Grid image after applying "Orthogonalize 1D"

6.5 Reverse Grid

The *Reverse* tool is shown in Figure 6-29. When a grid is generated each node has an I and J coordinate. This tool allows the user to reverse the I and J coordinates either separately, or at the same time using *Transpose*.

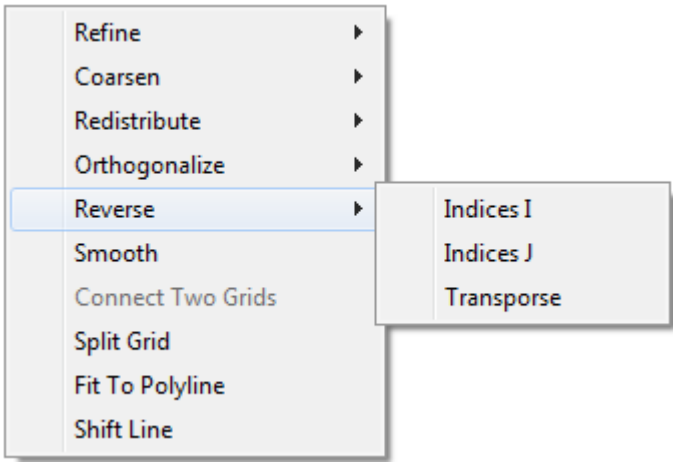


Figure 6-29 Options for reverse grid.

6.6 Smooth Grid

The *Smooth* tool is used to smooth the grid domain and improves the accuracy of the model. This tool is available with a RMC as shown in Figure 6-30.

When the *Smooth* feature is selected, the user must select two nodes on the grid domain to define a block on which to apply the smoothing. Figure 6-31 and Figure 6-32 show the grid before and after using the *Smooth* tool.

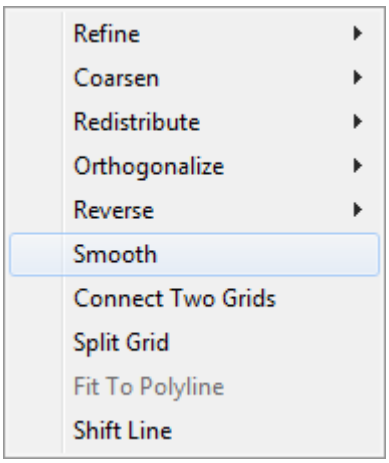


Figure 6-30 Smooth grid option.

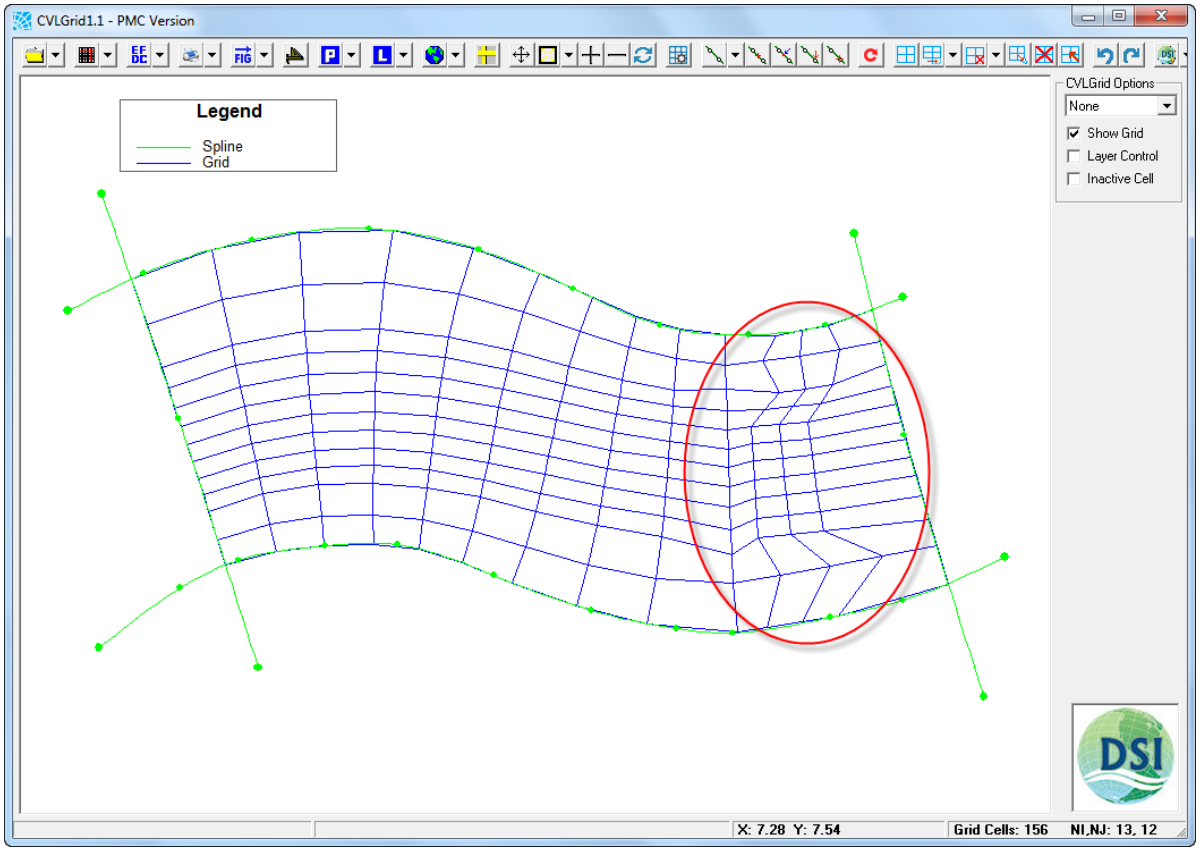


Figure 6-31 Grid image before applying smoothing.

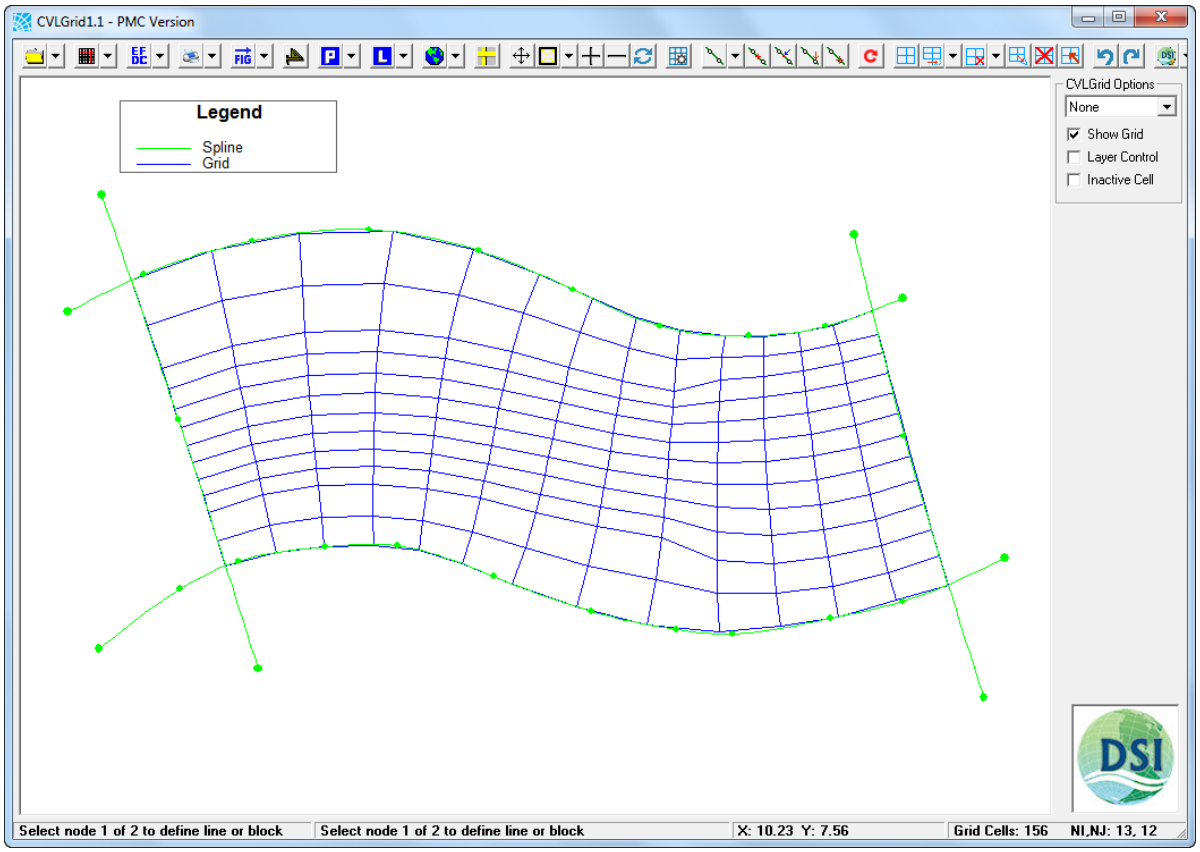


Figure 6-32 Grid image after applying smoothing.

6.7 Connect Two Grids

Often the model domain is too complex to create a single grid that accurately represents the area. Therefore, the user may have to create several grids and connect them using the *Connect* tool.

To connect two grids, the user must have the initial grid open. The user should then create a new layer for a new set of splines. These splines should extend from a location adjacent to the first grid and outline the new domain. Draw the farthest spline first, followed by the connection to the first grid and then the sides. When drawing the spline parallel to the first grid, ensure it is close to the edge. The user may then generate the grid making sure the number of cells in the I or J-axis of the new grid matches the number of cells connecting it to the first grid. Select the *Connect Two Grids* tool and click two nodes on the grid line of the second grid where it is adjacent to the first grid. An example of this is shown in Figure 6-33, Figure 6-34, and Figure 6-35.

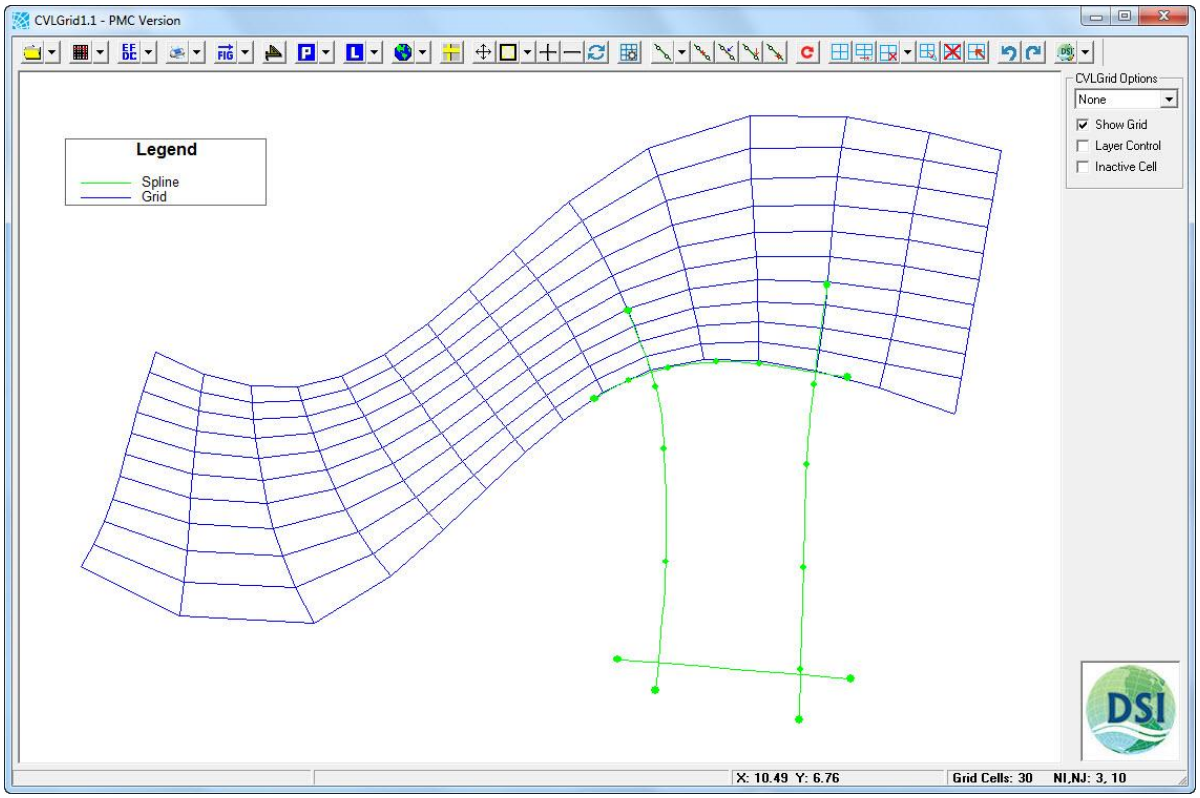


Figure 6-33 Splines for new grid drawn next to a previously generated grid.

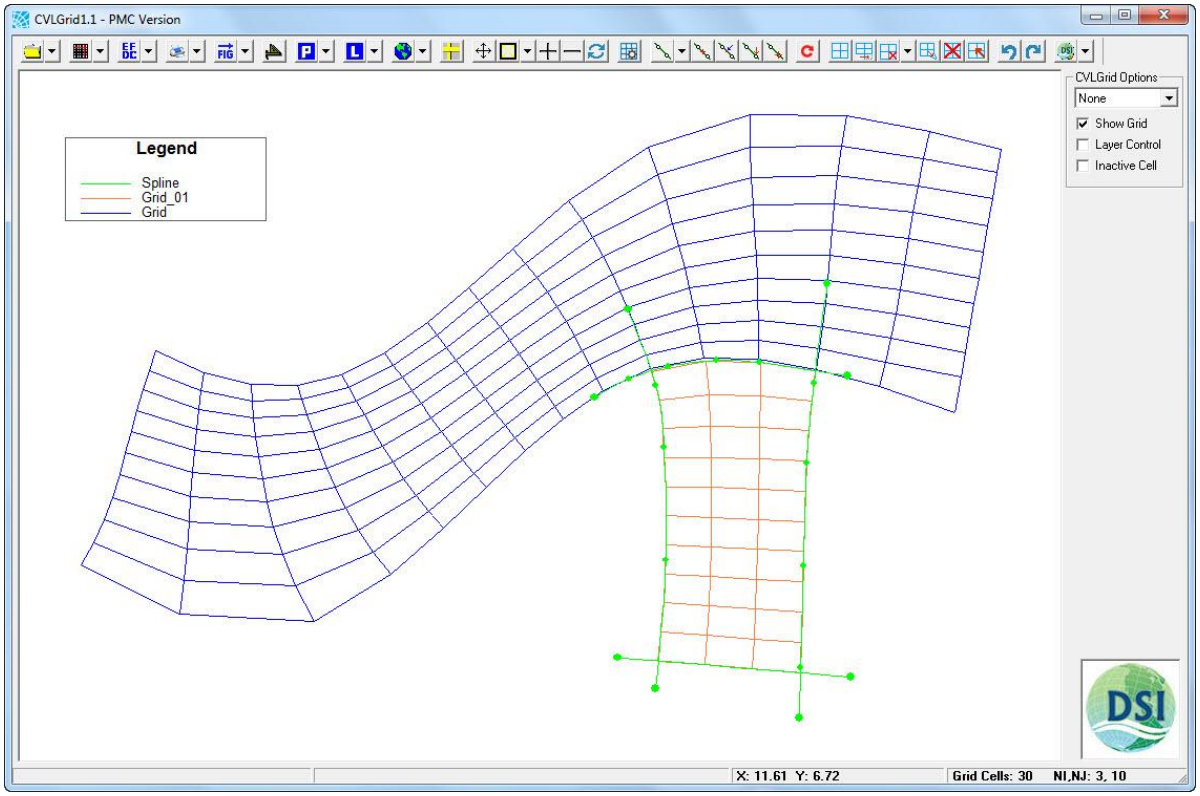


Figure 6-34 Second grid generated adjacent to the first grid.

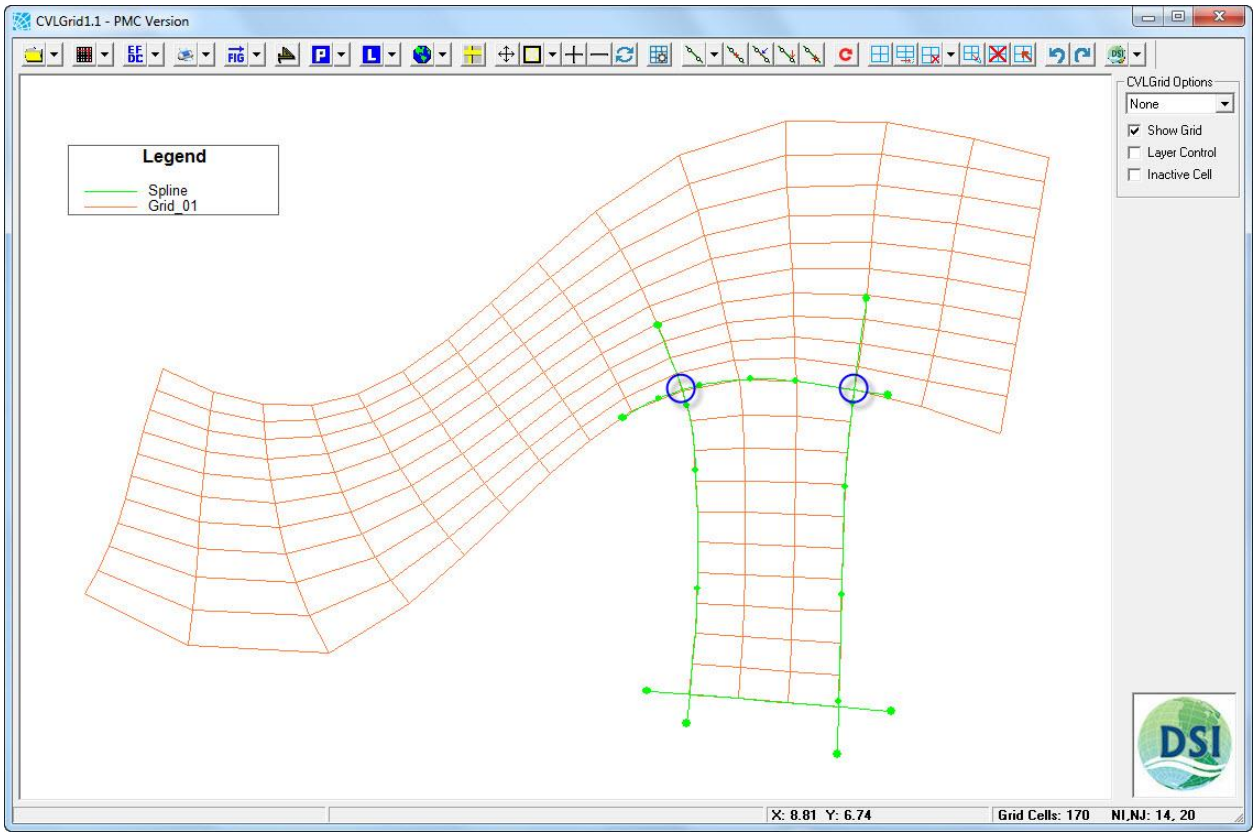


Figure 6-35 Two nodes on second grid selected using the *Connect* tool.

6.8 Split Grid

The *Split Grid* tool is used to split a grid domain into two sub-grid domains and may be selected with an RMC as shown in Figure 6-36.

Split Grid allows the user to create a polygon on an existing grid with LMC as shown in Figure 6-37. Closing the polygon with a RMC will generate a new grid inside the polygon as shown in Figure 6-38. The new grid layer will be created automatically on the *Layer Control* based on the existing grid layer

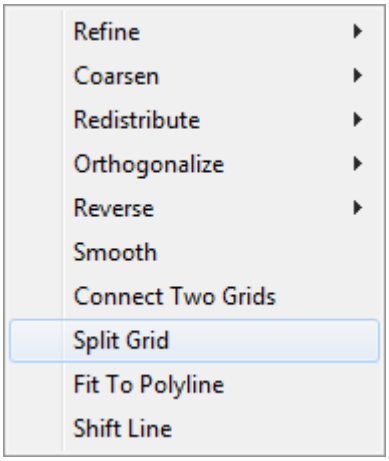


Figure 6-36 Split grid option.

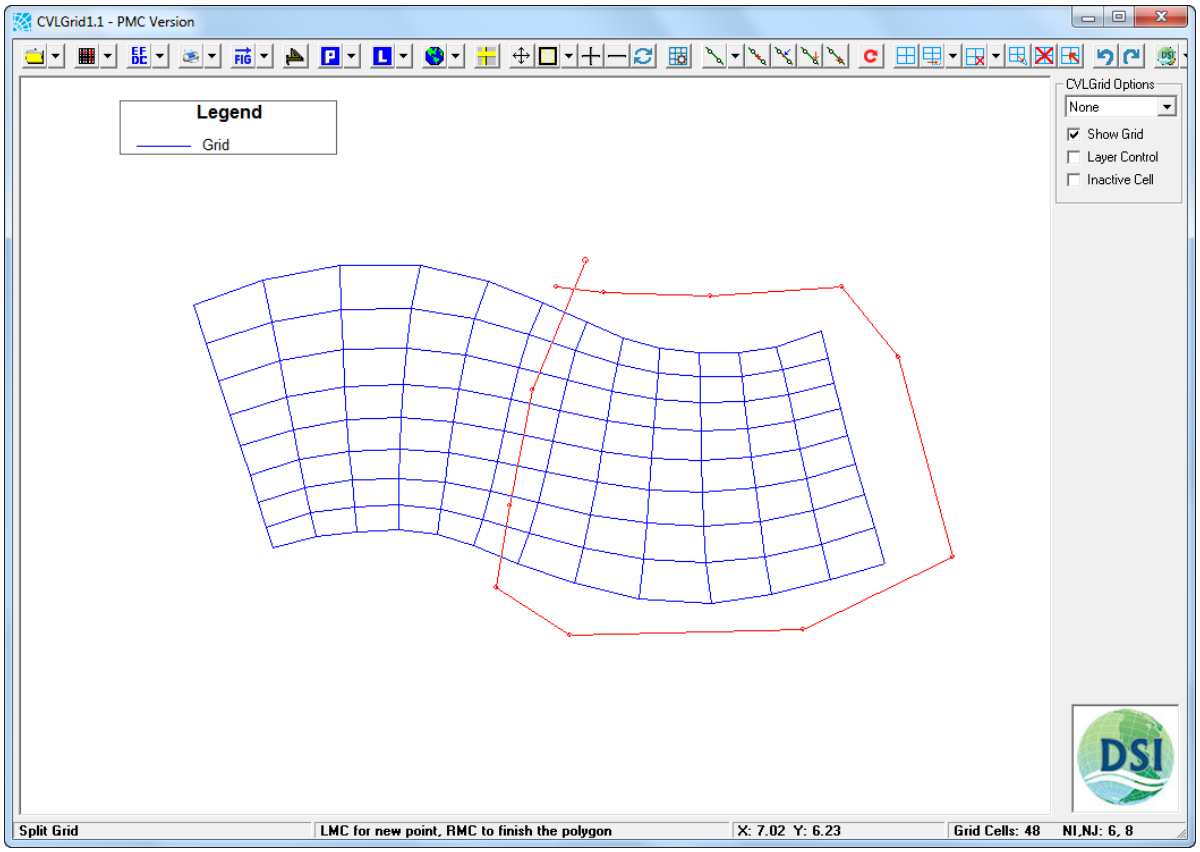


Figure 6-37 Split grid option: a polygon created

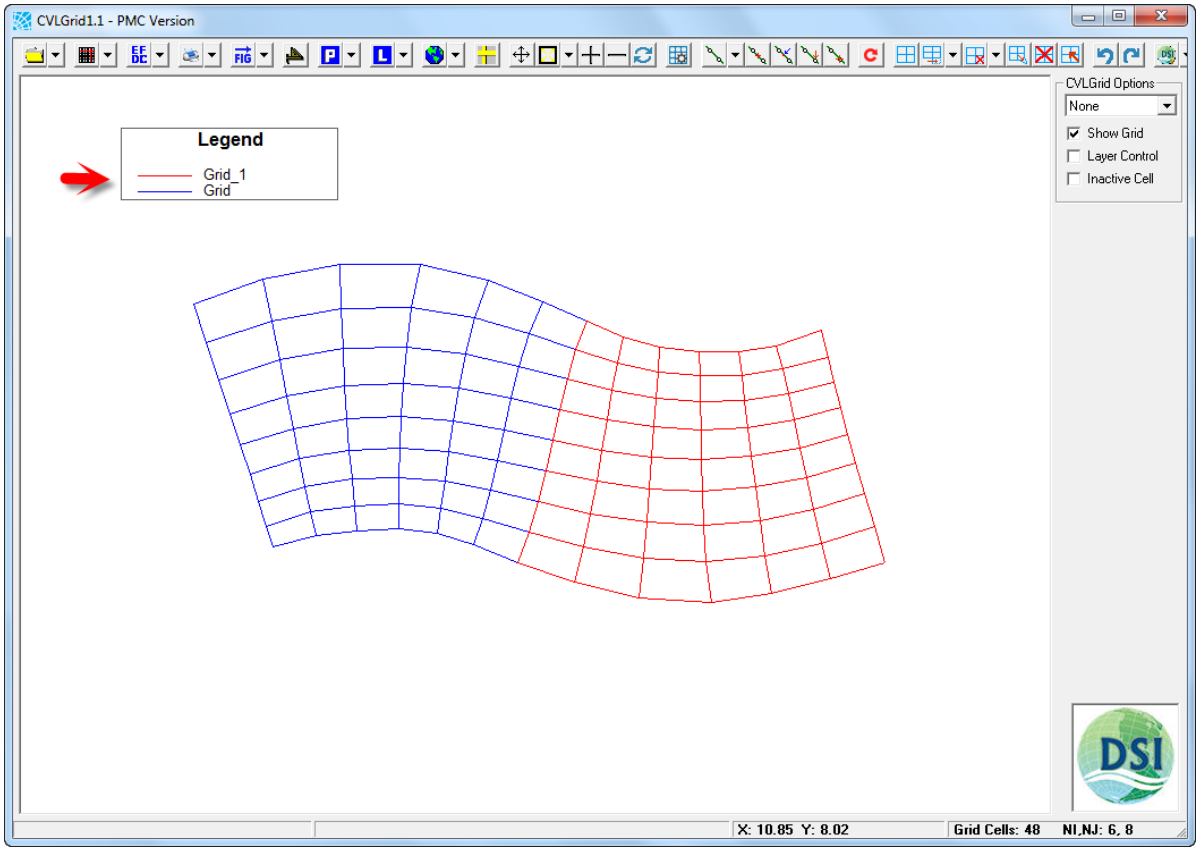


Figure 6-38 Split grid option: two sub-grid domains generated

6.9 Fit To Polyline

The *Fit to Polyline* tool allows the user to fit grid cells to a polyline (boundary) of the model domain. When a grid is generated and orthogonalized it will tend to pull away from the boundary to reposition itself. However, for accurate modeling the grid should fit to the domain boundary as closely as possible.

To use this tool simply select it from the *Grid and Cell Operation Menu* by RMC on the workspace and select two nodes along the same segment of the grid adjacent to the boundary. The boundary here can be of type *Overlay* or *Splines*. Select two nodes on the grid to indicate the section of grid that will be affected by this function. Figure 6-39, to Figure 6-42 demonstrate the use of this tool along with further explanation of the steps.

1. RMC to choose *"Fit to Polyline"* as shown in Figure 6-39.

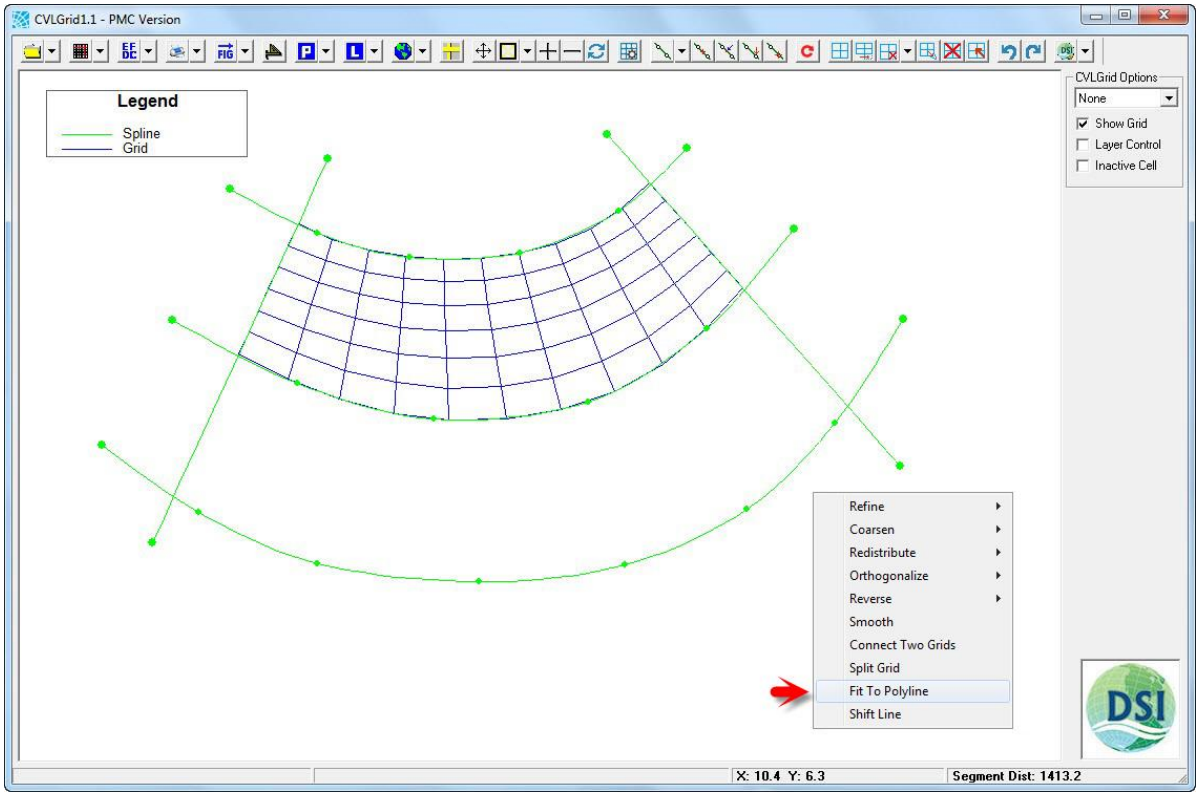


Figure 6-39 Fit to polyline tool.

2. Select two nodes from the edge of grid which does not fit to the boundary and requires moving as shown in Figure 6-40. This action defines the range of the grid cells that will be modified to fit to the boundary.

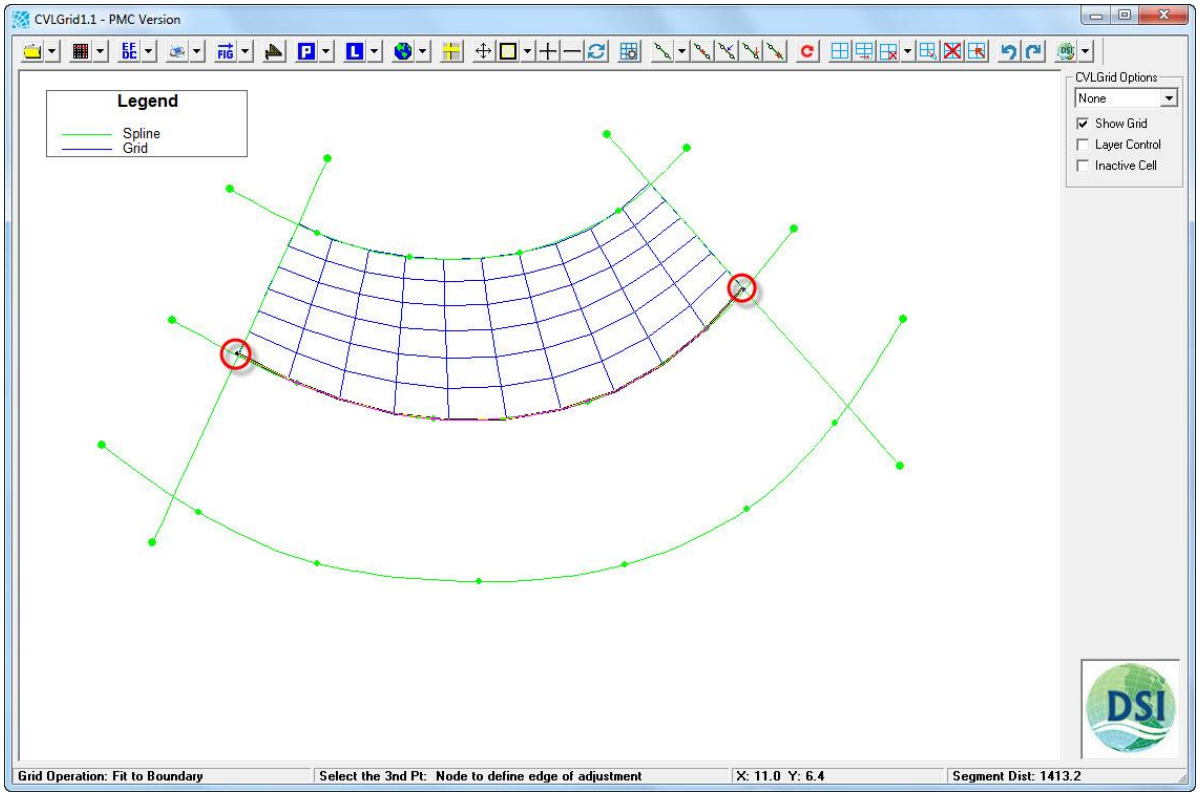


Figure 6-40 Fit to polyline tool: choose the first two nodes.

3. Select a node on the grid to determine the number of rows that will be modified. This effectively sets the depth of the impact of the shifting process. In Figure 6-41 this node is shown by the blue circle, indicating that four rows deep inside the grid will be modified (from the blue third node to the red edge that was created from the first two nodes).

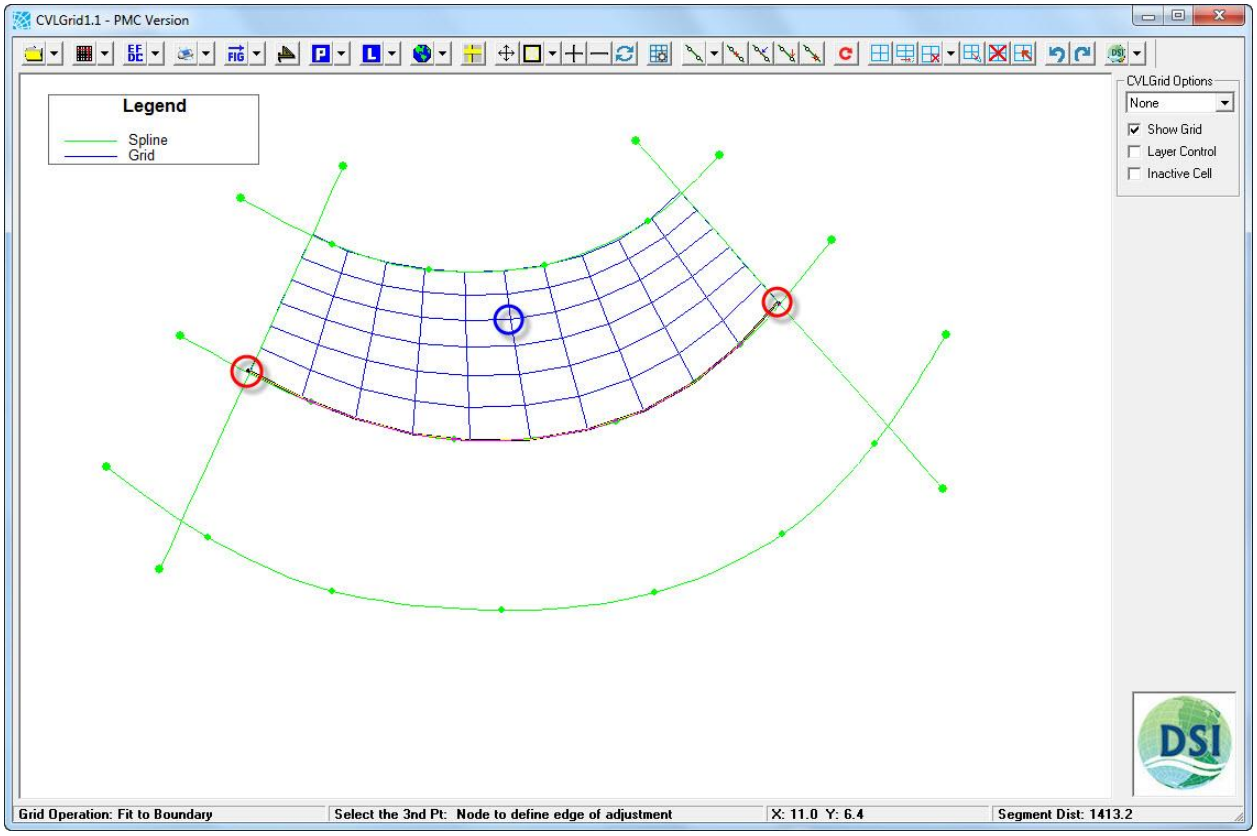


Figure 6-41 Fit to boundary tool: choose the third node

4. The last node will be placed on the boundary line to tell CVLGrid which polyline is being fitted to. Immediately after the user selects the four and final node, the *Fit to boundary* function will be automatically implemented as seen in Figure 6-42.

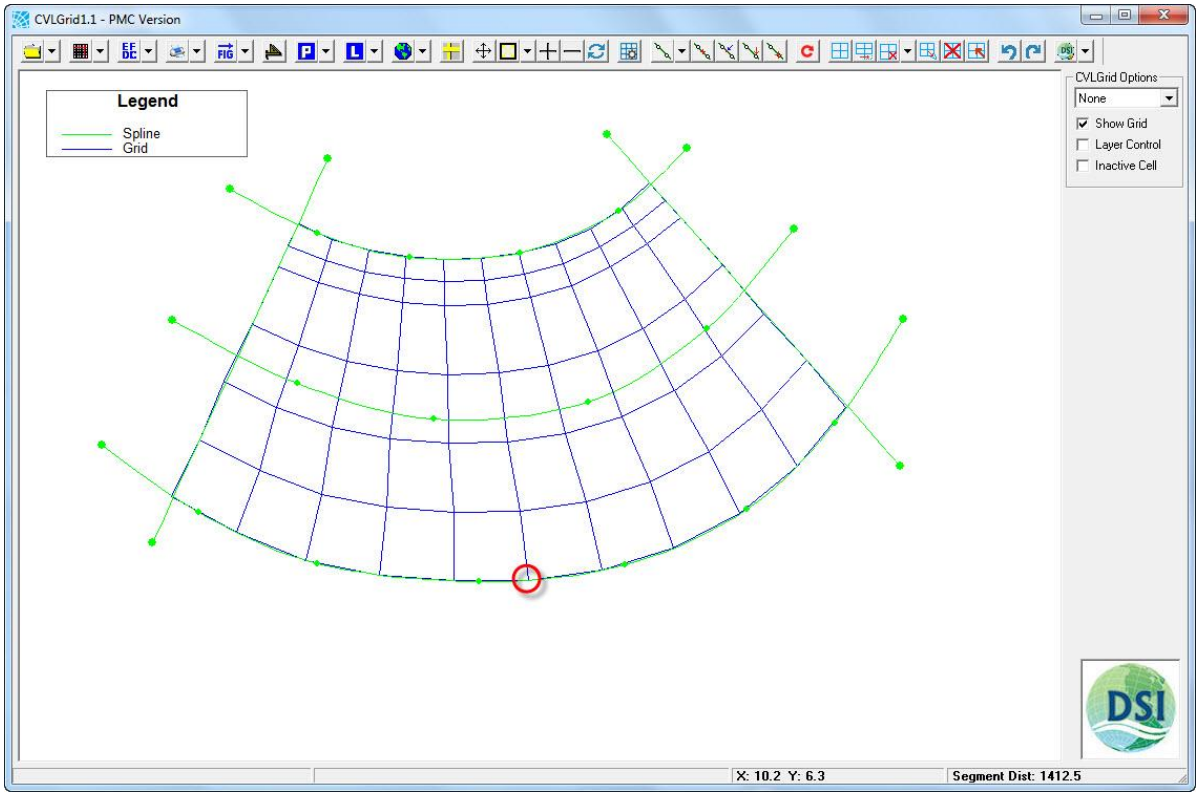


Figure 6-42 Fit to boundary tool: choose the final node.

5. When grid cells are fitted to a boundary only the closest cells are stretched. Thus, the grid can further be “regularized” to improve cell spacing shown in Figure 6-43.

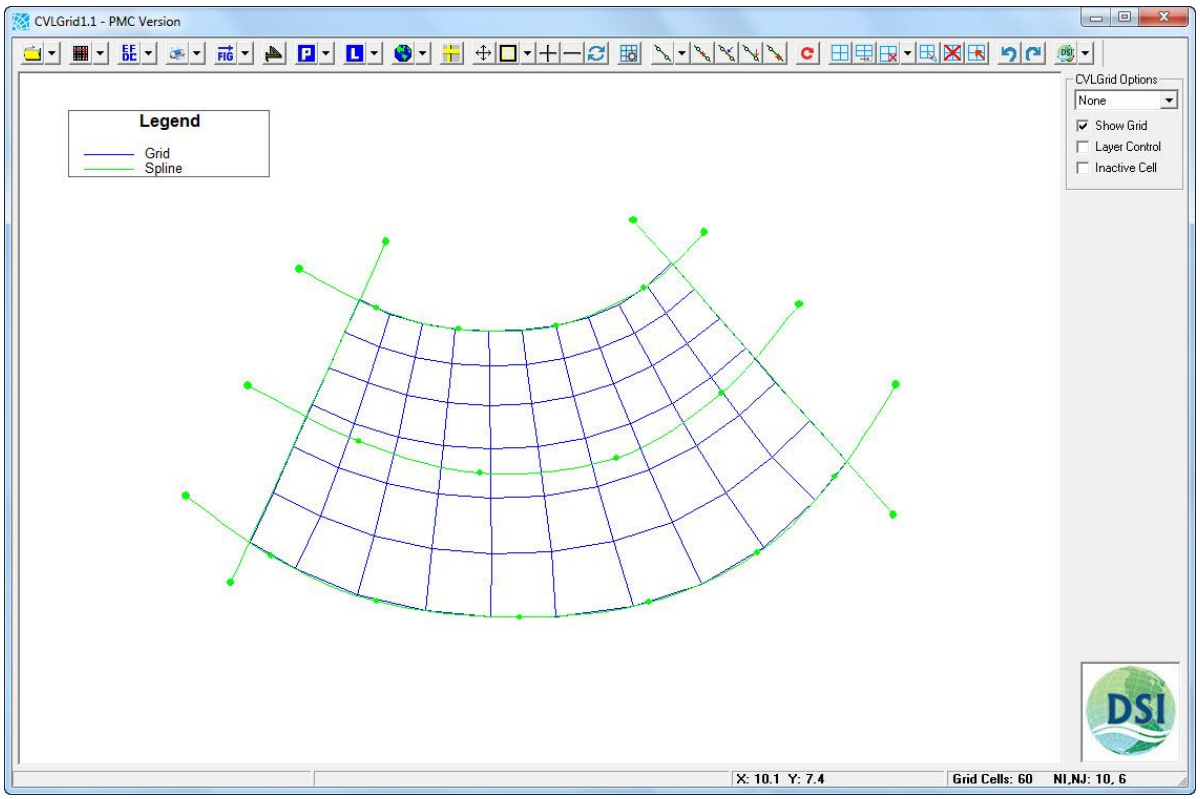


Figure 6-43 Grid after it has been regularized.

6.10 Shift Line

The *Shift Line* function allows the user to move a line (or multiple parallel lines) of the selected grid. To do this, the user must follow the steps below:

1. Select the first point and the second point on the boundary line of the grid. These two points determine the line that will be moved. The first point will act as a pivot when the grid line is moved. The second point will serve as a moveable point to determine where to move to. This process is shown in Figure 6-44.

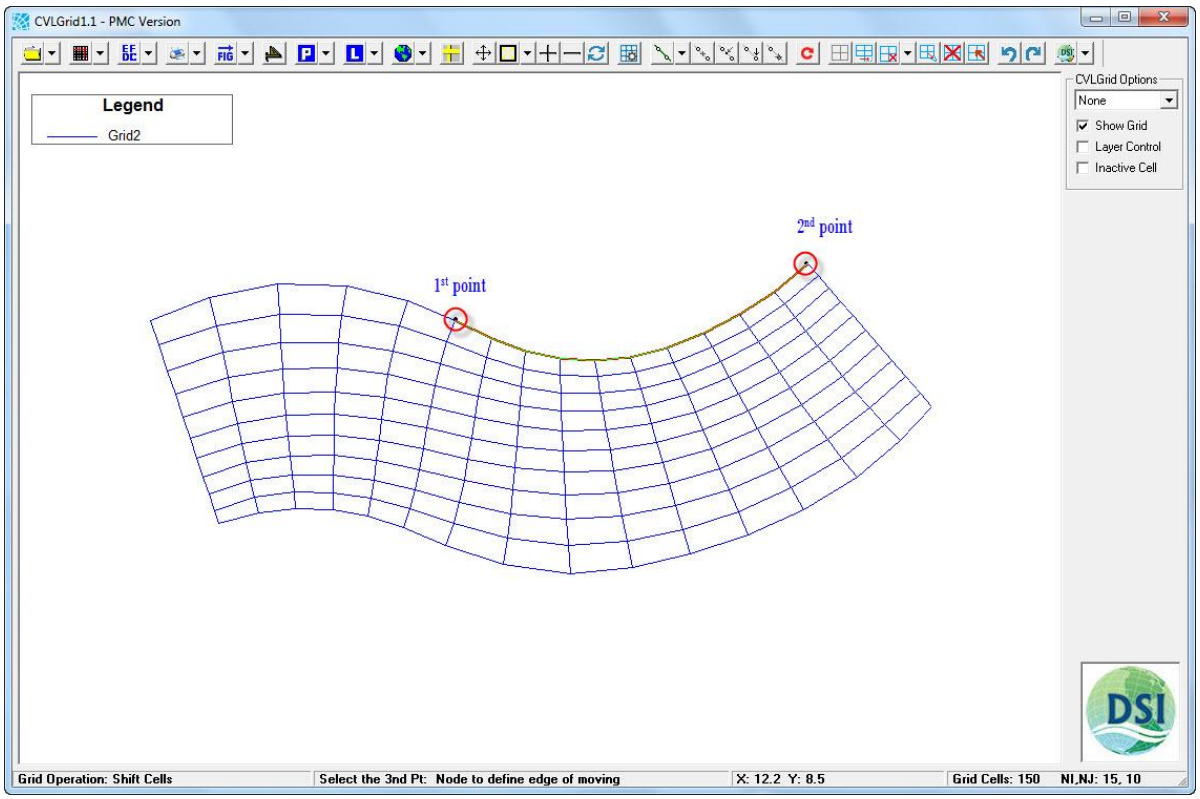


Figure 6-44 Shift line: choose the first two points.

2. Select a third node in the grid. This third point will determine the number of cell rows between this point and the selected line that will be moved. A line segment is connected by the first point and the second point will be resized after the line is moved. When the third point is chosen then the line segment will be highlighted automatically.

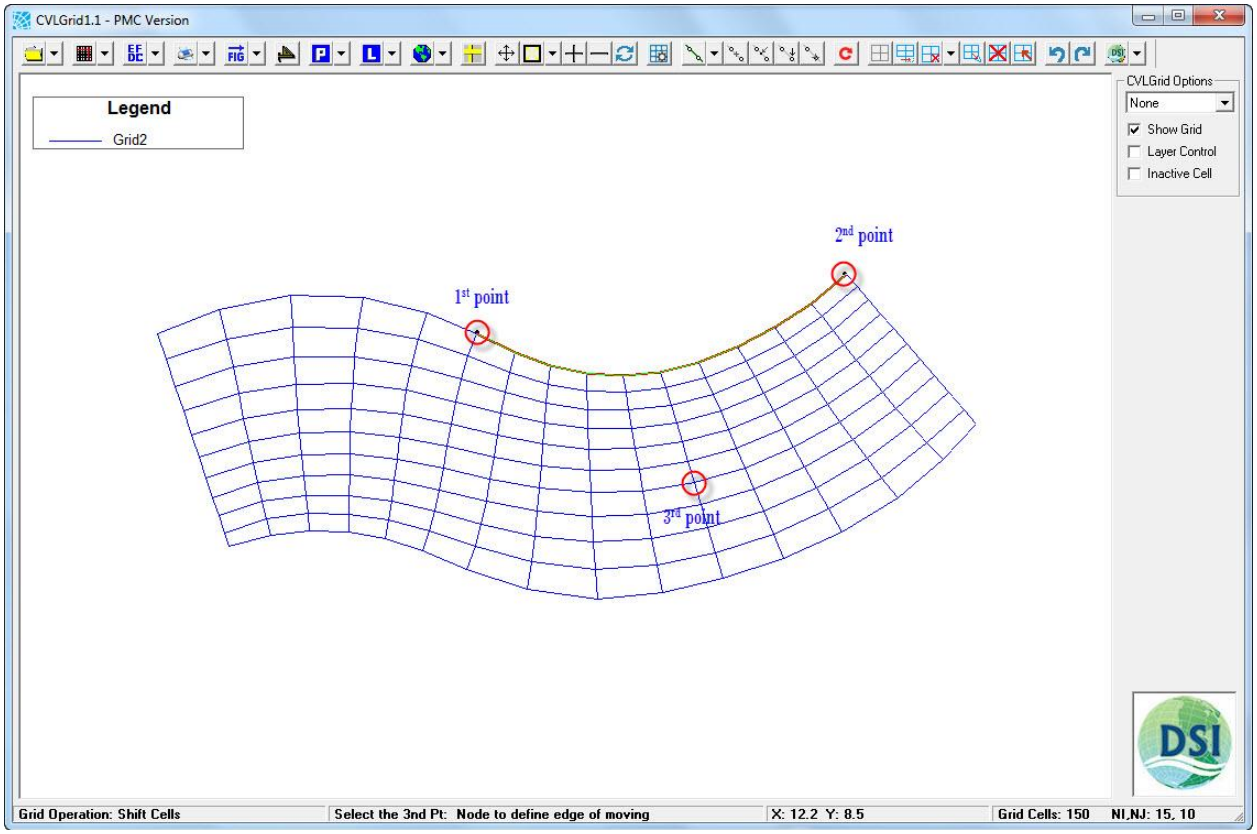


Figure 6-45 Shift line: choose the third point inside the grid

3. LMC on the second point to start moving then LMC again on the desired destination. The grid cells between the third point and the line segment will be resized.

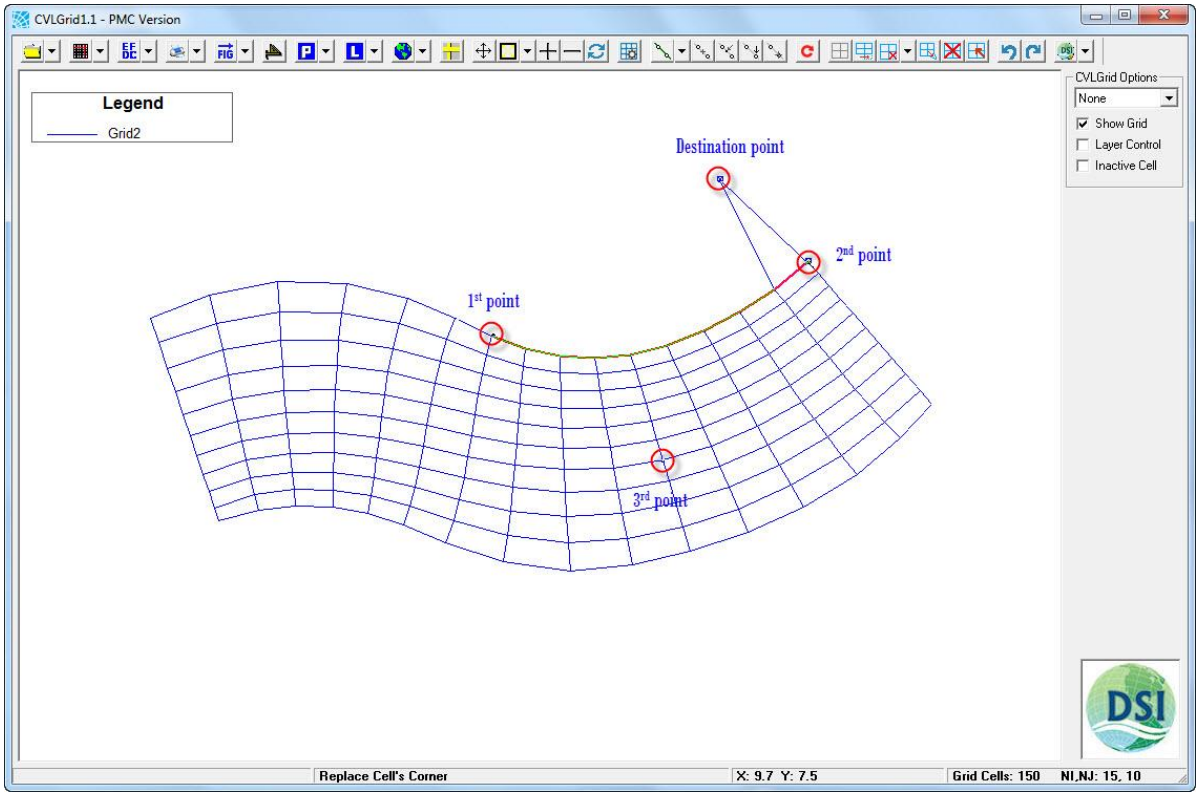


Figure 6-46 Shift line: move the second point to required destination

4. After clicking on the destination point the grid image will be changed as shown in Figure 6-47. Note that if the second point selected is not a corner node, then one cell to the side of the second is also moved along with the second point.

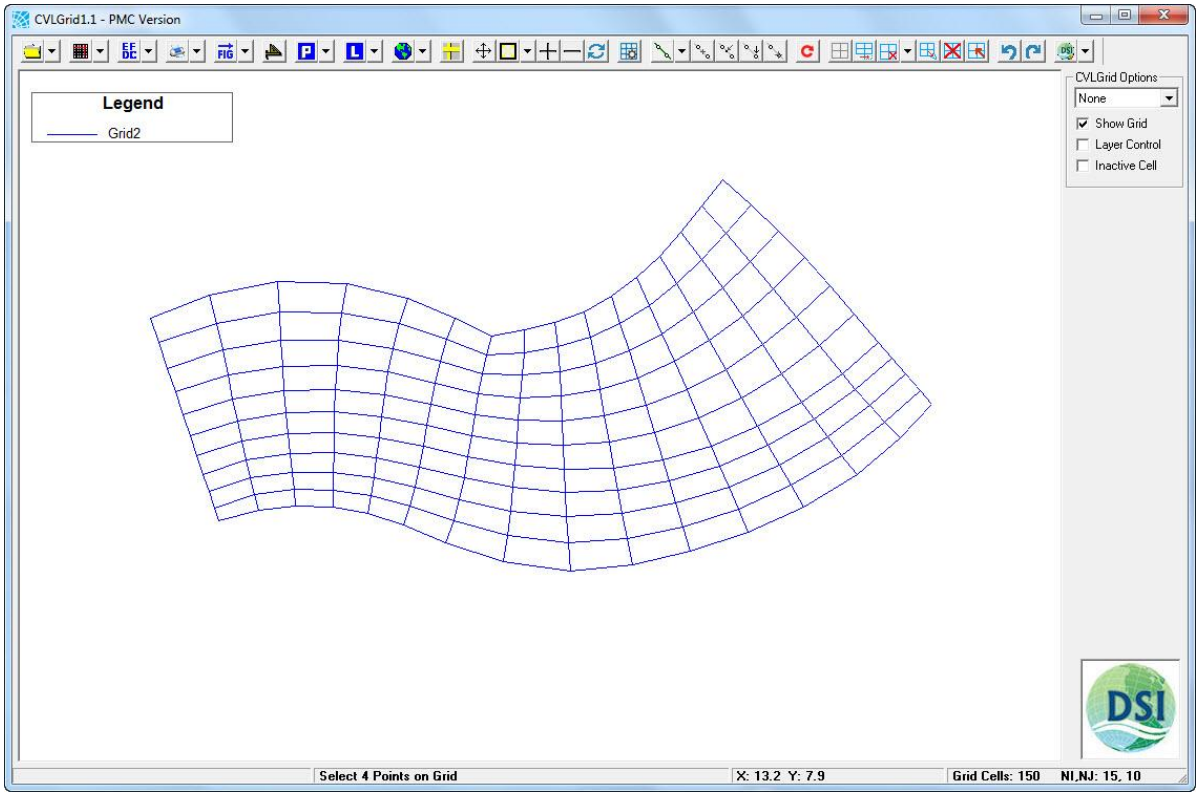


Figure 6-47 Shift line: grid after LMC on destination point.

7 Polyline/Polygon Operations

7.1 Overlay Operations

When an *Overlay* is loaded or has been generated, there are several functions available to the user to modify these polylines.

To access the operation the user should select the desired overlay layer in *Layer Control*. This makes this layer to be an active layer which may be seen from the bold font in the *Layer Control* space. In the workspace a hand icon should now be displayed instead of an arrow when the user hovers over this overlay line. The user should select this polyline with LMC. Now when the user RMCs a function list is displayed as shown in Figure 7-1 and Figure 7-2.

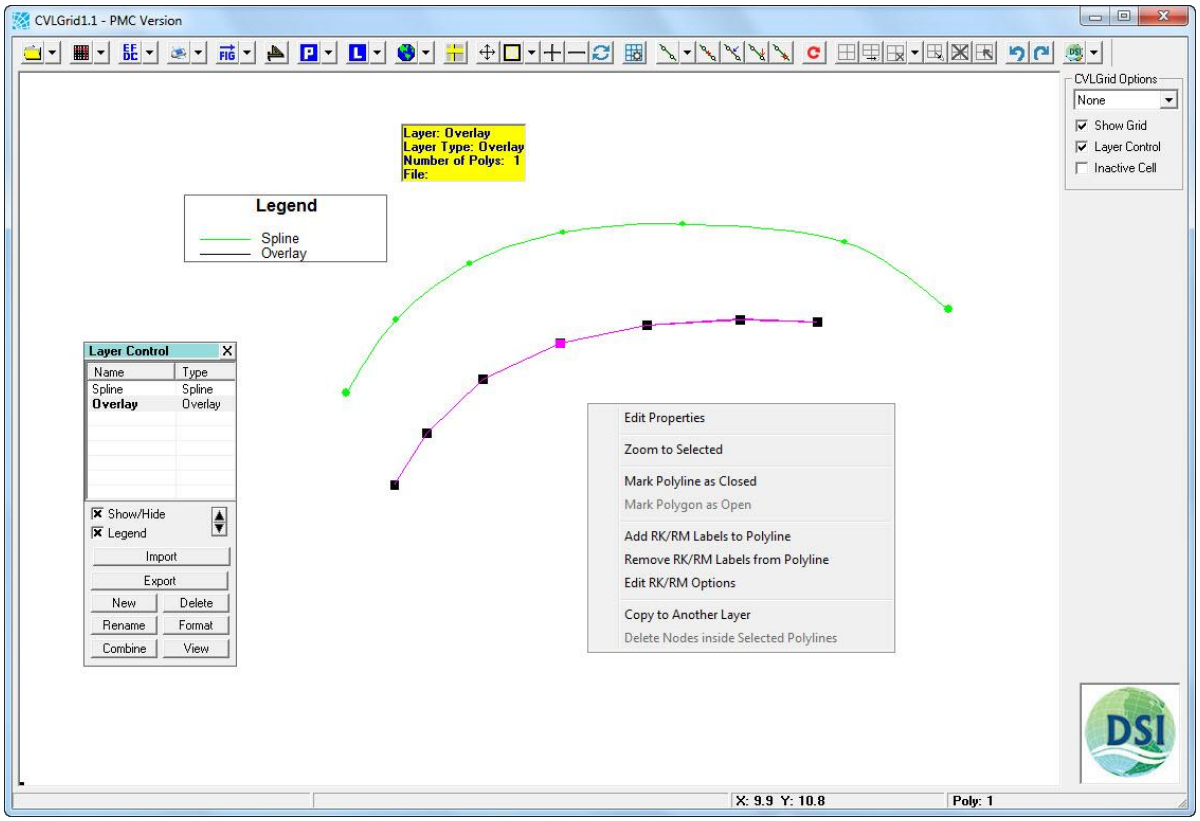


Figure 7-1 Overlay operations menu.

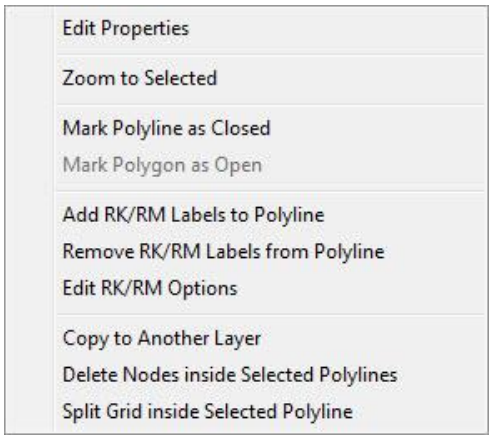


Figure 7-2 Overlay operations menu detailed.

Explanations of the options available for editing the overlay are as follows:

Edit Properties	Display the information from the current polygon/polylines selected and allow the user to edit their properties as shown in Figure 7-5.
Zoom to Selected	View the selected polyline entirely within the workspace.
Mark Polyline as Closed	Make the polyline become a polygon.
Mark Polyline as Open	Make the polygon become a polyline.
Add RK/RM Labels to Polyline	Add labels of River Kilometer/River Mile on polyline. Set distance to start labeling and distance between labels as shown in Figure 7-3
Remove RK/RM Labels from Polyline	Clear the RK/RM labels from the polyline.
Edit RK/RM Options	Settings for label and symbol distance as shown in Figure 7-4
Rename	Define another name for the selected polygon/polyline.
Copy to Another Layer	Copy the selected polygon/polyline to another available layer in *Layer Control*.
Delete Nodes inside Selected Polylines	Delete grid nodes inside the selected polygon/polyline.
Split Grid inside Selected Polylines	Split an original grid into two grids, a new grid inside selected polygon/polyline.

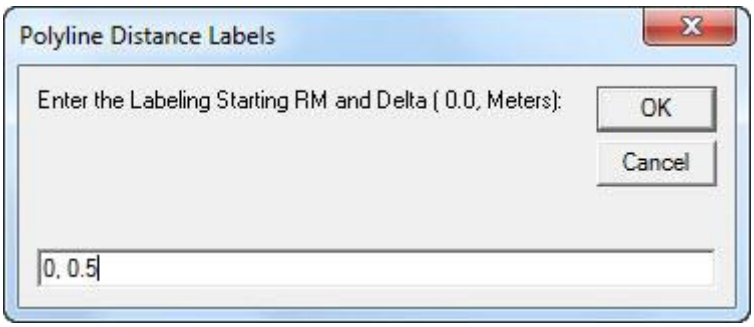


Figure 7-3 Polyline distance labels window.

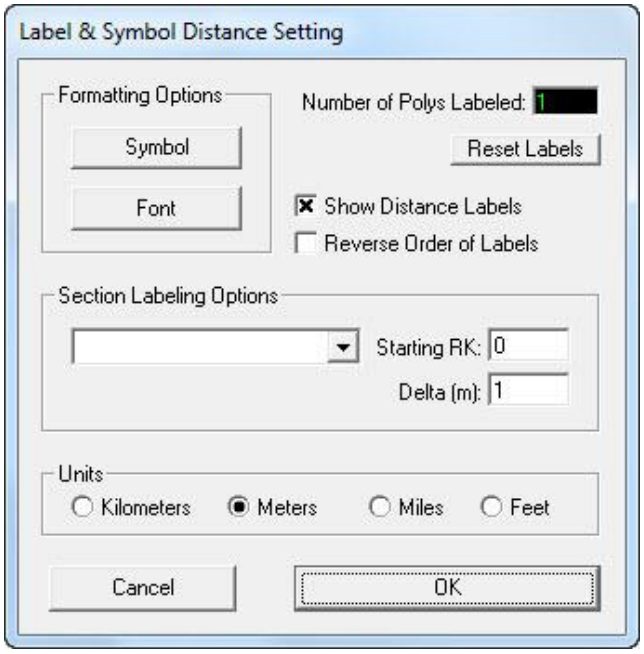


Figure 7-4 Edit RK/RM options window.

7.1.1 Polygon/Polyline Information Tool

The tool for editing the information related to each polyline is shown in Figure 7-5 and various options described in more detail below.

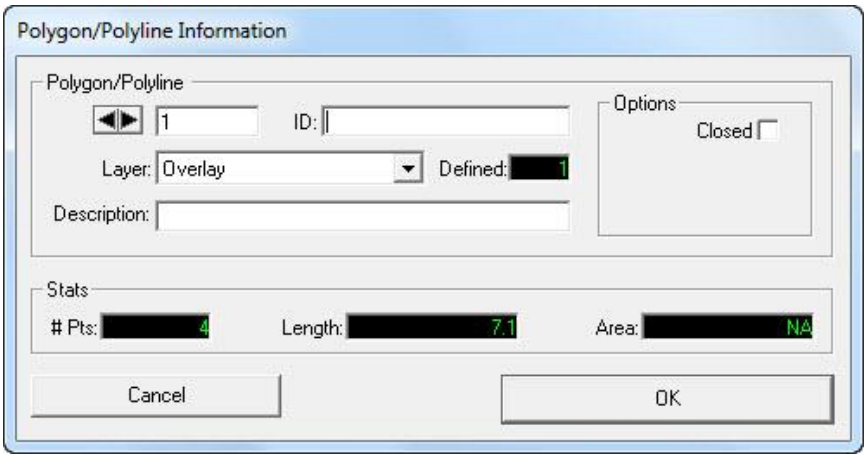


Figure 7-5 Polygon/polyline information window.

In the *Polygon/Polyline* sub frame the user can cycle through the current polygons and polylines using the arrow buttons. As the user cycles, the polygon number and ID are displayed on the right. The user can change a polyline ID by clicking in the text box and typing the desired ID.

The *Layer* drop down menu shows what layer the current polygon is on. *Defined* simply shows how many available layers are in the *Layer Control*. *Description* allows the user to describe the polygon/polyline.

The *Options* frame allows the user to close the polylines, thus creating polygons. When this box is checked the "*Area*" display box will show the area in meters within the current polygon.

The *Stats* window displays the polygon/polyline statistics. For the currently selected line it shows the number of points, the length, and the area. However, the area is only shown when the line is a polygon.

In the *Edit RK/RM Options Window* the user can LMC on *Symbol* and *Font* buttons to change the symbol and font as shown in Figure 7-6 and Figure 7-7 respectively. In addition, the user can modify the starting RK and Delta (m) and unit for display on the polyline. Figure 7-8 shows RK labels displayed on the polyline.

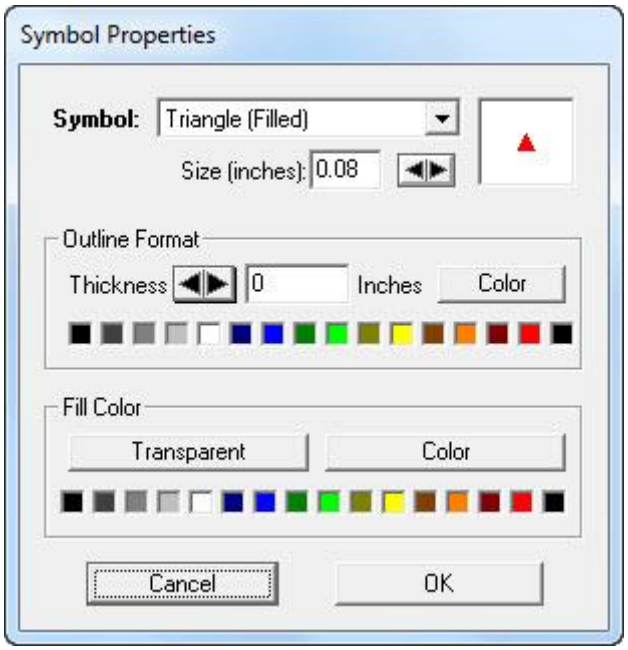


Figure 7-6 Symbol properties settings.

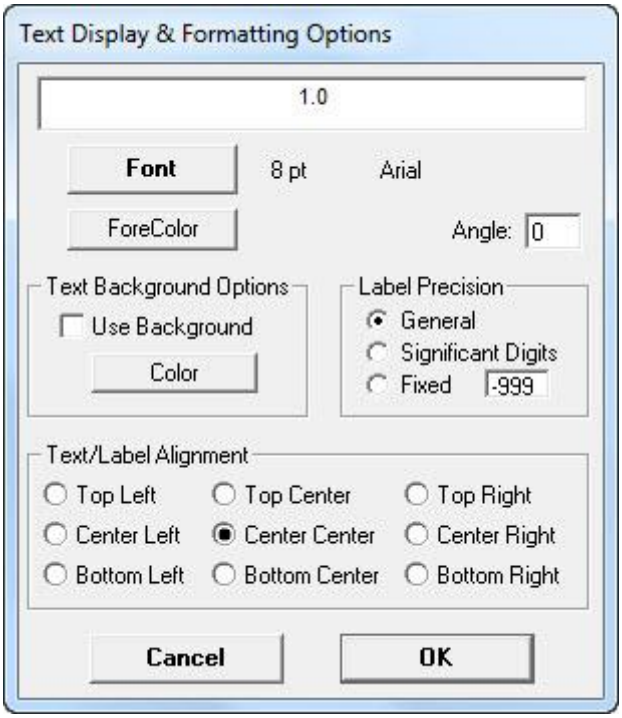


Figure 7-7 Font properties settings.

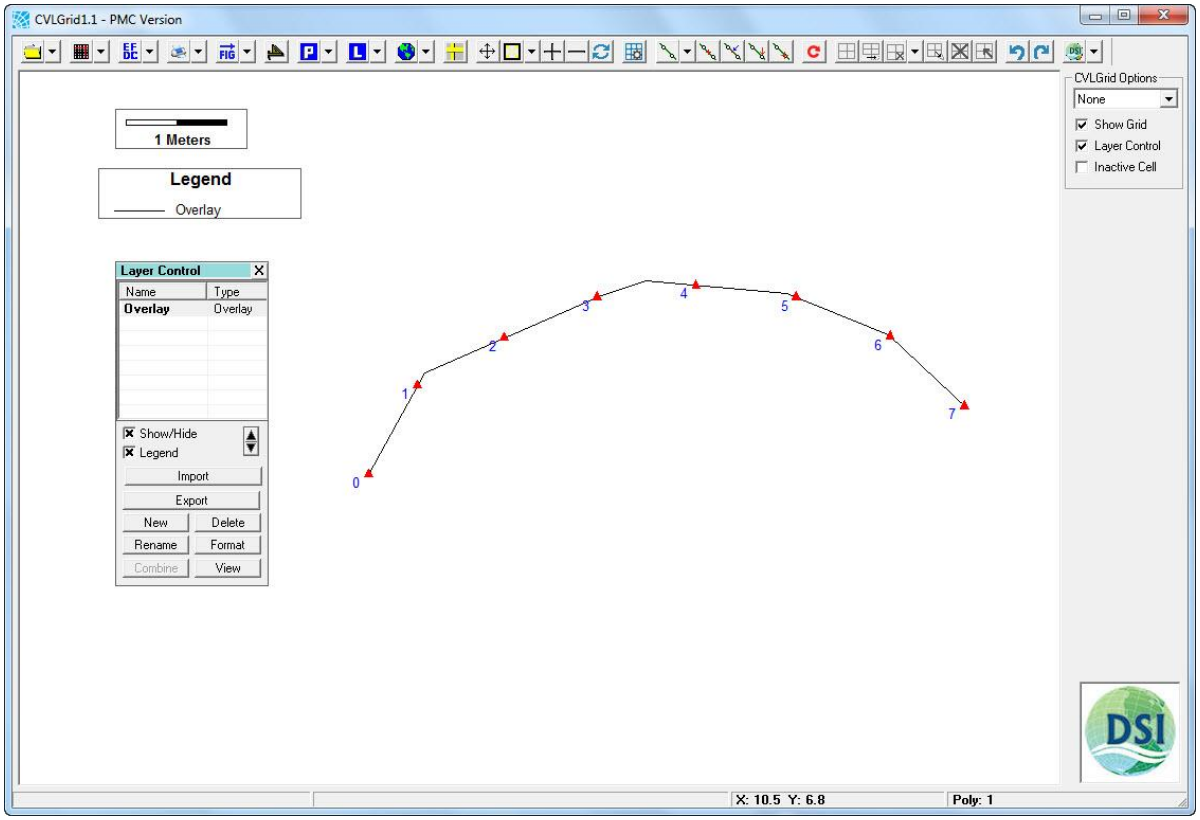


Figure 7-8 RK labels displaying on a polyline.

The *Copy to Another Layer* feature allows users to move spline to overlays and vice-versa. When there are from two layers types (both *Overlay* and *Spline*) in the *Layer Control*, the *Copy to Another Layer* function is enabled as shown in Figure 7-9. The user may then define a destination layer for copying the polygon/polyline to as shown in Figure 7-10. For an *Overlay* type, it can be copied to another *Overlay* or a *Spline* layer. Then it will have properties of the layer it is copied to.

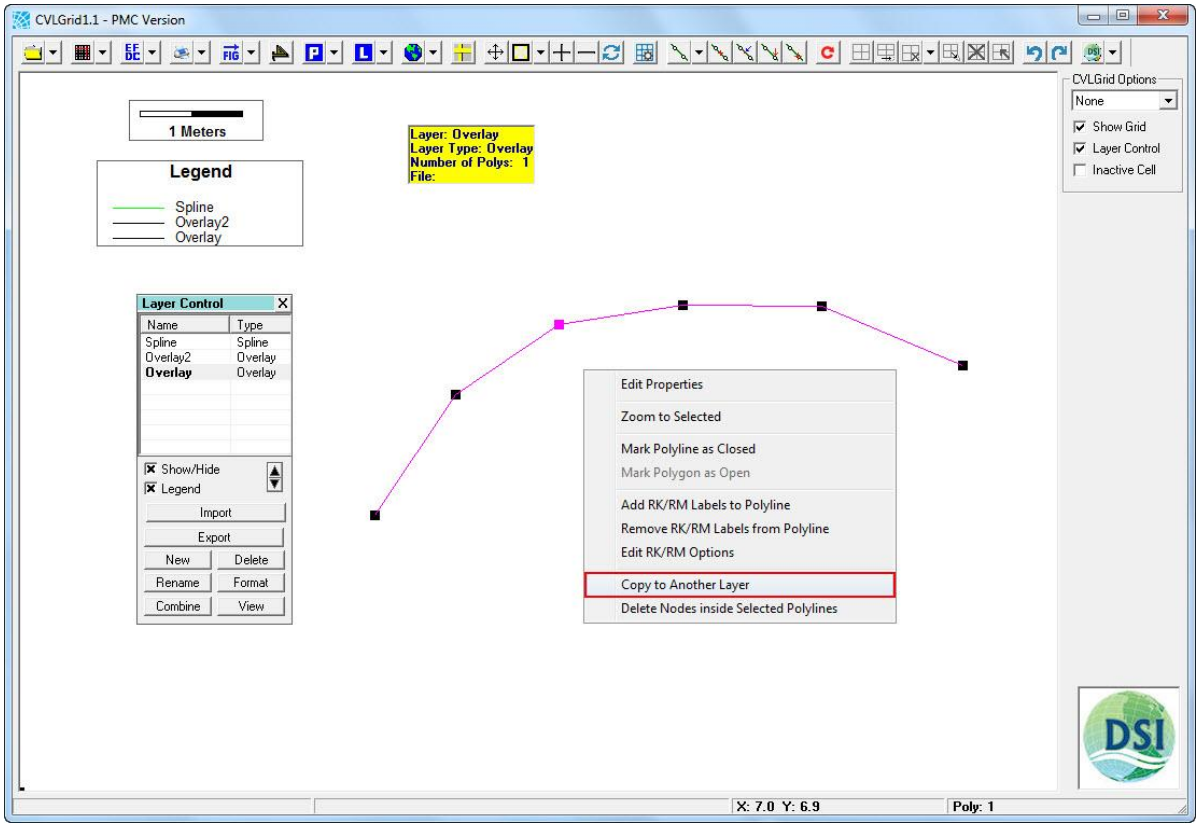


Figure 7-9 Copy to another layer function.

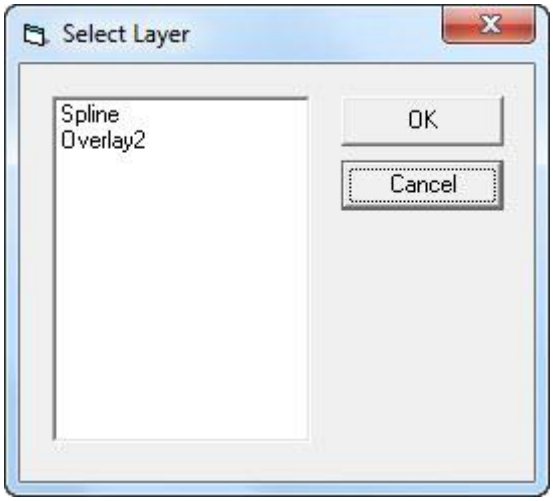


Figure 7-10 Destination layer selection.

The *Delete Nodes inside Selected Polylines* function is enabled when there is a *Grid* layer in *Layer Control*. It allows the user to delete all the grid nodes inside the polyline.

7.2 Spline Operations

When a *Spline* is loaded or generated, there are several functions available to the user to modify these polylines.

The user must first select the desired *Spline* layer in *Layer Control*. This makes the layer the active layer and it is displayed with bold font in the *Layer Control*. In the workspace a hand icon should now be displayed instead of an arrow when the user hovers over this spline. The user should select this polyline with LMC. Now when the user RMCs a function list is displayed as shown in Figure 7-11 and Figure 7-12.

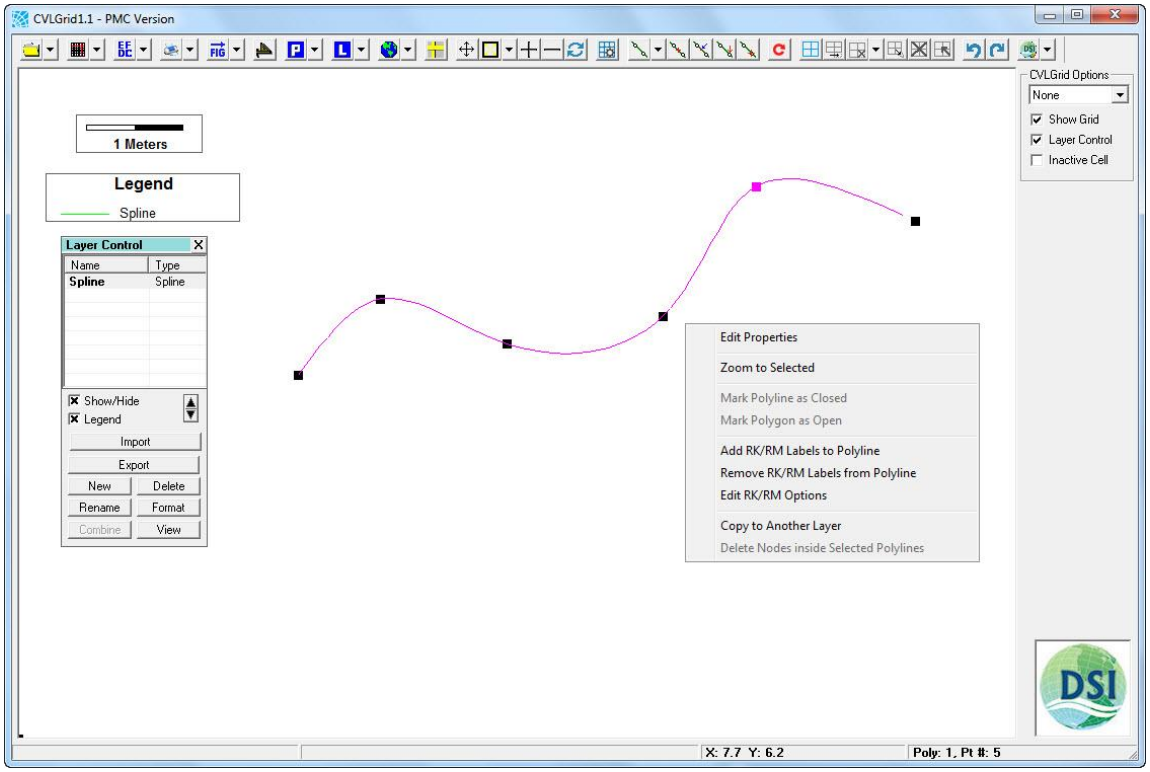


Figure 7-11 Spline operations menu.

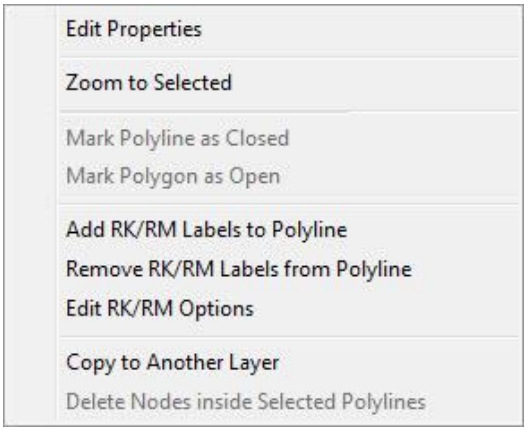


Figure 7-12 Spline operations menu detailed

Generally, most of enabled functions of *Spline Operations* are similar to those for *Overlay Operation* except *Copy to Another Layer.* For spline, this function is only applied to copy the selected spline to a spline layer. The other functions have been introduced in Overlay Operations in Section 7.1

8 References

Bieberbach, L. (1964). “Conformal mapping (4th ed.)”, Chelsea: New York, pp. 1–9, chap. 1.

Craig Paul M. and Chung D H “A Simple Algorithm for Grid Generation Based on Laplace Equations” [Report]. - 2012

Fornberg, B. (1980). “A numerical method for conformal mapping”, Journal on Scientific and Statistical Computing, 1, pp.386–400.

He Hong, Fan Jian-ren and Cen Ke-fa, (2000). “A numerical method for orthogonal grid generation by Laplace system”, Journal of Zhejiang University (Science), Vol. 1, No.2, pp. 125-128.

Hoffman, J.D. (2001). Numerical Methods for Engineers and Scientists, McGraw-Hill International Edition.

Hung Lee Nguyen, (1989). “On the applications of algebraic grid generation methods based on transfinite interpolation”, NASA Technical Memorandum 102095, pp.1-23.

Jeng, Y. N., Shu, Y. L, and Lin, W. W. (1995). “On grid generation for internal flow problems by methods solving hyperbolic equations”, Numerical heat transfer. Part B 27, pp. 43-61.

Patrick Knup and Stanly Steinberg, (1994). Fundamentals of grid generation, CRC Press

Reese L. Sorenson Joseph L. Steger, (?). “Numerical generation of two-dimensional grids by the use of Poisson equations with grid control at boundaries”, N81-14722, pp. 449 – 461.

Shih, T. I-P., Bailey, R. T., Nguyen, H. L., & Roelke, R. J. (1991). Algebraic grid generation for complex geometries. International Journal for Numerical Methods in Fluids, 13, 1–31.

Smith, R. E., Kudlimski, R. A., Verton, E. L. E., and Wiese, M. R. (1987). “Algebraic grid Generation”, Computer Methods in Appl. Mech. and Eng. 64, pp. 285-300.

Thompson, J. F., Thames, F. C., and Mastin, C. W., (1974). “Automatic numerical generation of body-fitted curvilinear coordinates for a field containing any number of arbitrary 2-D bodies”, Journal of Computational Physics, 15(3), pp. 30 - 37.

Thompson, J.F., Soni, B.K. and Weatherill, N.P. (1999). Handbook of Grid generation, CRC Press LLC

Vladimir D. Liseikin (2010). Grid generation methods, Springer

Yuan Chang Liou and Yih Nen Jeng, (1996). “A transfinite interpolation method of grid generation based on multipoints”, Journal of Scientific Computing, Vol. 13, No. 1, pp. 105-114.

APPENDIX A
CVL Formats

Native CVLGrid Formats

Data Format A-1 - CVL File

The CVL file is the native CVLGrid grid file. The header lines describe the format for the grid.

Example

```
* DSI CVLGrid Version 1.1
*
* The 1st line:  CVLGrid file format version
* The 2nd line: # row, # col, coordinate system (0-Cartesian),
*          UTM zone (default 0, > 0 for northern hemisphere, < 0 for southern hemisphere),
*          3 parameters (scale and shift for X and Y, default 1, 0, 0),
*          2 parameters (scale and shift, default 1 and 0) for vertical units conversion Z.
* Other lines:  I, J (corner nodes), Cell Type, X, Y, Z LL Corner Coordinate
CVLGridFormat=1.0
  136  280    0    0    1    0    0    1    0
    1  157    5  4.917964931152040E+05  5.242959347802510E+06    0
    1  158    5  4.912742546024100E+05  5.243034901705380E+06    0
    1  159    5  4.907515911957290E+05  5.243117804821200E+06    0
    1  160    5  4.902472357772360E+05  5.243210643889140E+06    0
    1  161    5  4.897719663253260E+05  5.243314696917570E+06    0
    1  162    5  4.893519862918540E+05  5.243423351823180E+06    0
```

Data Format A-2 P2D Polyline/Polygon File.

Any number of polylines/polygons can reside in the same file. Each one begins with a single line header that is used as the "ID" of the polyline/polygon. Next comes the data in either 2D or 3D format, the importing automatically handles either. Finally the polyline/polygon is finished (not necessarily "closed") by an "*" in column 1. There is no difference between polyline and polygon in a P2D file, only how the data are treated by the application reading the file.

Example

```
Polyline                                  Test
609115.69390674      3643612.72035394     0
608828.057738002     3642922.39387814     0
```

608569.186338242 3642232.06904821 0
608396.604307826 3640908.94563456 0
...
...
...
601493.350247946 3628713.19503985 0
601464.586301899 3627763.99747289 0
601522.11337106 3627016.14485373 0
601522.11337106 3626642.21854415 0 *

General Grid Formats

Data Format A-3 DXDY/LXLY

Example

C DXDY.INP FILE, IN FREE FORMAT ACROSS COLUMNS for 9322 Active Cells
C Project: AWWU BE

C	I	J	DX	DY	BOTTOM DEPTH	ELEV	Veg ZROUGH	TYPE
	3	41	713.7	1052.2	0.081	7.992	0.0010	
	3	42	693.1	1022.4	0.081	6.552	0.0010	
	3	43	674.6	996.8	0.081	8.050	0.0010	
	3	44	657.2	894.7	0.081	8.044	0.0010	
	3	45	640.1	1026.0	0.081	8.143	0.0010	
	3	46	622.5	942.8	0.081	6.302	0.0010	
	3	47	605.0	869.4	0.081	8.317	0.0010	
	4	27	879.6	1627.8	0.081	6.200	0.0010	
	4	40	690.6	1080.7	0.081	7.607	0.0010	

Example

C LXLY.INP FILE, IN FREE FORMAT ACROSS LINE for 9322 Active Cells
C Project: AWWU BE

C	I	J	X	Y	WIND CUE	CVE	CUN	CVN	SHELTER
	3	41	624703.9	6794905.0	0.23872	0.97109	-0.97109	0.23872	1.00
	3	42	625713.0	6795145.1	0.22543	0.97426	-0.97426	0.22543	1.00
	3	43	626698.3	6795365.6	0.21242	0.97718	-0.97718	0.21242	1.00
	3	44	627623.9	6795560.3	0.20017	0.97976	-0.97976	0.20017	1.00

3	45	628566.2	6795745.9	0.18848	0.98208	-0.98208	0.18848	1.00
3	46	629534.1	6795924.9	0.17670	0.98426	-0.98426	0.17670	1.00
3	47	630427.0	6796079.0	0.16587	0.98615	-0.98615	0.16587	1.00
4	27	610220.5	6788565.6	0.48437	0.87486	-0.87486	0.48437	1.00
4	40	623852.8	6793953.4	0.27212	0.96226	-0.96226	0.27212	1.00
4	41	624875.6	6794236.1	0.25910	0.96585	-0.96585	0.25910	1.00
4	42	625870.4	6794492.8	0.24376	0.96984	-0.96984	0.24376	1.00

Data Format A-4 Corners(4)

Example

```
* 07-Oct-07  3:49:15 PM
*    I     J       X       Y
     3     3 147268.3 211473.6 147427.3 211409.0 147474.5 211475.5 147312.4 211573.0
     4     3 147312.4 211573.0 147474.5 211475.5 147538.3 211555.3 147374.2 211687.1
     5     3 147374.2 211687.1 147538.3 211555.3 147622.8 211650.0 147452.3 211811.0
     6     3 147452.3 211811.0 147622.8 211650.0 147726.5 211758.3 147545.0 211943.4
     7     3 147545.0 211943.4 147726.5 211758.3 147843.6 211876.1 147649.1 212082.4
     8     3 147649.1 212082.4 147843.6 211876.1 147970.5 212002.1 147760.9 212226.5
     9     3 147760.9 212226.5 147970.5 212002.1 148102.9 212134.7 147876.0 212373.1
```

Grid Formats from Other Models Imported into CVLGrid

Data Format A-5 ECOMSED

ECOMSED is a three-dimensional hydrodynamic and sediment transport computer code developed by HydroQual.

Example of Grid

```
ECOMSED ST CHANNEL GRID   255X  161
 2 SIGMA LEVELS
   2
       0.00
      -1.00
 HORIZONTAL SEGMENTATION: 186.1318 M BY  283.3225 M
    257     163
 176    3   186.13180 283.32250      0.00    112.46     27.145591   -122.122904      0.00
```

177	3	187.18390	283.09130	0.00	112.44	27.147153	-122.123611	0.00
178	3	187.63180	282.70260	0.00	112.41	27.148721	-122.124321	0.00
179	3	188.10340	282.10040	0.00	112.38	27.150294	-122.125031	0.00
180	3	188.34230	281.25040	0.00	112.34	27.151870	-122.125741	0.00
181	3	188.27640	280.11520	0.00	112.29	27.153447	-122.126450	0.00
182	3	187.84130	278.66650	0.00	112.24	27.155023	-122.127157	0.00

Data Format A-6 Grid95

Grid95 is a public domain grid generator. The output from Grid95 can be directly imported into EFDC_Explorer's model generation function using the GRD saved results file. Refer to the user manual for Grid95 for details on the format.

Example of Grid

---

41 11

-17.500000000000000	-20.000000000000000
-19.489228298202900	-20.000000000000000
-21.687665562789850	-20.000000000000000
-24.117314492825130	-20.000000000000000
-26.802491831433730	-20.000000000000000
-29.770071735939730	-20.000000000000000
-33.049754743465460	-20.000000000000000
-36.674365023889780	-20.000000000000000
-40.680178895172740	-20.000000000000000
-45.107287888938710	-20.000000000000000
-50.000000000000000	-20.000000000000000
-17.388450867189730	-18.180965079699230
-19.352817413073470	-18.069105224173820

Data Format A-7 SEAGRID

SeaGrid is a Matlab 5.2+ application for generating an orthogonal grid within a curved perimeter, suitable for oceanographic modeling.

Example of Grid

Created by SeaGrid on 22-Dec-2015 14:52:03

Vertical Segmentation - Sigma Levels (KB)

2

0.00

-1.00

Horizontal Segmentation (IM & JM) -- Grid Information

257 163

176 3 186.13180 283.32250 0.00 112.46 0.00000 0.00000 -122.122904 27.145591

177 3 187.18390 283.09130 0.00 112.44 0.00000 0.00000 -122.123611 27.147153

178 3 187.63180 282.70260 0.00 112.41 0.00000 0.00000 -122.124321 27.148721

179 3 188.10340 282.10040 0.00 112.38 0.00000 0.00000 -122.125031 27.150294

180 3 188.34230 281.25040 0.00 112.34 0.00000 0.00000 -122.125741 27.151870

181 3 188.27640 280.11520 0.00 112.29 0.00000 0.00000 -122.126450 27.153447

Data Format A-8 CH3D

CH3D is a Curvilinear-grid Hydrodynamics 3D model developed originally by Dr. Y. Peter Sheng at the Aeronautical Research Associates of Princeton, Inc in the 1980s.

Example of Grid

CH3D GRID (FORT 15) FILE

404 761

0.733960125000E+06 0.530563984375E+05 0.274899997711E+02 1 1

0.734032812500E+06 0.531251015625E+05 0.275000000000E+02 2 1

0.734105375000E+06 0.531938007812E+05 0.274899997711E+02 3 1

0.734178125000E+06 0.532625000000E+05 0.274799995422E+02 4 1

0.734250812500E+06 0.533311992188E+05 0.274699993134E+02 5 1

0.734323375000E+06 0.533998984375E+05 0.274799995422E+02 6 1

0.734396125000E+06 0.534686015625E+05 0.274799995422E+02 7 1

Data Format A-9 Delft RGFGrid

Refer to the Delft RGFGrid user guide.

APPENDIX B
Mathematical Basis of CVLGrid

As mentioned previously, the Laplace equations are used by CVLGrid for grid generation. Therefore the family of orthogonal curves in a physical domain must be determined:

$$\xi = \xi(x,y), \eta = \eta(x,y) \tag{1}$$

They must satisfy the Laplace equations:

$$\frac{\partial^2 \xi}{\partial x^2} + \frac{\partial^2 \xi}{\partial y^2} = 0 \tag{2}$$

$$\frac{\partial^2 \eta}{\partial x^2} + \frac{\partial^2 \eta}{\partial y^2} = 0, \ \forall (x,y) \in \Omega \tag{3}$$

and the Dirichlet boundary conditions:

$$\xi(x,y)\big|_{(x,y)\in\Gamma} = \xi_0, \ \eta(x,y)\big|_{(x,y)\in\Gamma} = \eta_0 \tag{4}$$

in which Ω is the physical domain with the closed boundary. Since this domain is not rectangular, the equations are converted into the logical domain *D* of the rectangle. The family of orthogonal curves must then be determined:

$$x = x(\xi,\eta), y = y(\xi,\eta) \tag{5}$$

So that they satisfy the Laplace equations in the logical domain *D*:

$$a\frac{\partial^2 x}{\partial \xi^2} - 2b\frac{\partial^2 x}{\partial \xi \partial \eta} + c\frac{\partial^2 x}{\partial \eta^2} = 0 \tag{6}$$

$$a\frac{\partial^2 y}{\partial \xi^2} - 2b\frac{\partial^2 y}{\partial \xi \partial \eta} + c\frac{\partial^2 y}{\partial \eta^2} = 0, \ \forall (\xi,\eta) \in D \tag{7}$$

$$a = \left(\frac{\partial x}{\partial \eta}\right)^2 + \left(\frac{\partial y}{\partial \eta}\right)^2, \ b = \frac{\partial x}{\partial \xi}\frac{\partial x}{\partial \eta} + \frac{\partial y}{\partial \xi}\frac{\partial y}{\partial \eta}, \ c = \left(\frac{\partial x}{\partial \xi}\right)^2 + \left(\frac{\partial y}{\partial \xi}\right)^2$$

$$D = \{(\xi,\eta): 1 \le \xi \le I, 1 \le \eta \le J\}$$

with the boundary conditions on four sides of rectangle *D*:

$$x(\xi,\eta)\big|_{\xi=1} = x_1(\eta), \ y(\xi,\eta)\big|_{\xi=1} = y_1(\eta) \tag{8}$$

$$x(\xi,\eta)\big|_{\xi=I} = x_2(\eta), \ y(\xi,\eta)\big|_{\xi=I} = y_2(\eta) \tag{9}$$

$$x(\xi,\eta)\big|_{\eta=1} = x_3(\xi), \ y(\xi,\eta)\big|_{\eta=1} = y_3(\xi) \tag{10}$$

$$x(\xi,\eta)\big|_{\eta=J} = x_4(\xi), \ y(\xi,\eta)\big|_{\eta=J} = y_4(\xi) \tag{11}$$

in which (x,y) are the coordinates in physical space and being dependent variables, (ξ,η) the coordinates in the logical space *D*, x_i, y_i are given values on the boundaries.

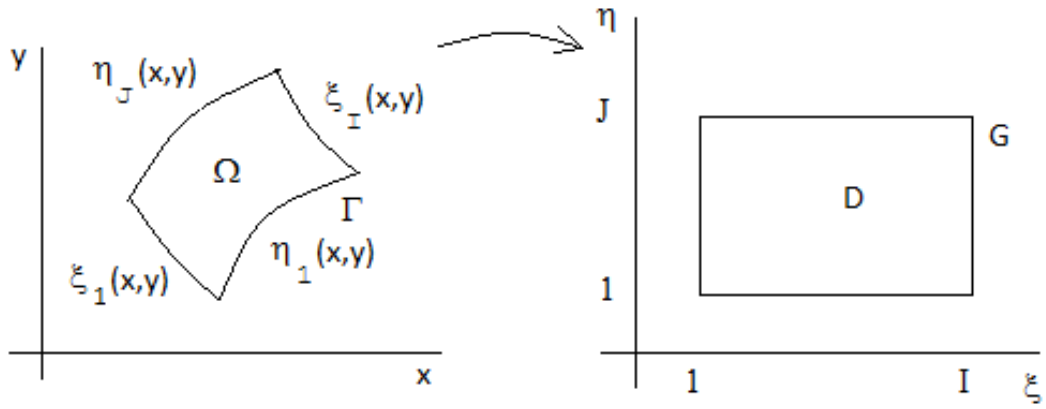


Figure 8-1 Sketches of physical and logical domains

The formulas of the scheme of central space difference for 9 points:

$$\left.\frac{\partial f}{\partial \xi}\right|_{(i,j)} = \frac{f_{i+1,j} - f_{i-1,j}}{2\Delta\xi}, \quad \left.\frac{\partial f}{\partial \eta}\right|_{(i,j)} = \frac{f_{i,j+1} - f_{i,j-1}}{2\Delta\eta} \tag{12}$$

$$\left.\frac{\partial^2 f}{\partial \xi^2}\right|_{(i,j)} = \frac{f_{i+1,j} - 2f_{i,j} + f_{i-1,j}}{\Delta\xi^2}, \quad \left.\frac{\partial^2 f}{\partial \eta^2}\right|_{(i,j)} = \frac{f_{i,j+1} - 2f_{i,j} + f_{i,j-1}}{\Delta\eta^2} \tag{13}$$

$$\left.\frac{\partial^2 f}{\partial \xi \partial \eta}\right|_{(i,j)} = \frac{f_{i+1,j+1} - f_{i+1,j-1} - f_{i-1,j+1} + f_{i-1,j-1}}{4\Delta\xi\Delta\eta} \tag{14}$$

Substituting the formulas (12)-(14) into Equations (6)-(7) we have the formulas of iteration solutions based on SOR method:

$$x_{i,j}^{(n+1)} = x_{i,j}^{(n)} + \omega\Delta x_{i,j}^{(n)} \tag{15}$$

$$y_{i,j}^{(n+1)} = y_{i,j}^{(n)} + \omega\Delta y_{i,j}^{(n)}, \quad \forall i \in [1, I], \forall j \in [1, J] \tag{16}$$

in which $x_{i,j}^{(n)}, y_{i,j}^{(n)}$ are known values at the previous iteration, $\Delta x_{i,j}^{(n)}, \Delta y_{i,j}^{(n)}$ depend on $x_{i,j}^{(n)}, y_{i,j}^{(n)}$ of 9 points (*i*, *j*), ω is the over-relaxation factor, the value of which is a constant, $1 < \omega < 2$.

The initial values for the interiors of grid $x_{i,j}^{(0)}, y_{i,j}^{(0)}$ can be directly obtained from transfinite interpolation based on the data sets of given boundary points (*x*, *y*).

For a rectangular domain *D* the optimum value of the over-relaxation factor is determined by Frankel's formula (Hoffman, J.D., 2001) as follows:

$$\omega_{opt} = 2\left(\frac{1-\sqrt{1-\lambda}}{\lambda}\right) \qquad (17)$$

$$\lambda = \left(\frac{\cos\left(\frac{\pi}{I-1}\right)+\beta^2\cos\left(\frac{\pi}{J-1}\right)}{1+\beta^2}\right)^2, \quad \beta = \Delta\xi/\Delta\eta$$

The solutions of *x* and *y* are obtained when the following conditions are satisfied:

$$\underset{i,j}{Max}\left|\Delta x_{i,j}^{(n)}\right| \le \varepsilon \text{ and } \underset{i,j}{Max}\left|\Delta y_{i,j}^{(n)}\right| \le \varepsilon \qquad (18)$$

Then we have:

$$x_{i,j} = x_{i,j}^{(n+1)} \text{ and } y_{i,j} = y_{i,j}^{(n+1)}, \quad \forall i \in [1,I], \forall j \in [1,J]$$